

Quantities, Units and Modelling

Mechanics is a model of the real world. Know the units, know which quantities have a direction, and know what each modelling word lets you assume.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Quantities, Units and Modelling

01 SI units

kg, m, s, N $\text{km h}^{-1} \div 3.6$ $\text{g} \rightarrow \text{kg}$

02 Scalars and vectors

distance / displacement speed / velocity signs

03 Modelling words

particle light, inextensible smooth, rough, uniform

04 Weight and accuracy

$W = mg$ $g = 9.8: 2 \text{ or } 3 \text{ s.f.}$

THE FOUR BASIC SI UNITS OF M1

mass: kg length: m time: s force: $N = kg \, m \, s^{-2}$
 change everything to these before you start $km \, h^{-1} \div 3.6 = m \, s^{-1}$

01 SI Units and Conversions

Every formula in M1 works only in SI units. Before you write an equation, change grams to kg, cm to m, km to m, tonnes to kg, minutes and hours to seconds, and $km \, h^{-1}$ to $m \, s^{-1}$.

WHEN YOU SEE → YOU DO

“54 $km \, h^{-1}$ ” → $\div 3.6$: 15 $m \, s^{-1}$

“mass 250 g”, “2 tonnes” → 0.25 kg, 2000 kg

“12 cm”, “4.8 km” → 0.12 m, 4800 m

“3 minutes” → 180 s

“stating the units of your answer” → impulse N s • moment N m • speed $m \, s^{-1}$

quantity	unit	formula
velocity	$m \, s^{-1}$	$\frac{\text{displacement}}{\text{time}}$
acceleration	$m \, s^{-2}$	$\frac{\text{change of velocity}}{\text{time}}$
force, weight	N	$F = ma$, $W = mg$
momentum, impulse	N s	mv , Ft
moment	N m	$F \times d$

IDEA 01

Practise: SI Units and Conversions

Questions written to practise exactly this idea.

Q1 DR ESLAM Speed units

M1 · TOPIC 1 2 marks

Convert (a) 72 km h^{-1} to m s^{-1} (b) 15 m s^{-1} to km h^{-1} .

$\text{km h}^{-1} \div 3.6 = \text{m s}^{-1}$.

Q2 DR ESLAM Mass and length

M1 · TOPIC 1 2 marks

Write in SI units: (a) 450 g (b) 2.4 tonnes (c) 35 cm (d) 1.2 km.

Q3 DR ESLAM Units of a quantity

M1 · TOPIC 1 2 marks

State the SI units of (a) acceleration (b) force (c) momentum (d) moment of a force.

02 Scalars and Vectors

A scalar has only a size (distance, speed, mass, time). A vector has a size **and** a direction (displacement, velocity, acceleration, force, momentum). In a straight line the direction is just a sign: choose a positive direction and keep it.

WHEN YOU SEE → YOU DO

“speed” vs “velocity”

→ speed is the size of the velocity (never negative)

“distance” vs “displacement”

→ distance adds every part; displacement is start to finish

“rebounds”, “reverses direction”

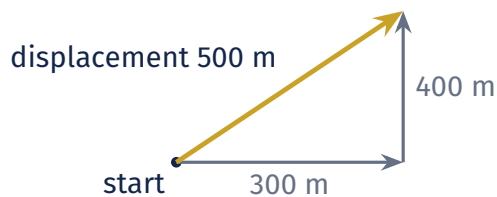
→ the velocity changes sign

“average speed”

→ total distance ÷ total time

a vector in i, j

→ size by Pythagoras (Topic 2)



distance = 700 m (scalar)

displacement = 500 m on 037° (vector)

IDEA 02

Practise: Scalars and Vectors

Questions written to practise exactly this idea.

Q4 DR ESLAM Scalar or vector?

M1 · TOPIC 1 2 marks

Sort into scalars and vectors: distance, displacement, speed, velocity, mass, weight, time, acceleration.

Q5 DR ESLAM Distance vs displacement

M1 · TOPIC 1 3 marks

A girl walks 300 m east then 400 m north in 500 s. Find her distance, her displacement (size and bearing), her average speed and the size of her average velocity.

Q6 DR ESLAM Velocity with a sign

M1 · TOPIC 1 2 marks

A ball moves up at 6 m s^{-1} and later down at 4 m s^{-1} . Taking up as positive, write both velocities and the change in velocity.

03 Modelling Words and What They Mean

The exam often asks: “state how you used the fact that ...”. Each modelling word gives you one thing in the working. Learn the pairs below — they are free marks.

WHEN YOU SEE → YOU DO

“modelled as a particle”

→ all the mass at one point; no size, no turning, no air resistance

“light string / rod”

→ no mass: the tension is the same all along it

“inextensible string”

→ both ends move with the same speed and acceleration

“smooth pulley” / “smooth surface”

→ no friction: same tension on both sides / $F = 0$

“rough surface”

→ friction acts: $F \leq \mu R$

“uniform rod”

→ weight acts at the midpoint

“rigid rod” / “beam stays straight”

→ it does not bend, so distances stay fixed

“state one way the model could be improved”

→ air resistance, size of the object, resistance varying with speed

ZONE THE BODY

particle, uniform, rigid

where the weight acts;
no size

ZONE THE STRING

light, inextensible

same T , same a

ZONE THE SURFACE

smooth or rough

$F = 0$ or $F \leq \mu R$

IDEA 03

Practise: Modelling Words and What They Mean

Questions written to practise exactly this idea.

Q7 **DR ESLAM** What does it mean?

M1 · TOPIC 1 4 marks

State what each modelling word means for your working: (a) particle (b) light string (c) inextensible string (d) smooth pulley (e) uniform rod (f) rough surface.

Q8 **DR ESLAM** Improve the model

M1 · TOPIC 1 2 marks

A car is modelled as a particle moving against a constant resistance. State two ways the model could be made more realistic.

04 Weight, g and Accuracy

Weight is a force: $W = mg$, acting straight down, in newtons. Mass (kg) never changes; weight depends on g . Take $g = 9.8 \text{ m s}^{-2}$ unless told otherwise, and because 9.8 has 2 significant figures, give final answers to 2 or 3 s.f.

WHEN YOU SEE → YOU DO

“a mass of 3 kg”

→ weight $3g = 29.4 \text{ N}$

“in terms of g ”

→ leave g as a letter

“give your answer to an appropriate degree of accuracy”

→ 2 or 3 s.f. when $g = 9.8$ was used

an exact form is asked (“show that $a = \frac{g}{5}$ ”)

→ keep g and fractions — no decimals

DR ESLAM’S TIP Keep full calculator values until the last line, then round once. Rounding in the middle is the most common way to lose the final accuracy mark.

IDEA 04

Practise: Weight, g and Accuracy

Questions written to practise exactly this idea.

Q9 DR ESLAM Weight and mass

M1 · TOPIC 1 2 marks

Find the weight of a 65 kg person. What is their mass on the Moon, where $g = 1.6 \text{ m s}^{-2}$, and their weight there?

Q10 DR ESLAM How many figures?

M1 · TOPIC 1 2 marks

A calculation using $g = 9.8$ gives $T = 23.5198 \text{ N}$. How should you give the answer, and why?

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------------|---|
| 01 | SI units | kg, m, s, N; $\text{km h}^{-1} \div 3.6$. |
| 02 | Scalars, vectors | Vectors have direction: choose + and keep it. |
| 03 | Modelling words | Particle, light, inextensible, smooth, rough, uniform. |
| 04 | Weight, accuracy | $W = mg$; $g = 9.8 \rightarrow 2 \text{ or } 3 \text{ s.f.}$ |

BOTTOM LINE SI units first • a sign for every direction • each modelling word gives one fact.

Answers

Quantities, Units and Modelling • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SI UNITS AND CONVERSIONS

- Q1** Dr Eslam • Speed units (a) 20 m s^{-1} (b) 54 km h^{-1}
- Q2** Dr Eslam • Mass and length (a) 0.45 kg (b) 2400 kg (c) 0.35 m (d) 1200 m
- Q3** Dr Eslam • Units of a quantity (a) m s^{-2} (b) $\text{N} (= \text{kg m s}^{-2})$ (c) $\text{N s} (= \text{kg m s}^{-1})$ (d) N m

IDEA 02 • SCALARS AND VECTORS

- Q4** Dr Eslam • Scalar or vector? scalars: distance, speed, mass, time; vectors: displacement, velocity, weight, acceleration
700 m; 500 m on 037° ; 1.4 m s^{-1} ; 1 m s^{-1}
- Q5** Dr Eslam • Distance vs displacement
- Q6** Dr Eslam • Velocity with a sign +6, -4; change = -10 m s^{-1}

IDEA 03 • MODELLING WORDS AND WHAT THEY MEAN

- Q7** Dr Eslam • What does it mean? (a) mass at one point, no size or rotation (b) no mass: same tension along it (c) same speed and acceleration at both ends (d) no friction: same tension on both sides (e) weight at the midpoint (f) friction acts, $F \leq \mu R$
- Q8** Dr Eslam • Improve the model e.g. resistance increasing with speed; include air resistance; car has size (not a particle); driving force not constant

IDEA 04 • WEIGHT, g AND ACCURACY

- Q9** Dr Eslam · Weight and mass 637 N; mass still 65 kg; weight 104 N
- Q10** Dr Eslam · How many figures? 23.5 N (3 s.f.) or 24 N (2 s.f.) — $g = 9.8$ is given to 2 s.f.



DR ESLAM AHMED
eliteigcse.com

Working with Vectors

Every vector is an $\mathbf{i}$ part and a $\mathbf{j}$ part. Sketch it, then Pythagoras for the size and $\tan$ for the direction. Position = start + velocity $\times$ time.

THE TOPIC ON ONE MAP

Working with Vectors

01 — Magnitude, direction, bearings

$|\mathbf{v}|$ angle with $\mathbf{i}$ or $\mathbf{j}$ bearing

02 — Forces as vectors

equilibrium: sum = 0 $\mathbf{F} = m\mathbf{a}$ parallel to a vector

03 — Constant velocity

$\mathbf{r}_0 + \mathbf{v}t$ due east / north of

04 — Two moving objects

$\vec{AB} = \mathbf{b} - \mathbf{a}$ collide closest

THE THREE VECTOR MOVES

size = $\sqrt{a^2 + b^2}$

direction: $\tan \theta = \frac{b}{a}$ from a sketch

parallel to

$p\mathbf{i} + q\mathbf{j}$: $\frac{b}{a} = \frac{q}{p}$

$\mathbf{i}$ east, $\mathbf{j}$ north

bearing: clockwise from north, three figures

01 Magnitude, Direction and Bearings

Always sketch the vector on a small $\mathbf{i}$ - $\mathbf{j}$ cross first. The size comes from Pythagoras, the angle from $\tan$ in the right-angled triangle, and the sketch tells you how to turn that angle into what the question asks (angle with $\mathbf{i}$, angle with $\mathbf{j}$, or a bearing).

WHEN YOU SEE → YOU DO

“exact value of the magnitude”

→ $\sqrt{a^2 + b^2}$ — leave the surd

“angle between $\mathbf{F}$ and $\mathbf{i} + \mathbf{j}$ ”

→ find each direction, then subtract

“ $\mathbf{F}$ acts on a bearing of 120° ”

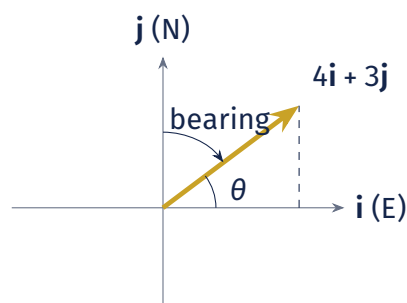
→ $F \sin 120^\circ \mathbf{i} + F \cos 120^\circ \mathbf{j}$

“give the direction as a bearing”

→ sketch; clockwise from north

two forces with an angle between them, no $\mathbf{i}, \mathbf{j}$

→ triangle of forces: cosine rule, sine rule



$$\text{size} = 5, \tan \theta = \frac{3}{4}, \theta = 36.9^\circ$$

$$\text{bearing} = 90^\circ - 36.9^\circ = 053^\circ$$

WORKED MODEL • JUNE 2024

P: 10 N due west. **Q:** 8 N on a bearing of 330° . Find $|\mathbf{P} + \mathbf{Q}|$.

1 $\mathbf{Q} = 8 \sin 330^\circ \mathbf{i} + 8 \cos 330^\circ \mathbf{j} = -4\mathbf{i} + 4\sqrt{3}\mathbf{j}$

bearing to $\mathbf{i}, \mathbf{j}$

2 $\mathbf{F} = -14\mathbf{i} + 4\sqrt{3}\mathbf{j}$

add

ANSWER $|\mathbf{F}| = \sqrt{196 + 48} = 2\sqrt{61} \approx 15.6 \text{ N}$

IDEA 01

Practise: Magnitude, Direction and Bearings

My questions first, then real past papers.

Q1 **DR ESLAM** Magnitude and bearing

M1 • TOPIC 2 3 marks

$\mathbf{i}$ is east, $\mathbf{j}$ is north. Find the magnitude of $\mathbf{F} = (-4\mathbf{i} + 3\mathbf{j})$ N and its bearing. Sketch it first: west 4, north 3.

Q2 **DR ESLAM** From a bearing to $\mathbf{i}, \mathbf{j}$

M1 • TOPIC 2 2 marks

A force of 10 N acts on a bearing of 060° . Write it in the form $a\mathbf{i} + b\mathbf{j}$.

Q3 **DR ESLAM** Angle between two vectors

M1 • TOPIC 2 3 marks

Find the angle between $3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{i} + \mathbf{j}$.

PAST PAPERS

Q4 PAST PAPER Pearson IAL WME01 • October 2019 • Q7

M1 • TOPIC 2 7 marks

7. Two forces, **F** and **G**, act on a particle. The force **F** has magnitude 4 N and acts in a direction with a bearing of 120° and the force **G** has magnitude 6 N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of **P**
- (ii) the direction of **P**, giving your answer as a bearing to the nearest degree.

(7)

Q5 **PAST PAPER** Pearson IAL WME01 • January 2020 • Q6**M1 • TOPIC 2** **11 marks**

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.

(a) Find the exact value of the magnitude of $\mathbf{F}$.

(2)

(b) Find, in degrees, the size of the angle between the direction of $\mathbf{F}$ and the direction of the vector $(\mathbf{i} + \mathbf{j})$.

(4)

The resultant of the force $\mathbf{F}$ and the force $(-15\mathbf{i} + a\mathbf{j})\text{N}$, where a is a constant, is parallel to, but in the opposite direction to, the vector $(2\mathbf{i} - 3\mathbf{j})$.

(c) Find the value of a .

(5)

Q6 **PAST PAPER** Pearson IAL WME01 • January 2021 • Q5**M1 • TOPIC 2 7 marks**

5. A particle is acted upon by two forces **F** and **G**. The force **F** has magnitude 8 N and acts in a direction with a bearing of 240° . The force **G** has magnitude 10 N and acts due South.

Given that $\mathbf{R} = \mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of **R**,
- (ii) the direction of **R**, giving your answer as a bearing to the nearest degree.

(7)

Q7**PAST PAPER**

Pearson IAL WME01 • January 2024 • Q4

M1 • TOPIC 2 8 marks

4.

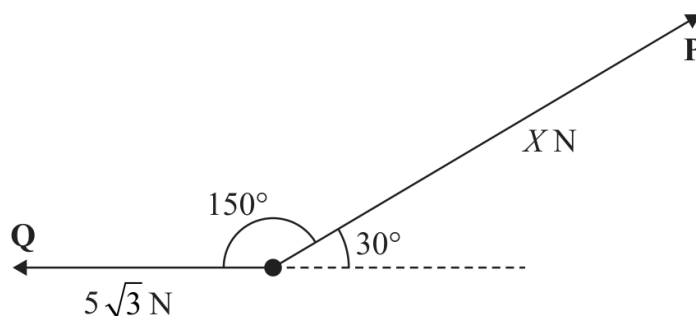
**Figure 3**

Figure 3 shows two horizontal forces **P** and **Q** acting on a particle.

The angle between the direction of **P** and the direction of **Q** is 150°

Force **P** has magnitude X newtons.

Force **Q** has magnitude $5\sqrt{3}$ N.

The resultant of **P** and **Q** has magnitude $\sqrt{129}$ N.

Find

- (i) the value of X .
- (ii) the angle between **Q** and the resultant, giving your answer to the nearest degree.

(8)

Q8 **PAST PAPER** Pearson IAL WME01 · June 2024 · Q2**M1 · TOPIC 2** **4 marks**

2. Two forces, **P** and **Q**, act on a particle.

- **P** has magnitude 10 N and acts due west
- **Q** has magnitude 8 N and acts on a bearing of 330°

Given that $\mathbf{F} = \mathbf{P} + \mathbf{Q}$, find the magnitude of **F**.

(4)

Q9**PAST PAPER**

Pearson IAL WME01 · June 2026 · Q6

M1 · TOPIC 2**9 marks**

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

(a) Find the magnitude of $\mathbf{R}$.

(2)

(b) Find, in degrees, the angle between the direction of $\mathbf{R}$ and the direction of the vector $(2\mathbf{i} + \mathbf{j})$. Show your method clearly.

(3)

Given that $\mathbf{F}_1$ is parallel to the vector $(-\mathbf{i} + 2\mathbf{j})$ and that $\mathbf{F}_2$ is parallel to the vector $(2\mathbf{i} - 3\mathbf{j})$,

(c) find $\mathbf{F}_1$ and $\mathbf{F}_2$

(4)

(Total for Question 6 is 9 marks)

02 Forces as Vectors: Equilibrium and $F = ma$

Add all the forces as vectors. In equilibrium the sum is $\mathbf{0}$ (the $\mathbf{i}$ total is 0 and the $\mathbf{j}$ total is 0). Otherwise the sum is $m\mathbf{a}$. “In the direction of $p\mathbf{i} + q\mathbf{j}$ ” gives a ratio equation.

WHEN YOU SEE → YOU DO

“rests in equilibrium under three forces”

$$\rightarrow \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$$

“the acceleration has magnitude 3.25”

$$\rightarrow |\mathbf{R}| = m \times 3.25$$

“the resultant is parallel to $(2\mathbf{i} - 3\mathbf{j})$ ”

$$\rightarrow \mathbf{R} = \lambda(2\mathbf{i} - 3\mathbf{j}) \text{ — compare parts}$$

“in the opposite direction to”

$$\rightarrow \lambda < 0$$

“moves in the direction of $(\mathbf{i} - 2\mathbf{j})$ ” from rest

$$\rightarrow \text{the resultant force has that direction}$$

Add the forces

Sum = 0 or $m\mathbf{a}$

Compare $\mathbf{i}$ and $\mathbf{j}$ parts

Solve

WORKED MODEL • JANUARY 2026

$\mathbf{F}_1 = 2p\mathbf{i} - 3p\mathbf{j}$, $\mathbf{F}_2 = q\mathbf{i} + \mathbf{j}$, $\mathbf{F}_3 = 3\mathbf{i} - q\mathbf{j}$. The resultant is parallel to $4\mathbf{i} - 5\mathbf{j}$. Show $2p - q = 19$.

1 $\mathbf{R} = (2p + q + 3)\mathbf{i} + (1 - 3p - q)\mathbf{j}$

add

2 $-5(2p + q + 3) = 4(1 - 3p - q)$

parts in the ratio 4 : -5

ANSWER $2p - q = 19$

× **TRAP** Writing “parallel” as “equal”. Parallel means a multiple: compare the ratio of the parts.

IDEA 02

Practise: Forces as Vectors: Equilibrium and $F = ma$

My questions first, then real past papers.

Q10 DR ESLAM Three forces in equilibrium

M1 · TOPIC 2 2 marks

$(2\mathbf{i} - \mathbf{j})$, $(p\mathbf{i} + 3\mathbf{j})$ and $(-5\mathbf{i} + q\mathbf{j})$ N are in equilibrium. Find p and q .

Q11 DR ESLAM Acceleration from forces

M1 · TOPIC 2 3 marks

A 2 kg particle is acted on by $(3\mathbf{i} + 2\mathbf{j})$ N and $(\mathbf{i} - 6\mathbf{j})$ N. Find its acceleration and the magnitude.

Q12 DR ESLAM Parallel to a vector

M1 · TOPIC 2 3 marks

The resultant of $(p\mathbf{i} + 2\mathbf{j})$ and $(3\mathbf{i} + p\mathbf{j})$ is parallel to $\mathbf{i} + 2\mathbf{j}$. Find p .
Parallel to $\mathbf{i} + 2\mathbf{j}$: the $\mathbf{j}$ part is twice the $\mathbf{i}$ part.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 • June 2021 • Q3

M1 • TOPIC 2 9 marks

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

Given that P rests in equilibrium on a smooth horizontal surface under the action of these three forces,

- (a) find the size of the angle between the direction of $\mathbf{F}_3$ and the direction of $-\mathbf{j}$. (4)

The force $\mathbf{F}_3$ is now removed and replaced by the force $\mathbf{F}_4$ given by $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})\text{N}$, where λ is a positive constant.

When the three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_4$ act on P , the acceleration of P has magnitude 3.25 ms^{-2}

- (b) Find the value of λ . (5)

Q14 PAST PAPER Pearson IAL WME01 • January 2022 • Q6

M1 • TOPIC 2 12 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 2 kg moves under the action of two forces, $(p\mathbf{i} + q\mathbf{j})\text{ N}$ and $(2q\mathbf{i} + p\mathbf{j})\text{ N}$, where p and q are constants.

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j})\text{ m s}^{-2}$

(a) find the value of p and the value of q .

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector $\mathbf{j}$.

(2)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

At $t = T$ seconds, P is moving in the direction of the vector $(11\mathbf{i} - 13\mathbf{j})$.

(c) Find the value of T .

(5)

Q15**PAST PAPER**

Pearson IAL WME01 · October 2022 · Q6

M1 · TOPIC 2 9 marks

6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$, where p and q are constants, are applied to A .

Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$

(4)

Given that $p = 5$

(b) find the speed of A at time $t = 4$ seconds.

(5)

Q16 PAST PAPER Pearson IAL WME01 · January 2023 · Q3

M1 · TOPIC 2 10 marks

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

- (a) Find the speed of P at time $t = 2$ seconds.

(3)

- (b) Find the size of the angle between the direction of $\mathbf{i}$ and the direction of motion of P at time $t = 2$ seconds.

(3)

At time $t = T$ seconds, P is moving in the direction of vector $(2\mathbf{i} - 3\mathbf{j})$

- (c) Find the value of T

(4)

Q17 PAST PAPER Pearson IAL WME01 • June 2023 • Q2

M1 • TOPIC 2 10 marks

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1 \text{ N}$, $\mathbf{F}_2 \text{ N}$ and $\mathbf{F}_3 \text{ N}$ where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

Given that P remains in equilibrium,

- (a) find $\mathbf{F}_3$ in terms of c , $\mathbf{i}$ and $\mathbf{j}$.

(2)

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

- (b) find the size of the angle between the direction of $\mathbf{i}$ and the direction of the resultant force acting on P .

(4)

The mass of P is $m \text{ kg}$.

Given that the magnitude of the acceleration of P is 8.5 m s^{-2}

- (c) find the value of m .

(4)

Q18 **PAST PAPER** Pearson IAL WME01 • January 2025 • Q1**M1 • TOPIC 2 8 marks**

1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{N}$$

- (a) Find the magnitude of the acceleration of the particle.

(4)

A fourth force, $\mathbf{F}_4 = (p\mathbf{i} + p\mathbf{j})\text{N}$, where p is a constant, is added.

The resultant of the four forces acts in the direction of the vector $(7\mathbf{i} + 2\mathbf{j})$.

- (b) Find the value of p .

(4)

Q19**PAST PAPER**

Pearson IAL WME01 · October 2025 · Q3

M1 · TOPIC 2 14 marks

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg moves on a smooth horizontal surface under the action of two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (-2\mathbf{i} + 3\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (4\mathbf{i} + 2\mathbf{j})\text{ N}$.

(a) Find the acceleration of P .

(3)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

(b) Find the speed of P when $t = 3$ seconds.

(4)

An additional force, $\mathbf{F}_3 = (b\mathbf{i} + c\mathbf{j})\text{ N}$, is applied to P .

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ is equal to $\lambda(\mathbf{i} + \mathbf{j})\text{ N}$, where λ is a constant.

(c) Show that $b - c = 3$

(3)

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ has magnitude $10\sqrt{2}\text{ N}$.

(d) Find the two possible $\mathbf{F}_3$ forces.

(4)

Q20 **PAST PAPER** Pearson IAL WME01 • January 2026 • Q3**M1 • TOPIC 2** **8 marks**

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The resultant force acts in the direction of the vector $(4\mathbf{i} - 5\mathbf{j})$

- (a) Show that

$$2p - q = 19 \quad (4)$$

The mass of the particle is 0.5 kg.

Given that $p = 7$

- (b) find the acceleration of the particle. (4)

03 Constant Velocity: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

Position at time t = where it started + velocity $\times t$. If you know two positions, the velocity is (change of position) $\div$ (time between them). “Due east of O ” means the $\mathbf{j}$ part is 0.

WHEN YOU SEE $\rightarrow$ YOU DO

“at 7 am ... at 7.36 am”

$\rightarrow$ time in hours: 0.6 h

“find the velocity” from two positions

$$\rightarrow \mathbf{v} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}$$

“due east / due north of O ”

$\rightarrow$ $\mathbf{j}$ part = 0 / $\mathbf{i}$ part = 0

“north-east of O ”

$\rightarrow$ $\mathbf{i}$ part = $\mathbf{j}$ part (both positive)

“distance from O is 9 km or less”

$\rightarrow |\mathbf{r}|^2 \leq 81$ — a quadratic in t

“ OP perpendicular to the direction of motion”

$\rightarrow \mathbf{r} \cdot \mathbf{v} = 0$, or the closest-point method

WORKED MODEL • JUNE 2025

$\mathbf{r} = -5\mathbf{i} + 16\mathbf{j}$ at $t = 2$ and $10\mathbf{i} + 4\mathbf{j}$ at $t = 5$. Find $\mathbf{v}$ and $\mathbf{r}$ at time t .

1 $\mathbf{v} = \frac{15\mathbf{i} - 12\mathbf{j}}{3} = 5\mathbf{i} - 4\mathbf{j}$

change $\div$ time

2 $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j}) + (t - 2)(5\mathbf{i} - 4\mathbf{j})$

start at $t = 2$

ANSWER $\mathbf{r} = (5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}$

IDEA 03

Practise: Constant Velocity: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

My questions first, then real past papers.

Q21 DR ESLAM Where and when

M1 • TOPIC 2 3 marks

A boat starts at $(2\mathbf{i} + 5\mathbf{j})$ km with velocity $(3\mathbf{i} - \mathbf{j})$ km h⁻¹. Find its position after t hours and when it is due east of O .

Q22 DR ESLAM Velocity from two positions

M1 • TOPIC 2 2 marks

A particle passes $(\mathbf{i} + 2\mathbf{j})$ m at $t = 0$ and $(7\mathbf{i} - 6\mathbf{j})$ m at $t = 2$. Find its velocity and speed.

PAST PAPERS

Q23 PAST PAPER Pearson IAL WME01 • October 2021 • Q4

M1 • TOPIC 2 8 marks

4. The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O , is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

- (a) Find, to the nearest degree, the size of the angle between $\mathbf{r}$ and the vector $\mathbf{j}$, when $t = 2$
(3)

- (b) Find the values of t for which the distance of P from O is 2.5 m.
(5)

Q24 **PAST PAPER** Pearson IAL WME01 • October 2021 • Q8**M1 • TOPIC 2 12 marks**

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

- (a) Show that the velocity of the ship is $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$ (2)

- (b) Find the position vector of the ship t hours after leaving port. (2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse. (4)

- (d) Find the position vector of the ship when it is due south of the lighthouse. (4)

(Total 12 marks)

Q25**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q3

M1 · TOPIC 2**11 marks**

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin.]

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

At time t hours after 1 pm, the position vector of A is $\mathbf{r}_A$ km.

- (a) Show that $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$ (4)

The speed of A is $V \text{ ms}^{-1}$

- (b) Find the value of V . (2)

A ship B is moving with constant velocity $(20\mathbf{i} - 6\mathbf{j}) \text{ km h}^{-1}$

At 1 pm, the position vector of B is $(35\mathbf{i} + 51\mathbf{j})$ km.

At 2:30 pm, B passes through the point P .

- (c) Show that A also passes through P . (5)

Q26 **PAST PAPER** Pearson IAL WME01 · January 2025 · Q5**M1 · TOPIC 2** **7 marks**

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target.
The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

- (a) Find the velocity of the ball.

(3)

The position vector of the target is $(13\mathbf{i} - 5\mathbf{j})$ m.

- (b) Use the model to find the distance of B from the target at time $t = 7$ s.

(4)

Q27**PAST PAPER**

Pearson IAL WME01 · June 2025 · Q2

M1 · TOPIC 2 7 marks

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of A .

(2)

(b) Find an expression for $\mathbf{r}$ at time t seconds.

Give your answer in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are functions of t

(2)

At time t seconds, the position vector of B is given by $\mathbf{s}$ metres where

$$\mathbf{s} = -5\mathbf{i} + 7\mathbf{j} + t(2\mathbf{i} - 3\mathbf{j})$$

(c) Find, to the nearest degree, the bearing of B from A when $t = 5$

(3)

Q28 PAST PAPER Pearson IAL WME01 · June 2026 · Q7

M1 · TOPIC 2 15 marks

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At time $t = 1.5$ hours after noon, the boat passes through the point with position vector $(11\mathbf{i} + 3.5\mathbf{j}) \text{ km}$.

- (a) Show that, at time t hours after noon, the position vector of the boat is

$$[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}] \text{ km} \quad (3)$$

The boat is visible from O when it is 9 km or less from O .

- (b) Find the length of time for which the boat is visible from O , showing your method clearly. (4)

- (c) Find the value of t when the boat is northwest of O . (2)

At the instant when the boat is at the point P , OP is perpendicular to the direction in which the boat is moving.

- (d) Show that $OP = \frac{18\sqrt{5}}{5} \text{ km}$

(Solutions relying on calculator technology are not acceptable.) (6)

(Total for Question 7 is 15 marks)

04 Two Moving Objects: Meet, Closest, Bearing

Write both position vectors in t . The vector from A to B is $\vec{AB} = \mathbf{b} - \mathbf{a}$. They collide only if *both* parts are zero at the *same* time. Their distance is $|\vec{AB}|$; the closest approach is the minimum of $|\vec{AB}|^2$ (a quadratic in t — complete the square).

WHEN YOU SEE → YOU DO

“show that $\vec{AB} = \dots$ ”

→ $\mathbf{b} - \mathbf{a}$, collect $\mathbf{i}$ and $\mathbf{j}$

“verify that they would collide at 13:30”

→ same position at $t = 1.5$

“show that they never collide”

→ the two parts are not zero together

“shortest distance between the ships”

→ minimise $|\vec{AB}|^2$: complete the square

“the bearing of B from A ”

→ sketch $\vec{AB}$, measure from north at A

“ B is north-east of A ”

→ $\vec{AB}$ has equal positive parts

WORKED MODEL • JANUARY 2022

$\vec{AB} = (12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}$. Find the shortest distance.

$$1 \quad |\vec{AB}|^2 = (12t - 10)^2 + (24t - 10)^2 = 720t^2 - 720t + 200$$

square and add

$$2 \quad = 720\left(t - \frac{1}{2}\right)^2 + 20$$

complete the square

ANSWER least $|AB| = \sqrt{20} = 2\sqrt{5} \approx 4.5 \text{ km}$ at $t = \frac{1}{2}$

× **TRAP** Saying they collide because the $\mathbf{i}$ parts are equal at one time. Both parts must match at that same time.

IDEA 04

Practise: Two Moving Objects: Meet, Closest, Bearing

My questions first, then real past papers.

Q29 DR ESLAM Collide or not? Closest?

M1 • TOPIC 2 5 marks

A starts at O with velocity $(4\mathbf{i} + 2\mathbf{j})$; B starts at $10\mathbf{i}$ with velocity $(2\mathbf{i} + 4\mathbf{j})$ (km, h). Find $\vec{AB}$, decide whether they collide, and find their least distance apart.

$\vec{AB} = \mathbf{b} - \mathbf{a}$. Collide: both parts zero at the SAME t .

PAST PAPERS

Q30 **PAST PAPER** Pearson IAL WME01 • October 2019 • Q8**M1 • TOPIC 2** **14 marks**

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (a) Find the velocity of B in the form $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$ (3)
- (b) Find expressions for $\mathbf{r}$ and $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

Q31 PAST PAPER Pearson IAL WME01 • January 2021 • Q6

M1 • TOPIC 2 12 marks

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{m s}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{m s}^{-1}$

- (a) Find the position vector of Agatha at time t seconds.

(2)

At time $t = 6$ seconds, Agatha passes through the point P .

- (b) Show that Brionie also passes through P and find the value of t when this occurs.

(4)

At time t seconds, Agatha is at the point A and Brionie is at the point B .

- (c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$

(2)

- (d) Find the distance between the two girls when they are closest together.

(4)

Q32 PAST PAPER Pearson IAL WME01 · January 2022 · Q8

M1 · TOPIC 2 14 marks

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ km h}^{-1}$

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j}) \text{ km}$.

- (a) Find $\mathbf{r}$ in terms of t .

(2)

Another ship B moves with constant velocity $(15\mathbf{i} + 14\mathbf{j}) \text{ km h}^{-1}$

At time $t = 0$, B is at the point with position vector $(3\mathbf{i} - 5\mathbf{j}) \text{ km}$.

- (b) Show that, at time t hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km}$$

(4)

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

- (d) find the bearing of ship B from ship A when the ships are closest.

(2)

(Total 14 marks)

Q33**PAST PAPER**

Pearson IAL WME01 • June 2022 • Q8

M1 • TOPIC 2 17 marks

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ m s}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the direction in which Q is travelling, giving your answer as a bearing.

(2)

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j} \text{ m}$.

At time t seconds, the position vector of P is $\mathbf{p} \text{ m}$ and the position vector of Q is $\mathbf{q} \text{ m}$.

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (c) Find the bearing of P from Q when $t = 10$

(2)

- (d) Find the distance between P and Q when Q is north east of P

(5)

- (e) Find the times when P and Q are 200 m apart.

(3)**(Total 17 marks)**

Q34 PAST PAPER Pearson IAL WME01 · October 2022 · Q8

M1 · TOPIC 2 16 marks

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of A .

(2)

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$ and B is at the point with position vector $(16\mathbf{i} + 6\mathbf{j})\text{ km}$.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of p and the value of q .

(7)

- (c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

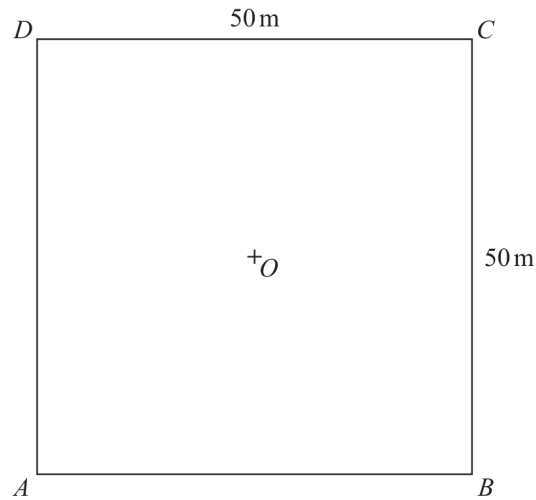
(Total 16 marks)

Q35**PAST PAPER**

Pearson IAL WME01 • June 2023 • Q8

M1 • TOPIC 2 9 marks

8.

**Figure 5**

A square floor space $ABCD$, with centre O , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O .

A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})$ m
- At time $t = 11$ s, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})$ m
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

A second robot S is at the point C .

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j})$ m s⁻¹
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

Q36 PAST PAPER Pearson IAL WME01 · October 2023 · Q6

M1 · TOPIC 2 15 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})$ km.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})$ km.

Given that P moves with constant velocity,

- (a) show that the velocity of P is $(10\mathbf{i} - 4\mathbf{j}) \text{ km h}^{-1}$ (2)

At time t hours after 12:00, the position vector of P is $\mathbf{p}$ km.

- (b) Find an expression for $\mathbf{p}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t . (2)

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}$ km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships P and Q are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30
(ii) find the position vector of the point at which the collision would occur. (4)

At 12:30 Q changes speed and direction to avoid the collision.

Ship Q now travels due north with a constant speed of 15 km h^{-1}

Ship P maintains the same constant velocity throughout.

- (d) Find the exact distance between P and Q at 14:30 (7)

Q37 PAST PAPER Pearson IAL WME01 • January 2024 • Q7

M1 • TOPIC 2 11 marks

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At midnight, a ship S is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of S .

(2)

At time t hours after midnight, the position vector of S is $\mathbf{s}\text{ km}$.

- (b) Find an expression for $\mathbf{s}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector $(26\mathbf{i} + 15\mathbf{j})\text{ km}$.

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of t when S is closest to the lighthouse.

- (ii) Hence determine whether it is safe for S to continue its course.

(7)

Q38 PAST PAPER Pearson IAL WME01 • June 2024 • Q7

M1 • TOPIC 2 13 marks

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity.

- the velocity of A is 40 km h^{-1} due east
- the velocity of B is 20 km h^{-1} on a bearing of angle α ($0^\circ < \alpha < 90^\circ$), where $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B in km h^{-1} (2)

At noon

- the position vector of A is $20\mathbf{j}$ km
- the position vector of B is $(10\mathbf{i} + 5\mathbf{j})$ km

At time t hours after noon

- the position vector of A is $\mathbf{r}$ km, where $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of B is $\mathbf{s}$ km

- (b) Find an expression for $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (2)

- (c) Show that at time t hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km} \quad (2)$$

- (d) Show that the boats will never collide. (3)

- (e) Find the distance between the boats when the bearing of B from A is 225° (4)

Q39**PAST PAPER**

Pearson IAL WME01 · January 2026 · Q6

M1 · TOPIC 2 16 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 9 am, a ship S leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j}) \text{ km h}^{-1}$

- (a) Find the exact speed of S

(2)

The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j}) \text{ km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j}) \text{ km h}^{-1}$

- (b) Find the direction in which B is moving, giving your answer as a bearing to the nearest degree.

(3)

- (c) Show that at time t hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}] \text{ km}$$

(4)

- (d) Show that B and S do not collide.

(3)

When B is north-east of S , the distance between S and B is $d \text{ km}$.

- (e) Find the value of d .

(4)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | Size and direction | Sketch; Pythagoras; $\tan$; bearing clockwise from north. |
| 02 | Forces | Sum = 0 or ma ; parallel $\rightarrow$ ratio of parts. |
| 03 | Constant velocity | $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$; due east $\rightarrow$ $\mathbf{j}$ part 0. |
| 04 | Two objects | $\mathbf{b} - \mathbf{a}$; collide: both parts 0; closest: complete the square. |

BOTTOM LINE Sketch every vector $\cdot \mathbf{i}$ with $\mathbf{i}$, $\mathbf{j}$ with $\mathbf{j}$ $\cdot$ closest approach = minimum of $|\vec{AB}|^2$.

Answers

Working with Vectors • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MAGNITUDE, DIRECTION AND BEARINGS

- Q1** Dr Eslam • Magnitude and bearing
5 N; bearing $360^\circ - 53.1^\circ = 306.9^\circ$
- Q2** Dr Eslam • From a bearing to $\mathbf{i}, \mathbf{j}$
 $10 \sin 60^\circ \mathbf{i} + 10 \cos 60^\circ \mathbf{j} \approx 8.66\mathbf{i} + 5\mathbf{j}$
- Q3** Dr Eslam • Angle between two vectors
directions $\tan^{-1} \frac{4}{3} = 53.1^\circ$ and 45° : angle 8.1°
- Q4** Pearson IAL WME01 • October 2019 • Q7
(i) $\sqrt{52} \approx 7.21$ N (ii) 074°
- Q5** Pearson IAL WME01 • January 2020 • Q6
(a) $\sqrt{101}$ N (b) 39.3° (c) $a = 6.5$
- Q6** Pearson IAL WME01 • January 2021 • Q5
(i) $\sqrt{244} \approx 15.6$ N (ii) 206°
- Q7** Pearson IAL WME01 • January 2024 • Q4
(i) $X = 18$ (ii) 128°
- Q8** Pearson IAL WME01 • June 2024 • Q2
 $|\mathbf{F}| = 2\sqrt{61} \approx 15.6$ N
- Q9** Pearson IAL WME01 • June 2026 • Q6
(a) $\sqrt{61}$ N (b) 103° (c) $\mathbf{F}_1 = 3\mathbf{i} - 6\mathbf{j}$, $\mathbf{F}_2 = -8\mathbf{i} + 12\mathbf{j}$

IDEA 02 • FORCES AS VECTORS: EQUILIBRIUM AND $\mathbf{F} = m\mathbf{a}$

- Q10** Dr Eslam • Three forces in equilibrium
 $p = 3, q = -2$
- Q11** Dr Eslam • Acceleration from forces
 $\mathbf{a} = 2\mathbf{i} - 2\mathbf{j}; 2\sqrt{2} \approx 2.83 \text{ m s}^{-2}$
- Q12** Dr Eslam • Parallel to a vector
 $2(p + 3) = p + 2 \Rightarrow p = -4$
- Q13** Pearson IAL WME01 • June 2021 • Q3
 (a) 33.7° (b) $\lambda = 3$
- Q14** Pearson IAL WME01 • January 2022 • Q6
 (a) $p = -6, q = 4$ (b) 135° (c) $T = 2.5$
- Q15** Pearson IAL WME01 • October 2022 • Q6
 (a) $2p + q - 4 = 0$ (b) $16\sqrt{5} \approx 35.8 \text{ m s}^{-1}$
- Q16** Pearson IAL WME01 • January 2023 • Q3
 (a) $3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$ (b) 27° (c) $T = 3.2$
- Q17** Pearson IAL WME01 • June 2023 • Q2
 (a) $\mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}$ (b) 118° (c) $m = 2$
- Q18** Pearson IAL WME01 • January 2025 • Q1
 (a) $\frac{12\sqrt{13}}{5} \approx 8.7 \text{ m s}^{-2}$ (b) $p = -30$
- Q19** Pearson IAL WME01 • October 2025 • Q3
 (a) $\mathbf{i} + 2.5\mathbf{j}$ (b) $\frac{\sqrt{193}}{2} \approx 6.95 \text{ m s}^{-1}$ (c) $b - c = 3$
 (d) $8\mathbf{i} + 5\mathbf{j}$ or $-12\mathbf{i} - 15\mathbf{j}$
- Q20** Pearson IAL WME01 • January 2026 • Q3
 (a) $2p - q = 19$ (b) $24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$

IDEA 03 • CONSTANT VELOCITY: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

- Q21** Dr Eslam • Where and when
 $\mathbf{r} = (2 + 3t)\mathbf{i} + (5 - t)\mathbf{j}$; due east when $t = 5$: $17\mathbf{i}$
- Q22** Dr Eslam • Velocity from two positions
 $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$; 5 m s^{-1}
- Q23** Pearson IAL WME01 • October 2021 • Q4
 (a) 162° (b) $t = \frac{1}{2}$ or $\frac{3}{2}$
- Q24** Pearson IAL WME01 • October 2021 • Q8
 (a) $10\mathbf{i} - 5\mathbf{j}$ (b) $(-2 + 10t)\mathbf{i} + (9 - 5t)\mathbf{j}$ (c) $19\mathbf{i}$ (d) $19\mathbf{i} - 1.5\mathbf{j}$
- Q25** Pearson IAL WME01 • October 2024 • Q3
 (a) $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$ (b) $v = \frac{5\sqrt{41}}{6} \approx 5.34$ (c) A is at P at $t = \frac{8}{3}$
- Q26** Pearson IAL WME01 • January 2025 • Q5
 (a) $1.5\mathbf{i} - \mathbf{j} \text{ m s}^{-1}$ (b) $2\sqrt{13} \approx 7.2 \text{ m}$
- Q27** Pearson IAL WME01 • June 2025 • Q2
 (a) $5\mathbf{i} - 4\mathbf{j}$ (b) $(5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}$ (c) 203°
- Q28** Pearson IAL WME01 • June 2026 • Q7
 (a) $\mathbf{r} = (14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}$ (b) 3.6 hours (c) $t = 16$ (d) $\frac{18\sqrt{5}}{5} \text{ km}$

IDEA 04 • TWO MOVING OBJECTS: MEET, CLOSEST, BEARING

- Q29** Dr Eslam • Collide or not? Closest?
 $\vec{AB} = (10 - 2t)\mathbf{i} + 2t\mathbf{j}$; no collision; $|AB|^2 = 8t^2 - 40t + 100$, least at $t = 2.5$: $\sqrt{50} \approx 7.07$ km
- Q30** Pearson IAL WME01 • October 2019 • Q8
 (a) $20\mathbf{i} - 20\sqrt{3}\mathbf{j}$ (b) $\mathbf{r} = (60 - 20t)\mathbf{i}$, $\mathbf{s} = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$ (c) 13:43
- Q31** Pearson IAL WME01 • January 2021 • Q6
 (a) $(11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}$ (b) $t = 5.5$ (c) shown (d) $\sqrt{0.5} \approx 0.707$ m
- Q32** Pearson IAL WME01 • January 2022 • Q8
 (a) $(13 + 3t)\mathbf{i} + (5 - 10t)\mathbf{j}$ (b) shown (c) $2\sqrt{5} \approx 4.5$ km (d) 297°
- Q33** Pearson IAL WME01 • June 2022 • Q8
 (a) 135° (b) shown (c) 270° (d) $40\sqrt{2}$ m (e) $t = 0$ or $\frac{320}{17}$
- Q34** Pearson IAL WME01 • October 2022 • Q8
 (a) $3\sqrt{17}$ km h⁻¹ (b) $p = -9$, $q = 3$ (c) 323°
- Q35** Pearson IAL WME01 • June 2023 • Q8
 (a) $(t - 2)\mathbf{i} + (2t + 1)\mathbf{j}$ (b) $(25 - t)\mathbf{i} + (25 - t)\mathbf{j}$ (c) shown (d) $t = \frac{126}{13} \approx 9.7$ s
- Q36** Pearson IAL WME01 • October 2023 • Q6
 (a) $10\mathbf{i} - 4\mathbf{j}$ (b) $(15 + 10t)\mathbf{i} + (36 - 4t)\mathbf{j}$ (c)(i) both at $30\mathbf{i} + 30\mathbf{j}$ when $t = 1.5$ (ii) $30\mathbf{i} + 30\mathbf{j}$ (d) $2\sqrt{101}$ km
- Q37** Pearson IAL WME01 • January 2024 • Q7
 (a) 20 km h⁻¹ (b) $(19 + 12t)\mathbf{i} + (22 - 16t)\mathbf{j}$ (c)(i) $t = 0.49$ (ii) least distance 1.4 km > 1.3: safe
- Q38** Pearson IAL WME01 • June 2024 • Q7
 (a) $16\mathbf{i} + 12\mathbf{j}$ (b) $(10 + 16t)\mathbf{i} + (5 + 12t)\mathbf{j}$ (c) shown (d) never both zero (e) $\frac{20\sqrt{2}}{3} \approx 9.43$ km
- Q39** Pearson IAL WME01 • January 2026 • Q6
 (a) $5\sqrt{13}$ km h⁻¹ (b) 321° (c) shown (d) do not collide (e) $d = \frac{70\sqrt{2}}{13} \approx 7.6$ km



DR ESLAM AHMED
eliteigcse.com

Kinematics Graphs

Sketch the graph first, label every number, then read it: the gradient is the acceleration and the area is the distance.

THE TOPIC ON ONE MAP

Kinematics Graphs

01 Gradient and area

gradient = a

area = distance

below the axis

02 One journey

trapezium

find T or V

quadratic in T

03 Two journeys, same axes

same speed

catches up

overtakes

04 Other graphs

$a-t$

$s-t$

impulse at a jump

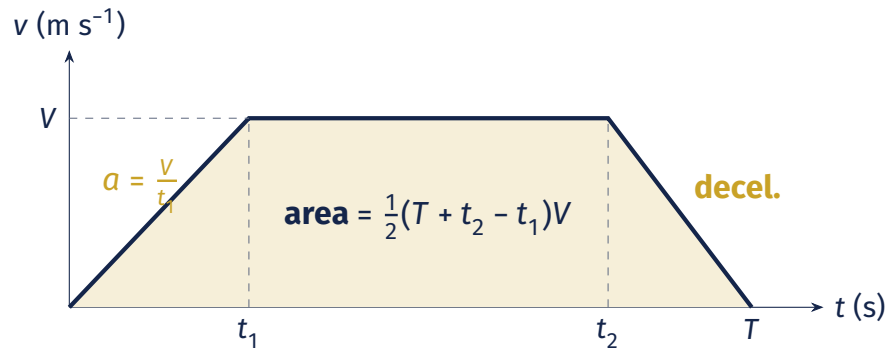
THE TWO FACTS OF EVERY SPEED-TIME GRAPH

gradient = acceleration

steeper line = bigger acceleration

area = distance

split into triangles, rectangles, trapezia



01 Gradient and Area

Every question on a speed–time or velocity–time graph uses one of two facts. Gradient = acceleration (negative gradient = deceleration). Area = distance; for velocity, area **below** the axis is motion backwards.

WHEN YOU SEE → YOU DO

“find the acceleration / deceleration”

→ gradient = $\frac{\text{rise}}{\text{run}}$; deceleration is the size

“find the distance”

→ area — split into simple shapes

the velocity line crosses the t -axis

→ at rest there; area below = moving backwards

“thrown vertically upwards ... velocity–time graph”

→ gradient = -9.8 ; area above the axis = height gained

“total distance” vs “displacement”

→ add all areas vs areas with signs

WORKED MODEL • OCTOBER 2021

A ball is thrown upwards from A; its velocity reaches 0 at $t = 2.5$ and it hits the ground at $t = 7$. Find the launch speed and the height of A.

1 $u = 9.8 \times 2.5 = 24.5 \text{ m s}^{-1}$

gradient = $-g$

2 at $t = 7$: $v = 24.5 - 9.8(7) = -44.1$

velocity line

ANSWER $\frac{1}{2}(4.5)(44.1) - \frac{1}{2}(2.5)(24.5) = 99.2 - 30.6 = 68.6 \text{ m}$

IDEA 01

Practise: Gradient and Area

My questions first, then real past papers.

Q1 DR ESLAM Gradient and area

M1 · TOPIC 3 3 marks

A particle speeds up uniformly from 4 m s^{-1} to 10 m s^{-1} in 3 s, then moves at 10 m s^{-1} for 5 s. Find its acceleration and the total distance.

Gradient = acceleration; area = distance (trapezium + rectangle).

Q2 DR ESLAM Below the axis

M1 · TOPIC 3 4 marks

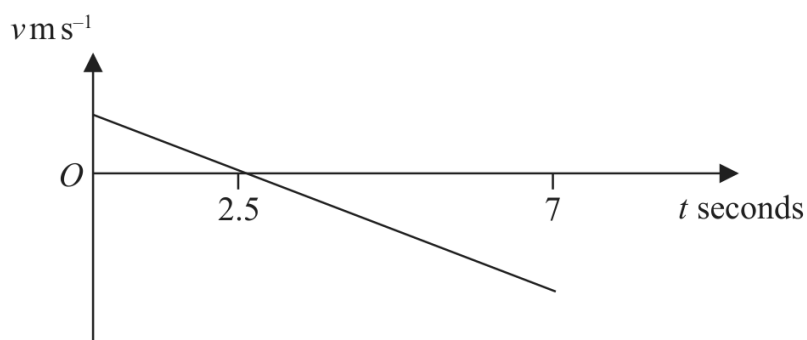
The velocity of a particle changes at a constant rate from 12 m s^{-1} to -6 m s^{-1} in 9 s. Find when it is at rest, its displacement and the distance it travels.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 • October 2021 • Q6

M1 • TOPIC 3 5 marks

6.

**Figure 3**

A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown.

(2)

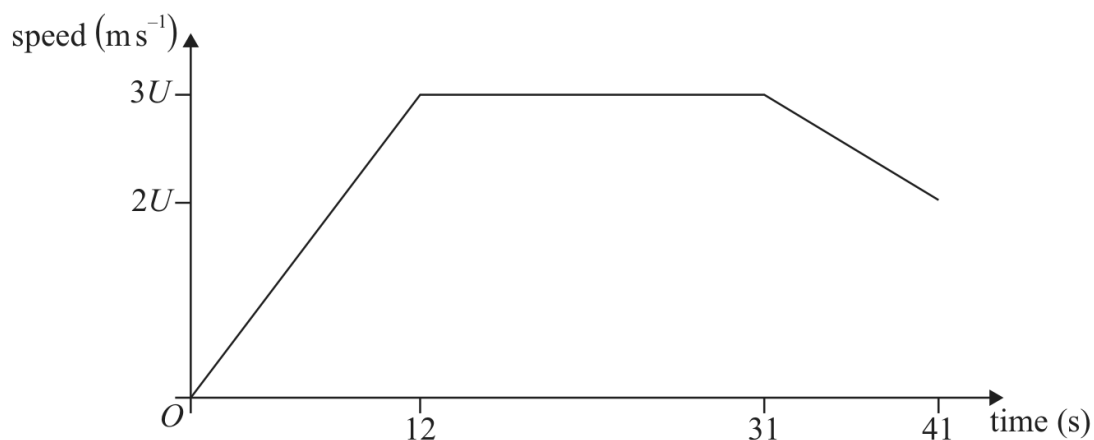
(b) Find the height of A above the ground.

(3)

Q4 PAST PAPER Pearson IAL WME01 • January 2026 • Q2

M1 • TOPIC 3 10 marks

2.

**Figure 1**

A particle P starts from rest at time $t = 0$ and moves in a straight line

- from $t = 0$ to $t = 12$ s, P accelerates uniformly until it reaches a speed of $3U \text{ m s}^{-1}$
- from $t = 12$ s to $t = 31$ s, P moves with constant speed $3U \text{ m s}^{-1}$
- from $t = 31$ s to $t = 41$ s, P decelerates uniformly until it has speed $2U \text{ m s}^{-1}$

as shown on the speed-time graph in Figure 1.

The distance travelled by P between $t = 0$ and $t = 41$ s is 600 m.

- (a) Show that $U = 6$ (3)
- (b) Find the distance travelled by P whilst moving from rest to a speed of 6 m s^{-1} (2)
- (c) Find the acceleration of P between $t = 0$ and $t = 12$ s. (1)
- (d) Find the deceleration of P between $t = 31$ s and $t = 41$ s. (2)
- (e) Sketch an acceleration-time graph to represent the motion of P from $t = 0$ to $t = 41$ s. (2)

02 One Journey: Sketch, Then Use the Area

Draw the journey as straight lines: speeding up, steady, slowing down. Put every number from the question on the graph. Then one equation from the area (the distance) and one from the times.

WHEN YOU SEE → YOU DO

“sketch a speed–time graph”

→ straight segments, label every speed and time

“the total distance is 4.8 km”

→ area = 4800 — change units first

“the deceleration is twice the acceleration”

→ deceleration time = half the acceleration time

“show that $T^2 - kT + 500 = 0$ ”

→ area in T , then tidy into the quadratic

“hence find T , justifying”

→ reject the root that breaks the story ($T > \text{total time}$)

Sketch and label

Area = distance

Solve for T or V

Check the story

WORKED MODEL • JANUARY 2023

From rest to 20 m s^{-1} in $T \text{ s}$, 20 m s^{-1} for 180 s , then deceleration twice the acceleration. Distance 4.8 km . Find T .

1 slowing down takes $\frac{T}{2}$

twice the rate, half the time

2 $\frac{1}{2}(20)T + 20(180) + \frac{1}{2}(20)\frac{T}{2} = 4800$

area

3 $15T = 1200$

collect

ANSWER $T = 40 \text{ s}$

× **TRAP** Forgetting to change 4.8 km into 4800 m , or minutes into seconds, before writing the area.

IDEA 02

Practise: One Journey: Sketch, Then Use the Area

My questions first, then real past papers.

Q5 DR ESLAM Trapezium: find V

M1 • TOPIC 3 3 marks

A train accelerates from rest to $V \text{ m s}^{-1}$ in 20 s, travels at V for 60 s, then decelerates to rest in 10 s. It travels 1.35 km. Find V .

Trapezium area = $\frac{1}{2}(\text{top} + \text{bottom}) \times V$.

$V =$ _____

Q6 DR ESLAM Times from accelerations

M1 • TOPIC 3 4 marks

A car accelerates from rest at 2 m s^{-2} to 20 m s^{-1} , keeps this speed, then decelerates at 4 m s^{-2} to rest. The whole journey takes 60 s. Find the distance.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WME01 · October 2019 · Q6

M1 · TOPIC 3 12 marks

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for V in terms of T . (1)
- (c) Show that $T^2 - kT + 500 = 0$, where k is a constant to be found. (4)
- (d) Hence find the value of T , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

Q8 PAST PAPER Pearson IAL WME01 • January 2021 • Q7

M1 • TOPIC 3 12 marks

7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he hits the ground. (2)
- (c) Find the value of T . (5)
- (d) Find, to the nearest metre, the height of the point H above the ground. (4)

Q9**PAST PAPER**

Pearson IAL WME01 · January 2023 · Q1

M1 · TOPIC 3**10 marks**

1. A train travels along a straight horizontal track between two stations A and B .

The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station B .

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

- (a) On the axes below, sketch a speed–time graph to illustrate the motion of the train as it moves from station A to station B .



If you need to redraw your graph, use the axes on page 3

(3)

Stations A and B are 4.8 km apart.

- (b) Find the value of T

(5)

- (c) Find the acceleration of the train during the first T seconds of its motion.

(2)

Only use these axes if you need to redraw your graph.



(Total for Question 1 is 10 marks)

Q10**PAST PAPER**

Pearson IAL WME01 · January 2025 · Q2

M1 · TOPIC 3 11 marks

2. The fixed points A , B and C lie in a straight line on a horizontal road.

- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
- From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
- The motorbike takes T_1 seconds to travel from A to B
- From B , the motorbike decelerates uniformly until it comes to rest at C
- The motorbike takes T_2 seconds to travel from B to C

(a) Sketch a speed-time graph for the motion of the motorbike as it moves from A to C .

(3)

The distance AB is 132 m and the distance BC is 136 m.

(b) Find, in terms of V , an expression for

(i) T_1

(ii) T_2

(4)

Given that the motorbike takes 28 s to travel from A to C ,

(c) find the value of V ,

(2)

(d) find the deceleration of the motorbike.

(2)

03 Two Journeys on the Same Axes

Two vehicles on one graph. **Same speed** is where the two lines cross. **Same distance** (catches up, overtakes) is where the two **areas** are equal — not where the lines cross.

WHEN YOU SEE → YOU DO

“sketch on the same axes”

→ both graphs, one diagram, key points labelled

“at the instant they are moving with the same speed”

→ the lines cross: equal v

“catches up with” / “overtakes” / “passes”

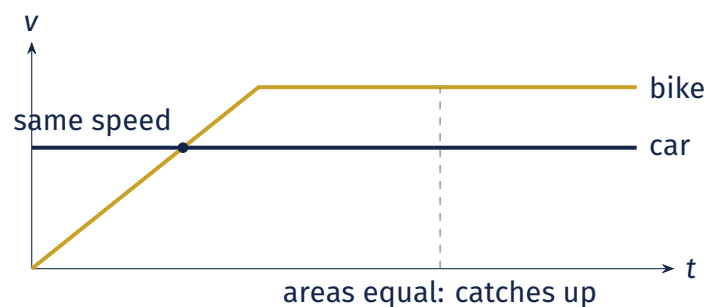
→ equal areas from the start

“starts 20 seconds later”

→ its graph starts at $t = 20$ on the axis

“how far in front is A?”

→ difference of the two areas



WORKED MODEL • JANUARY 2020

A car at a constant 40 m s^{-1} passes a motorcyclist at rest. She accelerates at 4 m s^{-2} for 15 s. When are their speeds equal?

1 motorcyclist: $v = 4t$

gradient 4

ANSWER $4t_1 = 40 \Rightarrow t_1 = 10 \text{ s}$

× **TRAP** Answering “catches up” with the time the lines cross. That is only when the speeds are equal.

IDEA 03

Practise: Two Journeys on the Same Axes

My questions first, then real past papers.

Q11 **DR ESLAM** Same speed, then catch up

M1 · TOPIC 3 5 marks

At $t = 0$ a car passes a motorcycle at rest, moving at a constant 15 m s^{-1} . The motorcycle accelerates at 3 m s^{-2} to 24 m s^{-1} , then keeps that speed. Find (a) when they have the same speed (b) when the motorcycle catches the car. Same speed: where the lines cross. Catch up: where the AREAS are equal.

PAST PAPERS

Q12 PAST PAPER Pearson IAL WME01 · January 2020 · Q5

M1 · TOPIC 3 10 marks

5. A car travels at a constant speed of 40 ms^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 ms^{-2} for 15 s and then accelerates at 1 ms^{-2} for a further T seconds until he catches up with the car.

- (a) Sketch, on the same axes, the speed-time graph for the motion of the car and the speed-time graph for the motion of the motorcyclist, from time $t = 0$ to the instant when the motorcyclist catches up with the car.

(2)

At the instant when $t = t_1$ seconds, the car and the motorcyclist are moving at the same speed.

- (b) Find the value of t_1

(2)

- (c) Show that $T^2 + kT - 300 = 0$, where k is a constant to be found.

(6)

Q13**PAST PAPER**

Pearson IAL WME01 • June 2021 • Q8

M1 • TOPIC 3 13 marks

8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram A and a speed-time graph for the motion of tram B , from $t = 0$ to the instant when tram B overtakes tram A .

(3)

At the instant when the two trams are moving with the same speed, tram A is d metres in front of tram B .

- (b) Find the value of d .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram B overtakes tram A .

(5)**(Total 13 marks)**

Q14 PAST PAPER Pearson IAL WME01 · June 2022 · Q7

M1 · TOPIC 3 13 marks

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at 0.8 m s^{-2} up to a speed of 4 m s^{-1} and then maintains this speed until he crosses the finish line.

Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line. (3)

- (b) Find the time taken by Ajaz to complete the race. (4)

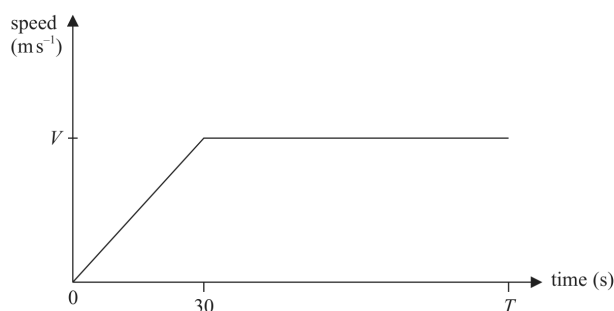
- (c) Find the value of T (4)

- (d) Find the difference in the speeds of the two children as they cross the finish line. (2)

Q15 PAST PAPER Pearson IAL WME01 • June 2023 • Q5

M1 • TOPIC 3 11 marks

5.

**Figure 2**

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ m s}^{-1}$.

The car then travels at a constant speed of $V \text{ m s}^{-1}$ until time $t = T$ seconds.

- (a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time $t = 10 \text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V \text{ m s}^{-1}$ at time T seconds

- (b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[If you need to redraw your sketch, there is a copy of Figure 2 on page 15.]

(2)

- (c) Show that the constant speed of the motorbike is $\frac{4V}{3} \text{ m s}^{-1}$

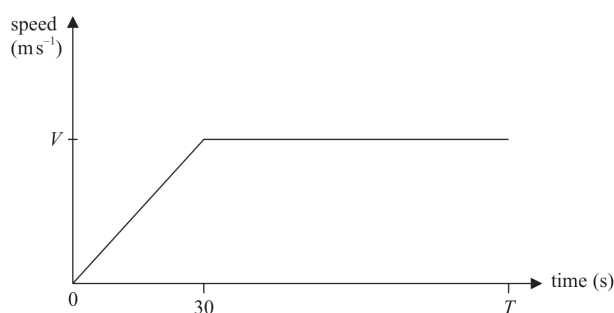
(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

- (d) Find the value of T

(5)

Only use this copy of Figure 2 if you need to redraw your sketch.

**Figure 2**

(Total for Question 5 is 11 marks)

Q16 PAST PAPER Pearson IAL WME01 • October 2024 • Q4

M1 • TOPIC 3 13 marks

4. The points A and B lie on the same straight horizontal road.

Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$.

He then travels at constant speed $V \text{ m s}^{-1}$.

When $t = 42$, cyclist P passes B .

Given that the distance AB is 120 m,

- (a) show that $V = 3.2$ (3)

- (b) Find the acceleration of cyclist P between $t = 0$ and $t = 9$ (2)

Cyclist P continues to cycle along the road in the same direction at the same constant speed, $V \text{ m s}^{-1}$.

When $t = 6$, a second cyclist Q sets off from A and travels in the same direction as P along the same road. She accelerates for T seconds until her speed is 3.6 m s^{-1} .

She then travels at constant speed 3.6 m s^{-1} .

Cyclist Q catches up with P when $t = 54$.

- (c) On Figure 2, on page 11, sketch a speed-time graph showing the journeys of both cyclists, for the interval $0 \leq t \leq 54$ (3)

- (d) Find the value of T (5)

Question 4 continued

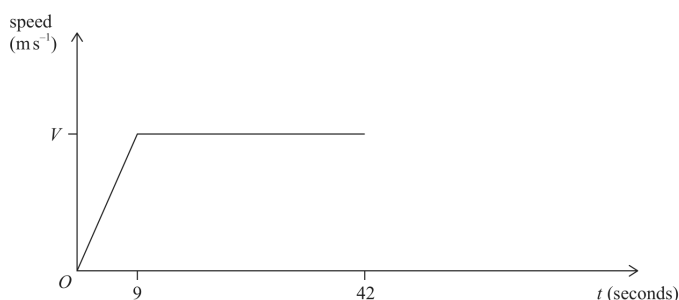
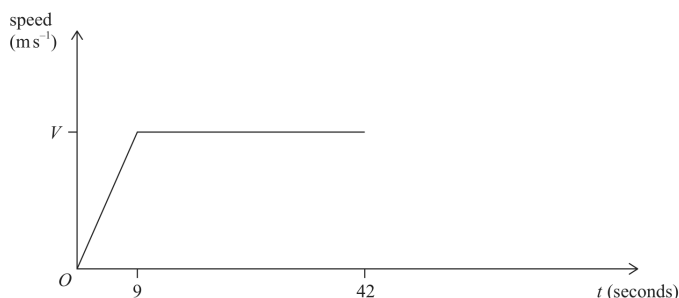


Figure 2

A copy of Figure 2 is on page 13 if you need to redraw your answer to part (c).

Question 4 continued

Only use this copy of Figure 2 if you need to redraw your answer to part (c).



Copy of Figure 2

04 Acceleration–Time and Displacement–Time Graphs

Change the graph back into a speed–time graph first. On an acceleration–time graph the **area** is the change in speed. On a displacement–time graph the **gradient** is the velocity.

WHEN YOU SEE → YOU DO

an acceleration–time graph

→ area = change of speed; build v step by step

“sketch an acceleration–time graph”

→ horizontal steps: gradient of each v – t part

a displacement–time graph

→ gradient = velocity on each straight part

“impulse at $t = 0.3$ ” on an s – t graph

→ $m(v_{\text{after}} - v_{\text{before}})$

graph	gradient means	area means
displacement–time	velocity	—
velocity–time	acceleration	displacement
acceleration–time	—	change of velocity

WORKED MODEL • OCTOBER 2023

$v = 8$ at $t = 0$; $a = -0.5$ for 4 s, $a = 0$ for 6 s, $a = 1.2$ for 10 s. Show $v = 18$ at $t = 20$.

1 $8 - 0.5(4) = 6$

area of the first step

2 $6 + 0 + 1.2(10) = 18$

add the other areas

ANSWER $v = 18 \text{ m s}^{-1}$

IDEA 04

Practise: Acceleration–Time and Displacement–Time Graphs

My questions first, then real past papers.

Q17 **DR ESLAM** From an acceleration–time graph

M1 • TOPIC 3 4 marks

Starting from rest, $a = 2 \text{ m s}^{-2}$ for $0 \leq t \leq 5$, $a = 0$ for $5 \leq t \leq 15$, $a = -1 \text{ m s}^{-2}$ for $15 \leq t \leq 20$. Find the speed at $t = 20$ and the distance travelled.

Area under the a – t graph = change of speed.

Q18 **DR ESLAM** From a displacement–time graph

M1 • TOPIC 3 3 marks

A displacement–time graph is made of straight lines through $(0, 0)$, $(4, 12)$, $(10, 12)$ and $(14, 0)$. Find the velocity on each part, the total distance and the final displacement.

PAST PAPERS

Q19 PAST PAPER Pearson IAL WME01 · October 2023 · Q2

M1 · TOPIC 3 10 marks

2.

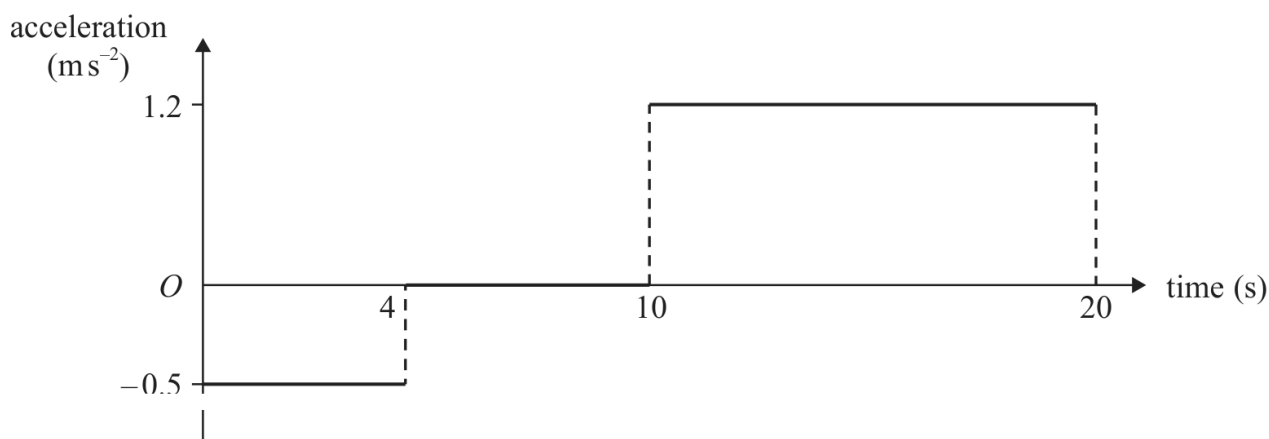


Figure 2

Two fixed points, A and B , are on a straight horizontal road.

The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B .

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

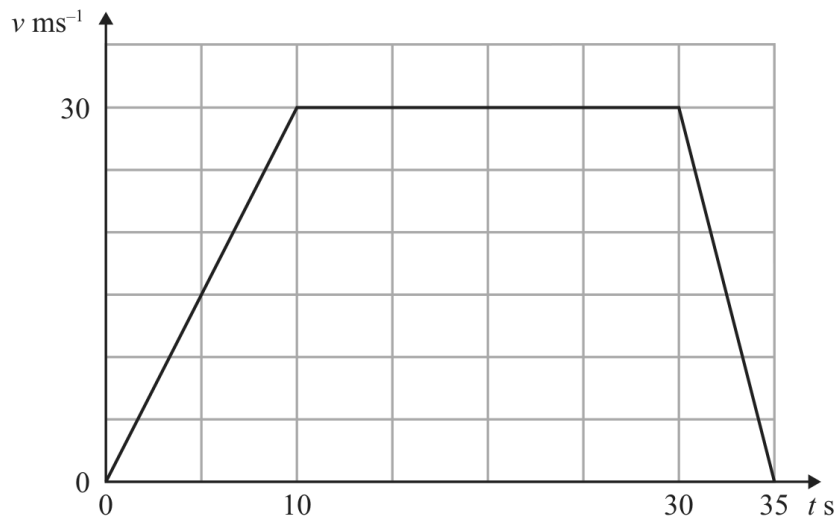
At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ m s}^{-1}$

- (a) Show that $v = 18$ (3)
- (b) Sketch a speed-time graph for the motion of the car from A to B . (3)
- (c) Find the distance AB . (4)

Q20 PAST PAPER Pearson IAL WME01 • October 2025 • Q6

M1 • TOPIC 3 10 marks

6.

**Figure 4**

The point O lies on a straight horizontal road.
At time $t = 0$, a car leaves O and travels along the road.

The velocity-time graph in Figure 4 shows the velocity, $v \text{ ms}^{-1}$, of the car at time t seconds for the first 35 seconds of its journey.

(a) Find

(i) the acceleration of the car for the period $0 \leq t \leq 10$ (ii) the deceleration of the car for the period $30 \leq t \leq 35$

(2)

(b) Sketch an acceleration-time graph for the car for the period $0 \leq t \leq 35$

(2)

(c) Find the distance travelled by the car for the period $0 \leq t \leq 35$

(2)

When $t = 5$, a motorcycle starts from rest at O .

The motorcycle travels along the same road as the car and in the same direction.

For the period $5 \leq t \leq 20$, the acceleration of the motorcycle is $A \text{ ms}^{-2}$, where A is a positive constant.

The motorcycle catches up with the car when $t = 20$

(d) Find the value of A .

(4)

3.

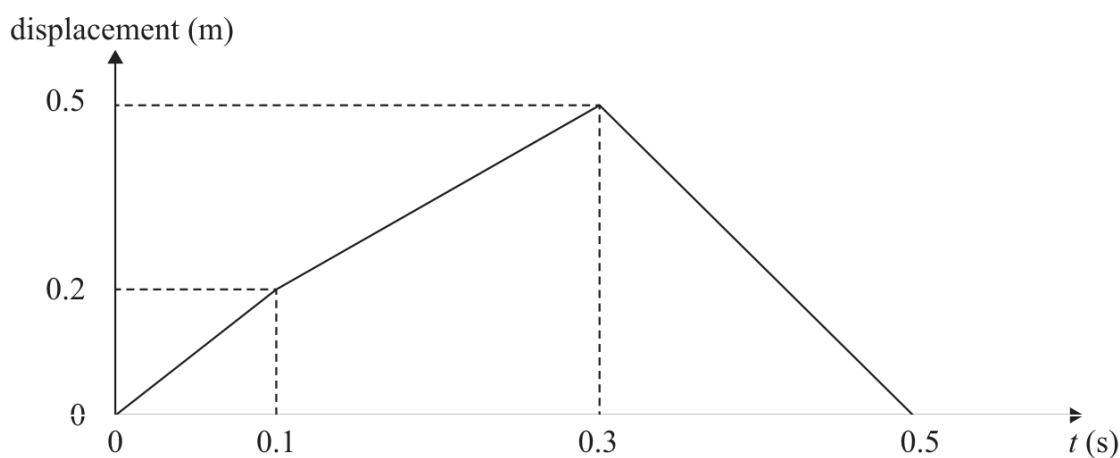


Figure 2

A small ball of mass 0.04 kg moves along a straight line on a pinball table.

At time $t = 0$, the ball passes through the point O

The displacement of the ball is measured from O

Figure 2 shows the **displacement-time** graph for the motion of the ball from $t = 0$ to $t = 0.5$

- (a) State the displacement of the ball from O at $t = 0.5$ (1)
- (b) Find the total distance moved by the ball between $t = 0$ and $t = 0.5$ (1)
- (c) Sketch a velocity-time graph for the motion of the ball from $t = 0$ to $t = 0.5$ (4)
- (d) Find the magnitude of the impulse exerted on the ball at $t = 0.3$, stating the units of your answer. (4)

(Total for Question 3 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|--|
| 01 | Gradient and area | Gradient = a ; area = distance; below the axis = backwards. |
| 02 | One journey | Sketch and label; area = distance; reject the impossible root. |
| 03 | Two journeys | Same speed: lines cross. Catch up: areas equal. |
| 04 | Other graphs | $a-t$ area = Δv ; $s-t$ gradient = v . |

BOTTOM LINE Sketch → label → gradient for a , area for distance. Catch up = equal areas.

Answers

Kinematics Graphs • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • GRADIENT AND AREA

- Q1** Dr Eslam • Gradient and area $a = 2 \text{ m s}^{-2}$; $21 + 50 = 71 \text{ m}$
- Q2** Dr Eslam • Below the axis at rest at $t = 6$; displacement $36 - 9 = 27 \text{ m}$; distance $36 + 9 = 45 \text{ m}$
- Q3** Pearson IAL WME01 • October 2021 • Q6 (a) 24.5 m s^{-1} (b) 68.6 m
- Q4** Pearson IAL WME01 • January 2026 • Q2 (a) $U = 6$ (b) 12 m (c) 1.5 m s^{-2} (d) 0.6 m s^{-2} (e) steps 1.5 on 0–12, 0 on 12–31, -0.6 on 31–41

IDEA 02 • ONE JOURNEY: SKETCH, THEN USE THE AREA

- Q5** Dr Eslam • Trapezium: find V $\frac{1}{2}(90 + 60)V = 1350 \Rightarrow V = 18$
- Q6** Dr Eslam • Times from accelerations 10 s and 5 s, so 45 s at 20: $\frac{1}{2}(60 + 45)(20) = 1050 \text{ m}$
- Q7** Pearson IAL WME01 • October 2019 • Q6 (a) straight line to (T, V) , then level to $t = 30$ (b) $V = 0.8T$ (c) $k = 60$ (d) $T = 10$ ($T = 50 > 30$ rejected) (e) e.g. reaction time; acceleration not constant / speed not constant at the end
- Q8** Pearson IAL WME01 • January 2021 • Q7 (a) 24.5 m s^{-1} (b) three stages: up to 24.5, down to V , level (c) $T = 5$ (d) 167 m
- Q9** Pearson IAL WME01 • January 2023 • Q1 (a) $(0, 0) \rightarrow (T, 20) \rightarrow (T + 180, 20) \rightarrow (3T + 180, 0)$ (b) $T = 40$ (c) 0.5 m s^{-2}
- Q10** Pearson IAL WME01 • January 2025 • Q2 (a) $(0, 5) \rightarrow (T_1, V) \rightarrow (T_1 + T_2, 0)$ (b)(i) $T_1 = \frac{264}{V + 5}$ (ii) $T_2 = \frac{272}{V}$ (c) $V = 17$ (d) $\frac{17}{16} = 1.0625 \text{ m s}^{-2}$

IDEA 03 • TWO JOURNEYS ON THE SAME AXES

- Q11** Dr Eslam • Same speed, then catch up
(a) $t = 5 \text{ s}$ (b) $96 + 24(t - 8) = 15t \Rightarrow t = \frac{32}{3} \approx 10.7 \text{ s}$
- Q12** Pearson IAL WME01 • January 2020 • Q5
(a) car level at 40; motorcycle up to 60 at $t = 15$, then less steep to $15 + T$ (b) $t_1 = 10$ (c) $T^2 + 40T - 300 = 0$, $k = 40$
- Q13** Pearson IAL WME01 • June 2021 • Q8
(a) A: $(0, 0) \rightarrow (5, 10)$ then level; B: rest to $t = 20$, $(24, 12)$ then level (b) $d = \frac{575}{3} \approx 192 \text{ m}$ (c) 1170 m
- Q14** Pearson IAL WME01 • June 2022 • Q7
(a) Ajaz: up to 4 at $t = 5$, level; Beth: up to T at $t = T$, level (b) 27.5 s (c) $T = 3.92$ (d) 0.085 m s^{-1}
- Q15** Pearson IAL WME01 • June 2023 • Q5
(a) $\frac{1}{2}(30)V + V(T - 30) = V(T - 15)$ (b) through $(10, 0)$, $(50, \frac{4}{3}V)$, $(60, \frac{4}{3}V)$, (T, V) (c) $\frac{4V}{3}$ (d) $T = 90$
- Q16** Pearson IAL WME01 • October 2024 • Q4
(a) $V = 3.2$ (b) $\frac{16}{45} = 0.356 \text{ m s}^{-2}$ (c) P level to $(54, 3.2)$; Q through $(6, 0)$, $(6 + T, 3.6)$, $(54, 3.6)$ (d) $T = 8$

IDEA 04 • ACCELERATION–TIME AND DISPLACEMENT–TIME GRAPHS

- Q17** Dr Eslam • From an acceleration–time graph
 $v(20) = 5 \text{ m s}^{-1}$; $25 + 100 + 37.5 = 162.5 \text{ m}$
- Q18** Dr Eslam • From a displacement–time graph
 $3, 0, -3 \text{ m s}^{-1}$; distance 24 m ; displacement 0
- Q19** Pearson IAL WME01 • October 2023 • Q2
(a) $8 - 0.5(4) + 1.2(10) = 18$ (b) $(0, 8) \rightarrow (4, 6) \rightarrow (10, 6) \rightarrow (20, 18)$ (c) $AB = 184 \text{ m}$
- Q20** Pearson IAL WME01 • October 2025 • Q6
(a)(i) 3 m s^{-2} (ii) 6 m s^{-2} (b) steps $3, 0, -6$ with breaks at $10, 30, 35$ (c) 825 m (d) $A = 4$
- Q21** Pearson IAL WME01 • June 2026 • Q3
(a) 0 (b) 1 m (c) $v = 2$ on $0-0.1$, 1.5 on $0.1-0.3$, -2.5 on $0.3-0.5$ (d) $0.04(1.5 + 2.5) = 0.16 \text{ N s}$

Constant Acceleration in 1D

Write down s, u, v, a, t . Choose the positive direction. The letter you are not given and not asked for chooses the equation.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Constant Acceleration in 1D

01 - The SUVAT toolkit

five equations

the missing letter

"in the 4th second"

02 - Journeys in stages

end of one = start of next

friction: $a = \mu g$

03 - Vertical motion

$a = -9.8$

greatest height: $v = 0$

from above the ground

04 - Bounces and meetings

speed halved

same height, same time

THE FIVE SUVAT EQUATIONS • each one misses one letter

$$v = u + at \text{ (no } s) \quad s = \frac{1}{2}(u + v)t \text{ (no } a) \quad v^2 = u^2 + 2as \text{ (no } t)$$

$$s = ut + \frac{1}{2}at^2 \text{ (no } v) \quad s = vt - \frac{1}{2}at^2 \text{ (no } u)$$

01 The SUVAT Toolkit: One Stage

Constant acceleration in a straight line. Write the five letters in a column, fill in what you know, mark what you want. The letter that is neither given nor wanted tells you which equation to use.

WHEN YOU SEE → YOU DO

“starts from rest” / “comes to rest”

→ $u = 0$ / $v = 0$ “decelerates at 2 m s^{-2} ”→ $a = -2$ (motion is positive)

“in the fourth second”

→ $s(4) - s(3)$ — two uses of $s = ut + \frac{1}{2}at^2$

“passes A ... 20 seconds later passes B”

→ $s = \frac{1}{2}(u + v)t$ when a is unknown km h^{-1} → $\div 3.6$ to get m s^{-1}

WORKED MODEL • JUNE 2022

A motorbike passes P , then Q 10 s later at 28 m s^{-1} . $PQ = 220 \text{ m}$. Find its acceleration.

1 $s = 220$, $v = 28$, $t = 10$, want a — no u

choose the equation

2 $220 = 28(10) - \frac{1}{2}a(100)$

 $s = vt - \frac{1}{2}at^2$

ANSWER $a = 1.2 \text{ m s}^{-2}$

IDEA 01

Practise: The SUVAT Toolkit: One Stage

My questions first, then real past papers.

Q1 DR ESLAM Pick the equation

M1 • TOPIC 4 3 marks

A car accelerates uniformly from 5 m s^{-1} to 25 m s^{-1} over 200 m. Find its acceleration and the time taken.

List s, u, v, a, t ; the one you are not given or asked for tells you the equation.

Q2 DR ESLAM Two intervals

M1 • TOPIC 4 4 marks

A particle with constant acceleration travels 20 m in the first 2 s and 30 m in the next 2 s. Find its initial speed and acceleration.

Use $s = ut + \frac{1}{2}at^2$ for 2 s and for 4 s (50 m).

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 · June 2021 · Q2

M1 · TOPIC 4 8 marks

2. A car moves along a straight horizontal road with constant acceleration $a \text{ ms}^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed $u \text{ ms}^{-1}$

In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 ms^{-1}

Find the time taken for the car to travel from A to B .

(8)

Q4 PAST PAPER Pearson IAL WME01 · June 2022 · Q2

M1 · TOPIC 4 7 marks

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

- (a) find the acceleration of the motorbike,

(3)

- (b) find the distance travelled by the motorbike during the fifth second after passing P

(4)

Q5**PAST PAPER**

Pearson IAL WME01 • January 2024 • Q3

M1 • TOPIC 4 10 marks

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

- (a) Show that $u = 12$

(2)

- (b) Find the time taken for the van to travel from A to the midpoint of AB .

(5)

The van has mass 1200 kg.

During its motion the van experiences a constant resistive force of magnitude 260 N

- (c) Find the magnitude of the driving force exerted by the engine of the van as it travels from A to B .

(3)

02 Journeys in Stages

When the acceleration changes, start a new SUVAT list. The speed at the end of one stage is the speed at the start of the next. Add the distances.

WHEN YOU SEE → YOU DO

“the acceleration changes instantaneously to ...”

→ new stage, new list

“first model ... second model”

→ solve each model separately and compare

“rough horizontal surface”, coefficient μ

→ deceleration = μg

“maintains this speed”

→ constant stage: distance = speed × time

Stage 1 list

Link speed

Stage 2 list

Add distances / times

WORKED MODEL • JANUARY 2023

PQ = 6x: from u to $2u$ in 12 s. Show $x = 3u$.

$$1 \quad 6x = \frac{1}{2}(u + 2u)(12)$$

$$\text{no } a: s = \frac{1}{2}(u + v)t$$

$$\text{ANSWER} \quad 6x = 18u \Rightarrow x = 3u$$

× **TRAP** Using one list for the whole journey when the acceleration changes halfway.

IDEA 02

Practise: Journeys in Stages

My questions first, then real past papers.

Q6 DR ESLAM Two stages

M1 • TOPIC 4 4 marks

A train starts from rest, accelerates at 0.5 m s^{-2} for 40 s, then decelerates at 1 m s^{-2} until it stops. Find the total distance.

The final speed of stage 1 is the initial speed of stage 2.

Q7 DR ESLAM Friction gives the deceleration

M1 • TOPIC 4 3 marks

A puck slides on rough horizontal ice with $\mu = 0.1$, starting at 6 m s^{-1} . How far does it slide?

PAST PAPERS

Q8 PAST PAPER Pearson IAL WME01 • October 2021 • Q3

M1 • TOPIC 4 10 marks

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

In a **first** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until the car reaches the road sign.

- (a) Use this first model to find the speed of the car as it reaches the sign.

(2)

The road sign indicates that the speed limit immediately after the sign is 13 m s^{-1} .

In a **second** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (b) Use this second model to find the value of t at which the car reaches the sign.

(4)

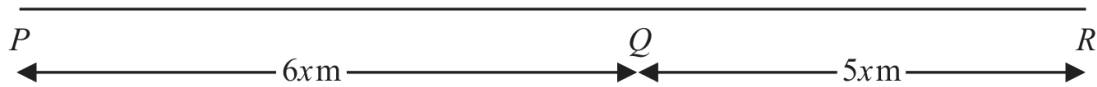
In a **third** model, the car is assumed to move with constant speed 25 m s^{-1} from A until time $t = 0.2$, the car then decelerates uniformly at 6 m s^{-2} until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (c) Use this third model to find the value of t at which the car reaches the sign.

(4)

Q9 PAST PAPER Pearson IAL WME01 · January 2023 · Q5**M1 · TOPIC 4** 9 marks

5.

**Figure 2**

Three points P , Q and R are on a horizontal road where PQR is a straight line.

The point Q is between P and R , with $PQ = 6x$ metres and $QR = 5x$ metres, as shown in Figure 2.

A vehicle moves along the road from P to Q with constant acceleration.

The vehicle is modelled as a particle.

At time $t = 0$, the vehicle passes P with speed $u \text{ m s}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ m s}^{-1}$

Using the model,

(a) show that $x = 3u$

(2)

As the vehicle passes Q , the acceleration of the vehicle changes instantaneously to 1.5 m s^{-2}

The vehicle continues to move with a constant acceleration of 1.5 m s^{-2} and passes R with speed $3u \text{ m s}^{-1}$

Using the model,

(b) find the value of u ,

(3)

(c) find the distance travelled by the vehicle during the first 14 seconds after passing P

(4)

Q10 PAST PAPER Pearson IAL WME01 · January 2026 · Q5

M1 · TOPIC 4 12 marks

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through A , B and C in that order.

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

- (a) Show that $U = 2.1$ to 2 significant figures.

(6)

The particle passes through C with speed $\frac{U}{3} \text{ m s}^{-1}$

- (b) Find the time taken to move from B to C .

(3)

- (c) Find the distance AC .

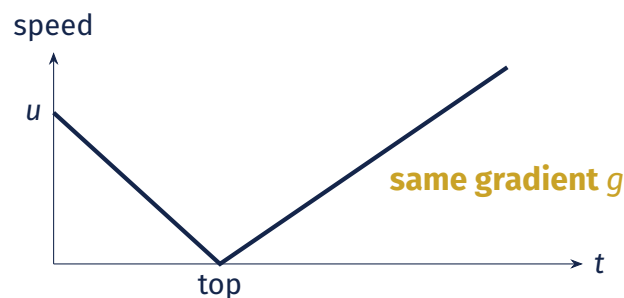
(3)

03 Vertical Motion Under Gravity

Moving freely under gravity means $a = -9.8$ if up is positive. Keep one list for the whole flight — up, top and down — and measure s from the point of projection. Below the start, s is negative.

WHEN YOU SEE → YOU DO

“projected vertically upwards”	→ up +, $a = -9.8$
“greatest height”	→ $v = 0$ at the top
“from a point 24.5 m above the ground”	→ the ground is at $s = -24.5$
“the time for which it is above C”	→ two roots of one quadratic, subtract
“sketch a speed–time graph”	→ V shape: down to 0 at the top, up again
“speed less than 14.7”	→ find where $v = \pm 14.7$



WORKED MODEL • JANUARY 2021

A stone is projected upwards at 20 m s^{-1} from O , 5 m above the ground. Find its speed when it is 2 m above the ground.

1 $s = -3$ (3 m below O), $u = 20$, $a = -9.8$

list with signs

2 $v^2 = 400 + 2(-9.8)(-3) = 458.8$

$v^2 = u^2 + 2as$

ANSWER $v = 21.4 \text{ m s}^{-1}$

IDEA 03

Practise: Vertical Motion Under Gravity

My questions first, then real past papers.

Q11 **DR ESLAM** Up and back down

M1 • TOPIC 4 3 marks

A ball is thrown vertically upwards at 14.7 m s^{-1} . Find the greatest height and the time until it returns to the point of projection.

Q12 **DR ESLAM** From above the ground

M1 • TOPIC 4 4 marks

A stone is thrown vertically upwards at 10 m s^{-1} from a point 20 m above the ground. Find the time until it hits the ground.

Up is +: the ground is at $s = -20$. One equation, then the positive root.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · January 2021 · Q1

M1 · TOPIC 4 6 marks

1. A small stone is projected vertically upwards with speed 20 m s^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (a) the speed of the stone at the instant when it is 2 m above the ground, (2)
- (b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground. (4)

Q14 PAST PAPER Pearson IAL WME01 · January 2022 · Q4

M1 · TOPIC 4 8 marks

4. At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B , where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

The ball is modelled as a particle moving freely under gravity.

- (a) Find the value of T .

(6)

- (b) Sketch a speed-time graph for the motion of the ball from $t = 0$ to $t = T$ seconds.

(No further calculations are needed in order to draw this sketch.)

(2)

Q15**PAST PAPER**

Pearson IAL WME01 · October 2022 · Q5

M1 · TOPIC 4**9 marks**

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1} (3)
- (b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1} (3)
- (c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from A to the instant when it hits the ground. Show clearly where your graph meets the axes. (3)

Q16**PAST PAPER**

Pearson IAL WME01 · January 2024 · Q6

M1 · TOPIC 4 12 marks

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of V .

(3)

- (b) Find the time taken for the particle to reach B .

(3)

The point C is 10 m vertically above A .

- (c) Find the length of time for which the particle is above C .

(4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches B . (No further calculations are required.)

(2)

Q17 PAST PAPER Pearson IAL WME01 • June 2024 • Q5

M1 • TOPIC 4 14 marks

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.
- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
 - the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
 - from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
 - the parachute and box continue to fall vertically downwards after the parachute opens
 - the parachute and box are modelled throughout the motion as a particle P of mass 250 kg
- (a) Find the distance fallen by P in the first 5 seconds. (2)
- (b) Find the speed with which P lands on the ground. (7)
- (c) Find the total time from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (3)
- (d) Sketch a speed-time graph for the motion of P from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (2)

Q18 **PAST PAPER** Pearson IAL WME01 • June 2025 • Q3**M1 • TOPIC 4** **8 marks**

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (a) Find the value of U . (3)
- (b) Find the speed of the particle as it hits the ground at B . (3)
- (c) Sketch a speed-time graph for the motion of the particle from the instant it leaves A to the instant it reaches B . (No further calculations are required.) (2)

04 Bounces and Two Particles Meeting

A bounce starts a new flight: find the speed at impact, change it (halved, or given), and start again with u upwards. Two particles meet when they are at the same height at the same time: write each height in t and set them equal.

WHEN YOU SEE → YOU DO

“the speed is instantaneously reduced by 50%”

→ new $u = \frac{1}{2}$ the impact speed

“rebounds to a height of 2 m”

→ $0 = u^2 - 2g(2)$ gives the rebound speed

“impulse received on impact”

→ $m(v + u)$ — the directions reverse

“projected 1 second later”

→ its time is $t - 1$

“collide” / “at the same height”

→ equal heights from one reference level

WORKED MODEL • OCTOBER 2019

Released from 40 m. Each bounce halves the speed. Height after the third bounce?

1 impact speed $\sqrt{2(9.8)(40)} = 28$

$$v^2 = 2gs$$

2 after 3 bounces: $28 \rightarrow 14 \rightarrow 7 \rightarrow 3.5$

halve each time

ANSWER $h = \frac{3.5^2}{19.6} = 0.625 \text{ m}$

IDEA 04

Practise: Bounces and Two Particles Meeting

My questions first, then real past papers.

Q19**DR ESLAM**

One bounce

M1 • TOPIC 4**3 marks**

A ball is dropped from 10 m. It rebounds with half the speed it hits the ground with. Find the greatest height after the bounce.

Q20**DR ESLAM**

Two particles meet

M1 • TOPIC 4**4 marks**

P is dropped from 40 m. At the same instant Q is thrown upwards from the ground below at 20 m s^{-1} . When and at what height do they meet?

Heights equal: $40 - 4.9t^2 = 20t - 4.9t^2$.

PAST PAPERS

Q21 PAST PAPER Pearson IAL WME01 · October 2019 · Q2

M1 · TOPIC 4 9 marks

2. A small ball is released from rest from a point that is 40 m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (a) Find the time from the instant when the ball is released from rest to the instant when it hits the ground for the second time. (5)
- (b) Find the maximum height reached by the ball above the ground after the ball's third bounce. (4)

Q22**PAST PAPER**

Pearson IAL WME01 · January 2020 · Q3

M1 · TOPIC 4**11 marks**

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .
- (a) Find H in terms of U and g . (2)
- (b) Find, in terms of U and g , the time between the instant when Q is projected and the instant when the two particles collide. (6)
- (c) Find where the two particles collide. (3)

Q23**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q7

M1 · TOPIC 4**11 marks**

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

- (a) Show that $T_1 = 1.63$ to 3 significant figures.

(2)

After the first impact with the ground, A rebounds to a height of 2 m above the ground.

Given that the mass of A is 0.1 kg,

- (b) find the magnitude of the impulse received by A as a result of its first impact with the ground.

(5)

At time $t = 1$ second, another small ball B is projected vertically upwards from another point on the ground with speed 5 m s^{-1}

Ball B is modelled as a particle moving freely under gravity.

At time $t = T_2$ seconds ($T_2 > 1$), A and B are at the same height above the ground for the first time.

- (c) Find the value of T_2

(4)

Q24 PAST PAPER Pearson IAL WME01 · October 2025 · Q4

M1 · TOPIC 4 9 marks

4. The point A is 10 m above horizontal ground.
At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A .
Particle P moves freely under gravity.

(a) Find the greatest height above A reached by P .

(3)

The point B is on the ground, vertically below A .

At time $t = 1$ second, a particle Q is projected vertically upwards with speed 7 m s^{-1} from B .

Particle Q moves freely under gravity.

Particles P and Q collide at time $t = T$ seconds.

(b) Find the value of T .

(4)

(c) Find the speed of P at the instant immediately before the particles collide.

(2)

Q25**PAST PAPER**

Pearson IAL WME01 • June 2026 • Q4

M1 • TOPIC 4 9 marks

4. The point O is 28.4 m above a horizontal plane. A small ball is projected vertically upwards from O with speed $U \text{ m s}^{-1}$ and first hits the plane with speed 24 m s^{-1}

Air resistance is assumed to be negligible.

- (a) Find the value of U .

(3)

Each time the ball hits the plane, the speed of the ball is instantaneously halved.

- (b) Find the **total** time, **after the ball has hit the plane for the first time**, for which the ball is at least 4 m above the plane.

(6)

(Total for Question 4 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 01 | SUVAT | List s, u, v, a, t ; the missing letter picks the equation. |
| 02 | Stages | New list each stage; end speed = next start speed. |
| 03 | Vertical | Up +, $a = -9.8$; below the start $s < 0$. |
| 04 | Bounces, meetings | New flight after a bounce; equal heights, same t . |

BOTTOM LINE Positive direction first • five letters • one equation • check the sign of every answer.

Answers

Constant Acceleration in 1D · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE SUVAT TOOLKIT: ONE STAGE

- Q1** Dr Eslam · Pick the equation $625 = 25 + 400a \Rightarrow a = 1.5 \text{ m s}^{-2}; t = \frac{40}{3} \approx 13.3 \text{ s}$
- Q2** Dr Eslam · Two intervals $20 = 2u + 2a, 50 = 4u + 8a: a = 2.5, u = 7.5$
- Q3** Pearson IAL WME01 · June 2021 · Q2 $t = 9.5 \text{ s}$
- Q4** Pearson IAL WME01 · June 2022 · Q2 (a) 1.2 m s^{-2} (b) 21.4 m
- Q5** Pearson IAL WME01 · January 2024 · Q3 (a) $400 = \frac{1}{2}(u + 28)(20) \Rightarrow u = 12$ (b) $5\sqrt{29} - 15 \approx 11.9 \text{ s}$ (c) 1220 N

IDEA 02 · JOURNEYS IN STAGES

- Q6** Dr Eslam · Two stages $v = 20; 400 + 200 = 600 \text{ m}$
- Q7** Dr Eslam · Friction gives the deceleration $a = -0.1g = -0.98; 0 = 36 - 1.96s \Rightarrow s = 18.4 \text{ m}$
- Q8** Pearson IAL WME01 · October 2021 · Q3 (a) 7 m s^{-1} (b) $t = \frac{36}{13} \approx 2.77 \text{ s}$ (c) $t = \frac{168}{65} \approx 2.58 \text{ s}$
- Q9** Pearson IAL WME01 · January 2023 · Q5 (a) $6x = \frac{1}{2}(u + 2u)(12) \Rightarrow x = 3u$ (b) $u = 9$ (c) 201 m
- Q10** Pearson IAL WME01 · January 2026 · Q5 (a) $a = \frac{g}{7} = 1.4; \frac{U}{2} = U - 1.4(0.75) \Rightarrow U \approx 2.1$ (b) 0.25 s (c) $AC = 1.4 \text{ m}$

IDEA 03 · VERTICAL MOTION UNDER GRAVITY

- Q11** Dr Eslam • Up and back down
 $H = \frac{14.7^2}{19.6} = 11.0 \text{ m}; t = 3 \text{ s}$
- Q12** Dr Eslam • From above the ground
 $-20 = 10t - 4.9t^2 \Rightarrow t = 3.28 \text{ s}$
- Q13** Pearson IAL WME01 • January 2021 • Q1
 (a) 21.4 m s^{-1} (b) 4.32 s
- Q14** Pearson IAL WME01 • January 2022 • Q4
 (a) $T = 5$ (b) V shape: down to 0 at $t = 2$, then up to 29.4 at $t = 5$
- Q15** Pearson IAL WME01 • October 2022 • Q5
 (a) $22.05 \approx 22.1 \text{ m}$ (b) 0.606 s (c) V shape from 29.4 to 0 at $t = 3$, up to the landing speed
- Q16** Pearson IAL WME01 • January 2024 • Q6
 (a) $V = 25$ (b) 5 s (c) 3.98 s (d) V shape: 24 down to 0 , up to 25 at $t = 5$
- Q17** Pearson IAL WME01 • June 2024 • Q5
 (a) $122.5 \approx 123 \text{ m}$ (b) 4 m s^{-1} (c) 20 s (d) $(0, 0) \rightarrow (5, 49) \rightarrow (20, 4)$
- Q18** Pearson IAL WME01 • June 2025 • Q3
 (a) $U \approx 16.6$ (b) 22.6 m s^{-1} (c) V shape through $(0, 16.6)$, $(1.69, 0)$, $(4, 22.6)$

IDEA 04 • BOUNCES AND TWO PARTICLES MEETING

- Q19** Dr Eslam • One bounce hits at 14 m s^{-1} , leaves at 7: $\frac{49}{19.6} = 2.5 \text{ m}$
- Q20** Dr Eslam • Two particles meet $t = 2 \text{ s}$, height 20.4 m
- Q21** Pearson IAL WME01 • October 2019 • Q2 (a) $\frac{40}{7} \approx 5.71 \text{ s}$ (b) 0.625 m
- Q22** Pearson IAL WME01 • January 2020 • Q3 (a) $H = \frac{U^2}{2g}$ (b) $\frac{U}{g}$ (c) at O
- Q23** Pearson IAL WME01 • October 2024 • Q7 (a) $T_1 = \frac{16}{g} \approx 1.63$ (b) $0.1(8 + \sqrt{4g}) \approx 1.43 \text{ N s}$
(c) $T_2 = \frac{99}{68} \approx 1.46$
- Q24** Pearson IAL WME01 • October 2025 • Q4 (a) $\frac{25}{19.6} \approx 1.28 \text{ m}$ (b) $T \approx 1.86$ (c) 13.2 m s^{-1}
- Q25** Pearson IAL WME01 • June 2026 • Q4 (a) $U = 4.4$ (b) 1.65 s

DR ESLAM AHMED
eliteigcse.com

Constant Acceleration in 2D

The same SUVAT, but every letter is a vector. Do the $\mathbf{i}$ part and the $\mathbf{j}$ part separately — then speed and direction come from Pythagoras and $\tan$.

THE TOPIC ON ONE MAP

Constant Acceleration in 2D

01 — Velocity vectors

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\text{speed} = |\mathbf{v}|$$

$\mathbf{a}$ from two velocities

02 — Direction of motion

parallel to $\mathbf{a}$ vector

bearing

same speed

03 — Position vectors

$$\mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

distance from O

VECTOR SUVAT • two equations you need

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\text{speed} = \sqrt{v_i^2 + v_j^2}$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

$$\text{direction: } \tan \theta = \frac{v_j}{v_i} \text{ with a sketch}$$

01 Velocity Vectors: $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

Everything is a vector: add and multiply the **i** parts together and the **j** parts together. Speed is the magnitude of the velocity; acceleration is the change of velocity divided by the time.

WHEN YOU SEE → YOU DO

“find the speed”

$$\rightarrow |\mathbf{v}| = \sqrt{a^2 + b^2}$$

velocities at two times

$$\rightarrow \mathbf{a} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{t_2 - t_1}$$

“ $\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$ ”

→ the bracket times t is **a**

“both boats have the same speed”

$$\rightarrow |\mathbf{v}_A|^2 = |\mathbf{v}_B|^2 - \text{a quadratic in } t$$

km h⁻¹, “hours after noon”

→ keep hours; $t = 0$ at noon

WORKED MODEL • JUNE 2022

$\mathbf{v} = -\mathbf{i} + 4\mathbf{j}$ at $t = 1$ and $5\mathbf{i} - 8\mathbf{j}$ at $t = 4$. Find the speed at $t = 3.5$.

1 $\mathbf{a} = \frac{(6\mathbf{i} - 12\mathbf{j})}{3} = 2\mathbf{i} - 4\mathbf{j}$

change ÷ time

2 $\mathbf{v} = -\mathbf{i} + 4\mathbf{j} + 2.5(2\mathbf{i} - 4\mathbf{j}) = 4\mathbf{i} - 6\mathbf{j}$

2.5 s after $t = 1$

ANSWER $\sqrt{16 + 36} = \sqrt{52} \approx 7.2 \text{ m s}^{-1}$

IDEA 01

Practise: Velocity Vectors: $\mathbf{v} = \mathbf{u} + \mathbf{at}$

My questions first, then real past papers.

Q1 DR ESLAM Velocity at a time

M1 • TOPIC 5 3 marks

A particle has initial velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and acceleration $(\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. Find its velocity and speed at $t = 4$.

Q2 DR ESLAM Acceleration from two velocities

M1 • TOPIC 5 3 marks

The velocity changes from $(3\mathbf{i} + \mathbf{j})$ to $(9\mathbf{i} - 7\mathbf{j}) \text{ m s}^{-1}$ in 2 s. Find the acceleration and its magnitude.

$\mathbf{a} = \frac{\mathbf{v} - \mathbf{u}}{t}$ — do $\mathbf{i}$ and $\mathbf{j}$ separately.

PAST PAPERS

Q3 **PAST PAPER** Pearson IAL WME01 • June 2022 • Q6**M1 • TOPIC 5** **6 marks**

6. A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j}) \text{ m s}^{-1}$

Find the speed of P at time $t = 3.5$ seconds.

(6)

Q4 PAST PAPER Pearson IAL WME01 · June 2025 · Q7

M1 · TOPIC 5 13 marks

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]Two boats, A and B , are each moving with constant acceleration.At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

(a) Find the magnitude of the acceleration of A .

(2)

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$ When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$ (b) Find the acceleration of B , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)

(c) Find the velocity of B at time $t = 0$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)

At time T_1 hours after noon, both boats are moving with the same speed.(d) Find the exact value of T_1

(4)

At time T_2 hours after noon, both boats are moving in the same direction.(e) Show that $3T_2^2 + pT_2 + q = 0$, where p and q are integers to be found.

(3)

02 Direction of Motion and Bearings

The direction of motion is the direction of the **velocity**. “Parallel to $p\mathbf{i} + q\mathbf{j}$ ” means the parts of $\mathbf{v}$ are in the ratio $p : q$. For a bearing, sketch the vector from north and measure clockwise.

WHEN YOU SEE → YOU DO

“moving in the direction of $(-\mathbf{i} + 2\mathbf{j})$ ”

→ $\frac{v_j}{v_i} = \frac{2}{-1}$, keep the signs

“moving due east / parallel to $\mathbf{i}$ ”

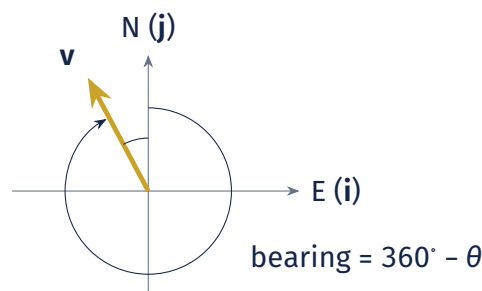
→ $\mathbf{j}$ part = 0

“as a bearing”

→ sketch; angle clockwise from north

“the velocity of P equals the velocity of Q ”

→ equal $\mathbf{i}$ parts and equal $\mathbf{j}$ parts



WORKED MODEL • JUNE 2021

$\mathbf{v} = (3 - 4T)\mathbf{i} + (5 + 4T)\mathbf{j}$ is parallel to $(-\mathbf{i} + 2\mathbf{j})$. Find T .

1 $5 + 4T = -2(3 - 4T)$

$\mathbf{j}$ part = $-2 \times \mathbf{i}$ part

ANSWER $T = \frac{11}{4}$ hours

× **TRAP** Using the position vector for the direction of motion. Direction of motion = direction of $\mathbf{v}$.

IDEA 02

Practise: Direction of Motion and Bearings

My questions first, then real past papers.

Q5 DR ESLAM Bearing of a velocity

M1 • TOPIC 5 2 marks

$\mathbf{i}$ is due east and $\mathbf{j}$ due north. Find the bearing of the velocity $(-3\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$.

Q6 DR ESLAM Parallel to a vector

M1 • TOPIC 5 4 marks

$\mathbf{v} = (2 + t)\mathbf{i} + (5 - 2t)\mathbf{j}$. Find when the particle moves (a) parallel to $\mathbf{i}$ (b) parallel to $\mathbf{i} + \mathbf{j}$.

Parallel to $\mathbf{i}$: the $\mathbf{j}$ part is 0. Parallel to $\mathbf{i} + \mathbf{j}$: the two parts are equal.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WME01 • June 2021 • Q5

M1 • TOPIC 5 9 marks

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2 pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30 pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2 pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

(a) Find the value of T .

(6)

Another particle, Q , has velocity $\mathbf{v}_Q \text{ km h}^{-1}$ at time t hours after 2 pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and μ is a constant.

Given that there is an instant when the velocity of P is equal to the velocity of Q ,

(b) find the value of μ .

(3)

Q8**PAST PAPER**

Pearson IAL WME01 • October 2023 • Q4

M1 • TOPIC 5**10 marks**

4. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ m s}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

- (a) Find the velocity of P when $t = 5\text{ s}$, giving your answer in terms of $\mathbf{i}$, $\mathbf{j}$ and λ .

(2)

The speed of P when $t = 5\text{ s}$ is 13 m s^{-1}

- (b) Show that

$$25\lambda^2 - 42\lambda - 16 = 0$$

(3)

- (c) Find the direction of motion of P when $t = 4\text{ s}$, giving your answer as a bearing to the nearest degree.

(5)

03 Position Vectors: $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

Start from where the particle is at $t = 0$, add $\mathbf{u}t$, add $\frac{1}{2}\mathbf{a}t^2$. The distance from O is the magnitude of $\mathbf{r}$; the distance between two particles is the magnitude of the difference.

WHEN YOU SEE → YOU DO

“the position vector at time t ” $\longrightarrow \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

“starts from rest at O ” $\longrightarrow \mathbf{r}_0 = \mathbf{0}, \mathbf{u} = \mathbf{0}$

“distance from O ” $\longrightarrow |\mathbf{r}|$

“due north of O ” $\longrightarrow \mathbf{i} \text{ part} = 0$

WORKED MODEL • A PARTICLE FROM REST

From rest at O with $\mathbf{a} = 2\mathbf{i} - \mathbf{j}$. Distance from O at $t = 3$?

1 $\mathbf{r} = \frac{1}{2}(2\mathbf{i} - \mathbf{j})(9) = 9\mathbf{i} - 4.5\mathbf{j}$

$\mathbf{r}_0 = \mathbf{u} = \mathbf{0}$

ANSWER $|\mathbf{r}| = \sqrt{81 + 20.25} \approx 10.1 \text{ m}$

IDEA 03

Practise: Position Vectors: $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

Questions written to practise exactly this idea.

Q9**DR ESLAM** Where is it?**M1 • TOPIC 5** 3 marks

$\mathbf{r}_0 = 2\mathbf{i} + \mathbf{j}$, $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j}$, $\mathbf{a} = \mathbf{i} + \mathbf{j}$. Find the position vector at $t = 2$.

Q10**DR ESLAM** Distance from the start**M1 • TOPIC 5** 3 marks

A particle starts from rest at O with acceleration $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-2}$. Find its distance from O after 3 s.

ALL The Whole Topic on One Page

Tear this page out before the exam.

01 Velocity

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t; \text{ speed} = |\mathbf{v}|.$$

02 Direction

Parts of $\mathbf{v}$ in the ratio of the vector; bearing from a sketch.

03 Position

$$\mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2; \text{ distance} = |\mathbf{r}|.$$

BOTTOM LINE $\mathbf{i}$ with $\mathbf{i}$, $\mathbf{j}$ with $\mathbf{j}$ • speed by Pythagoras • direction from $\mathbf{v}$, never from $\mathbf{r}$.

Answers

Constant Acceleration in 2D • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • VELOCITY VECTORS: $\mathbf{v} = \mathbf{u} + \mathbf{at}$

- Q1** Dr Eslam • Velocity at a time $\mathbf{v} = 6\mathbf{i} + 5\mathbf{j}$; speed $\sqrt{61} \approx 7.81 \text{ m s}^{-1}$
- Q2** Dr Eslam • Acceleration from two velocities $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$; $|\mathbf{a}| = 5 \text{ m s}^{-2}$
- Q3** Pearson IAL WME01 • June 2022 • Q6 $\mathbf{a} = 2\mathbf{i} - 4\mathbf{j}$; $\mathbf{v}(3.5) = 4\mathbf{i} - 6\mathbf{j}$; speed $\sqrt{52} = 2\sqrt{13} \approx 7.2 \text{ m s}^{-1}$
- Q4** Pearson IAL WME01 • June 2025 • Q7 (a) $\sqrt{17} \approx 4.1 \text{ km h}^{-2}$ (b) $-\mathbf{i} - 2\mathbf{j}$ (c) $6\mathbf{i} + 5\mathbf{j}$ (d) $T_1 = \frac{-1 + \sqrt{17}}{2}$ (e) $p = -14, q = 4$

IDEA 02 • DIRECTION OF MOTION AND BEARINGS

- Q5** Dr Eslam • Bearing of a velocity north-west: 315°
- Q6** Dr Eslam • Parallel to a vector (a) $t = 2.5$ (b) $2 + t = 5 - 2t \Rightarrow t = 1$
- Q7** Pearson IAL WME01 • June 2021 • Q5 (a) $T = \frac{11}{4} \text{ h}$ (b) $\mu = 8.5$
- Q8** Pearson IAL WME01 • October 2023 • Q4 (a) $(5 - 5\lambda)\mathbf{i} + (-8 + 10\lambda)\mathbf{j}$ (b) $25\lambda^2 - 42\lambda - 16 = 0$ (c) 339°

IDEA 03 • POSITION VECTORS: $\mathbf{r} = \mathbf{r}_0 + \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$

- Q9** Dr Eslam • Where is it? $\mathbf{r} = 10\mathbf{i} - \mathbf{j}$
- Q10** Dr Eslam • Distance from the start $\mathbf{r} = 9\mathbf{i} - 4.5\mathbf{j}$; $\sqrt{101.25} \approx 10.1 \text{ m}$



DR ESLAM AHMED
eliteigcse.com

Forces and Friction

Draw every force, then resolve in two directions at right angles. Friction is at most μR , and equals μR when limiting.

THE TOPIC ON ONE MAP

Forces and Friction

01 — Equilibrium of a particle

resolve twice

two strings

constant speed

02 — Friction on a horizontal surface

$F \leq \mu R$

pulled at an angle

does it move?

03 — Rings and beads

horizontal rod

vertical wire

which way is friction?

EQUILIBRIUM = RESULTANT FORCE ZERO

→ forces = ← forces

↑ forces = ↓ forces

“constant speed” and “at rest” are both equilibrium
limiting

friction: $F \leq \mu R$, $= \mu R$ when

01 Equilibrium of a Particle: Resolve Twice

Draw the particle as a dot with every force as an arrow. Choose two directions at right angles (usually horizontal and vertical) and write “forces one way = forces the other way” for each. Two equations, two unknowns.

WHEN YOU SEE → YOU DO

“in equilibrium”, “at rest”, “constant speed”

→ resultant force = 0: resolve twice

a force at θ to the horizontal

→ $F \cos \theta$ across, $F \sin \theta$ up

“ $\tan \theta = \frac{3}{4}$ ”

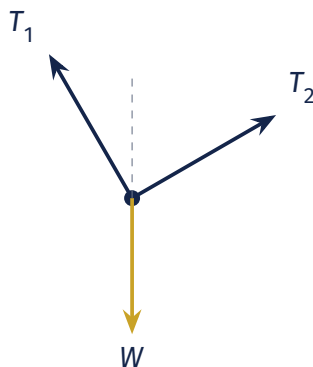
→ $\sin \theta = \frac{3}{5}$, $\cos \theta = \frac{4}{5}$ — exact

forces as vectors $a\mathbf{i} + b\mathbf{j}$

→ add the vectors, total = 0

“smooth ring on a string”

→ the tension is the same on both sides



$$\rightarrow: T_2 \cos 30^\circ = T_1 \cos 60^\circ$$

$$\uparrow: T_1 \sin 60^\circ + T_2 \sin 30^\circ = W$$

WORKED MODEL • JANUARY 2022

P (weight 5 N) on a string at 60° to the vertical, held by F perpendicular to the string. Find F and T .

1 resolve along the string: $T = 5 \cos 60^\circ$

perpendicular pair of directions

2 resolve along F : $F = 5 \sin 60^\circ$

ANSWER $F = \frac{5\sqrt{3}}{2} \approx 4.3 \text{ N}, \quad T = 2.5 \text{ N}$

IDEA 01

Practise: Equilibrium of a Particle: Resolve Twice

My questions first, then real past papers.

Q1 DR ESLAM Two strings

M1 • TOPIC 6 4 marks

A particle of weight 20 N hangs in equilibrium from two strings that make 30° and 60° with the vertical. Find the two tensions.

Resolve horizontally and vertically: two equations, two tensions.

Q2 DR ESLAM Vector forces

M1 • TOPIC 6 2 marks

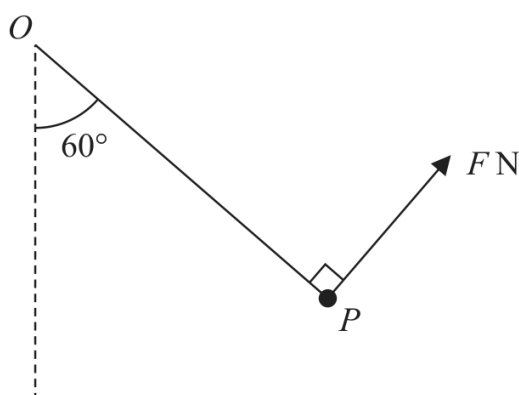
Forces $(4\mathbf{i}+3\mathbf{j})$ N, $(-2\mathbf{i}+5\mathbf{j})$ N and $\mathbf{F}$ are in equilibrium. Find $\mathbf{F}$ and its magnitude.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 • January 2022 • Q1

M1 • TOPIC 6 6 marks

1.

**Figure 1**

A particle P of weight 5 N is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O . The particle P is held in equilibrium by a force of magnitude F newtons. The direction of this force is perpendicular to the string and OP makes an angle of 60° with the vertical, as shown in Figure 1.

Find

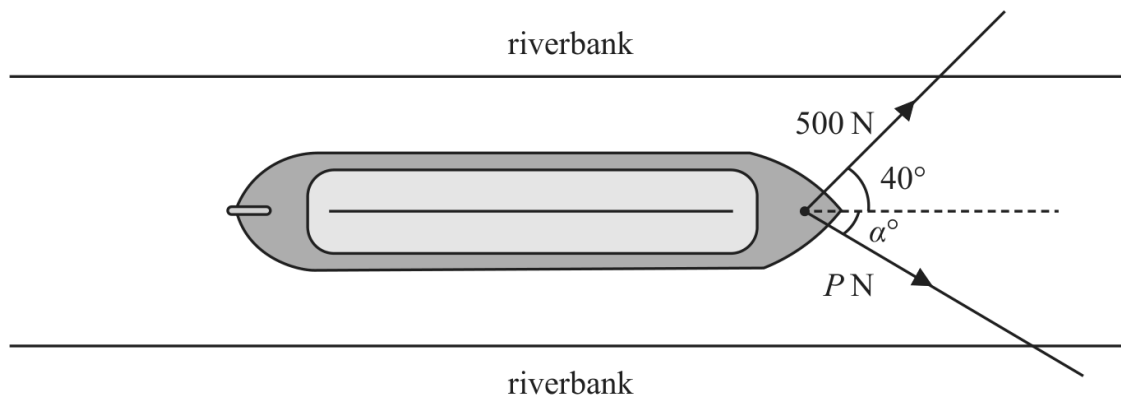
(a) the value of F (3)

(b) the tension in the string. (3)

Q4 PAST PAPER Pearson IAL WME01 • January 2023 • Q6

M1 • TOPIC 6 8 marks

6.

**Figure 3**

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of 40° to the direction of motion, as shown in Figure 3.
- The tension in the second rope is P newtons, acting at an angle of α° to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- the value of α
- the value of P

(8)

Q5 PAST PAPER Pearson IAL WME01 · January 2024 · Q1

M1 · TOPIC 6 6 marks

1.

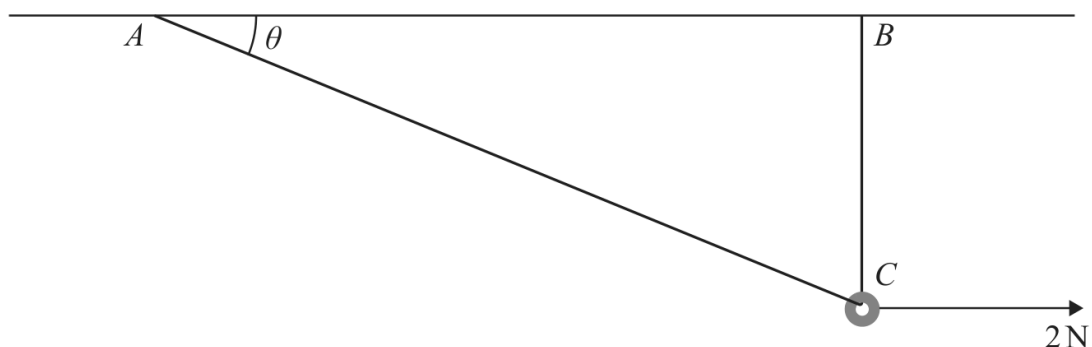
**Figure 1**

Figure 1 shows a small smooth ring **threaded** onto a light inextensible string.

One end of the string is attached to a fixed point A on a horizontal ceiling and the other end of the string is attached to a fixed point B on the ceiling.

A horizontal force of magnitude 2 N acts on the ring so that the ring rests in equilibrium at a point C , vertically below B , with the string taut.

The line of action of the horizontal force and the string both lie in the same vertical plane.

The angle that the string makes with the ceiling at A is θ , where $\tan \theta = \frac{3}{4}$

The tension in the string is T newtons. The mass of the ring is M kg.

(a) Find the value of T (3)

(b) Find the value of M (3)

02 Friction on a Horizontal Surface

First find R by resolving vertically — a pull at an angle makes R smaller, a push makes it bigger. Then friction: at rest, F is whatever is needed (at most μR); on the point of sliding, or moving, $F = \mu R$ exactly.

WHEN YOU SEE → YOU DO

“on the point of sliding” / “limiting equilibrium”

$$\rightarrow F = \mu R$$

“find the friction and describe what happens”

→ find the friction needed, compare with μR

a pull upwards at 30°

$$\rightarrow R = mg - P \sin 30^\circ$$

a push downwards at α

$$\rightarrow R = mg + P \sin \alpha$$

“show that $P \leq \dots$ ”

→ needed friction $\leq \mu R$, rearrange

moving: “find the acceleration”

$$\rightarrow F = \mu R, \text{ then } F = ma$$

Resolve vertically: R

Friction needed

Compare with μR

Conclude

WORKED MODEL • JUNE 2022

5 kg block, $\mu = \frac{3}{7}$, $P = 14$ N at 30° above the horizontal. Does it move?

1 $R = 5g - 14 \sin 30^\circ = 42$

resolve vertically

2 needed: $14 \cos 30^\circ = 12.1$; available: $\frac{3}{7}(42) = 18$

compare

ANSWER $F = 12.1$ N; the block stays at rest

× **TRAP** Writing $R = mg$ when the force is at an angle. Always resolve vertically first.

IDEA 02

Practise: Friction on a Horizontal Surface

My questions first, then real past papers.

Q6 DR ESLAM Does it move?

M1 • TOPIC 6 3 marks

A 10 kg box rests on a rough horizontal floor, $\mu = 0.4$. A horizontal force of 30 N acts on it. Does it move? Find the friction.

Compare the push with μR (the most friction can give).

Q7 DR ESLAM Pulled at an angle

M1 • TOPIC 6 4 marks

A 5 kg box, $\mu = 0.5$, is pulled by a force P at 30° above the horizontal and is on the point of sliding. Find P .

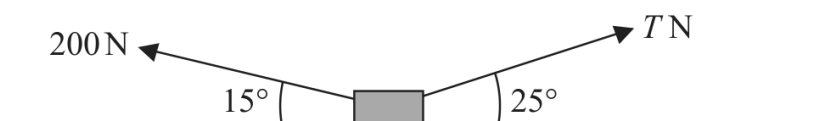
The pull lifts the box a little: $R = 5g - P \sin 30^\circ$.

PAST PAPERS

Q8 PAST PAPER Pearson IAL WME01 · January 2021 · Q3

M1 · TOPIC 6 9 marks

3.

**Figure 1**

A parcel of mass 20 kg is at rest on a rough horizontal floor. The coefficient of friction between the parcel and the floor is 0.3

Two forces, both acting in the same vertical plane, of magnitudes 200 N and T N are applied to the parcel. The line of action of the 200 N force makes an angle of 15° with the horizontal and the line of action of the T N force makes an angle of 25° with the horizontal, as shown in Figure 1. The parcel is modelled as a particle P .

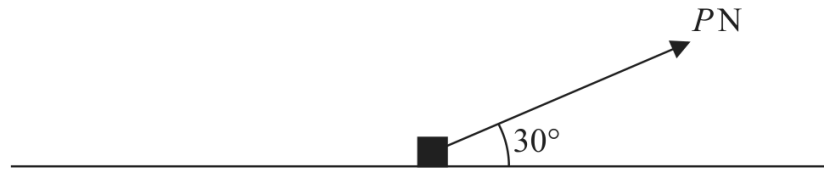
Find the smallest value of T for which P remains in equilibrium.

(9)

Q9 PAST PAPER Pearson IAL WME01 · June 2022 · Q4

M1 · TOPIC 6 12 marks

4.

**Figure 1**

A small block of mass 5 kg lies at rest on a rough horizontal plane.

The coefficient of friction between the block and the plane is $\frac{3}{7}$

A force of magnitude P newtons is applied to the block in a direction which makes an angle of 30° with the plane, as shown in Figure 1.

The block is modelled as a particle.

Given that $P = 14$

- (a) find the magnitude of the frictional force exerted on the block by the plane and describe what happens to the block, justifying your answer. (6)

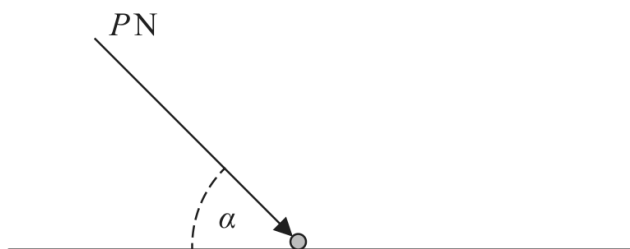
The value of P is now changed so that the block is on the point of slipping along the plane.

- (b) Find the value of P (6)

Q10 PAST PAPER Pearson IAL WME01 • June 2023 • Q6

M1 • TOPIC 6 7 marks

6.

**Figure 3**

A particle of weight W newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude P newtons is applied to the particle.

The force acts at an angle α to the horizontal, where $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is $\frac{1}{4}$

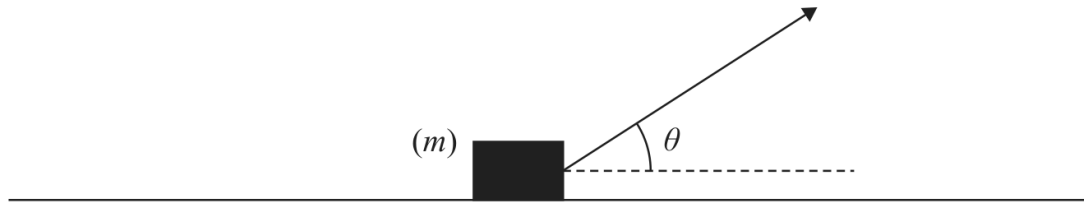
Given that the particle does not move, show that

$$P \leq \frac{5W}{8} \quad (7)$$

Q11 PAST PAPER Pearson IAL WME01 · June 2024 · Q6

M1 · TOPIC 6 12 marks

6.

**Figure 3**

A box of mass m lies on a rough horizontal plane. The box is pulled along the plane in a straight line at **constant speed** by a light rope. The rope is inclined at an angle θ to the plane, as shown in Figure 3.

The coefficient of friction between the box and the plane is $\frac{1}{3}$

The box is modelled as a particle.

Given that $\tan \theta = \frac{3}{4}$

(a) find, in terms of m and g , the tension in the rope.

(7)

The rope is now removed and the box is placed at rest on the plane. The box is then projected horizontally along the plane with speed u .

The box is again modelled as a particle.

When the box has moved a distance d along the plane, the speed of the box is $\frac{1}{2}u$.

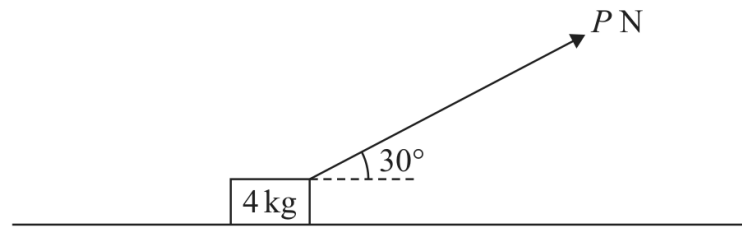
(b) Find d in terms of u and g .

(5)

Q12 PAST PAPER Pearson IAL WME01 • October 2025 • Q5

M1 • TOPIC 6 13 marks

5.

**Figure 3**

A box of mass 4 kg is placed on a rough horizontal surface.

A force of magnitude P newtons, acting at 30° to the horizontal, is applied to the box, as shown in Figure 3.

The coefficient of friction between the box and the surface is $\frac{2}{3}$

The box is modelled as a particle.

(a) Find the value of P when the box is on the point of sliding along the surface. (6)

The value of P is now increased to 25 and the box moves along the surface.

Find

(b) the acceleration of the box, (5)

(c) the speed of the box when it has moved 1.5 m. (2)

03 Rings and Beads on Rods and Wires

The normal reaction is **perpendicular to the rod**. On a horizontal rod it is vertical; on a vertical wire it is horizontal. Friction acts along the rod, against the way the ring would slide.

WHEN YOU SEE → YOU DO

“threaded on a rough horizontal rod/wire”

→ R vertical, friction horizontal

“threaded on a rough vertical rod/wire”

→ R horizontal, friction vertical

“state the direction of the friction”

→ opposite to the way it would move without friction

“the tension can vary — min and max”

→ about to slip down: friction up; about to slip up: friction down

two rings joined by a string with a mass below

→ resolve for the mass first, then each ring

WORKED MODEL • OCTOBER 2023

Ring 0.2 kg on a rough vertical rod; string at θ to the rod, $\tan \theta = \frac{12}{5}$, tension 10 N. Friction?

1 up the rod: $10 \cos \theta = 10 \cdot \frac{5}{13} = 3.85$

tension's vertical part

2 down: weight $0.2g = 1.96$

ANSWER $F = 3.85 - 1.96 \approx 1.89$ N, acting downwards

IDEA 03

Practise: Rings and Beads on Rods and Wires

My questions first, then real past papers.

Q13 DR ESLAM Ring on a horizontal rod

M1 · TOPIC 6 4 marks

A ring of mass 0.5 kg on a rough horizontal rod ($\mu = 0.2$) is pulled by a string at 40° above the rod and is about to slide. Find the tension.

Q14 DR ESLAM Bead on a vertical wire

M1 · TOPIC 6 3 marks

A bead of weight 3 N on a rough vertical wire is held at rest by a horizontal force of 12 N. Find the least possible value of μ .

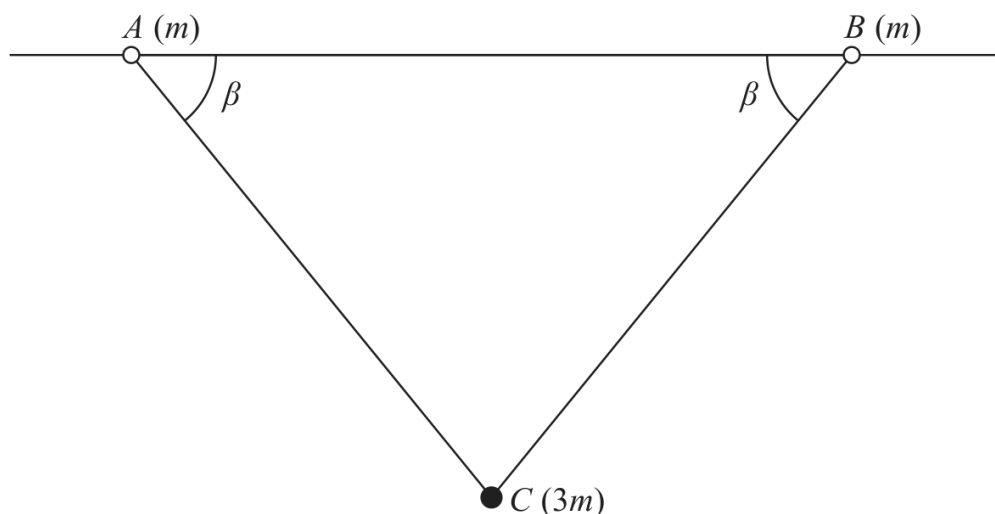
On a vertical wire the normal reaction is horizontal.

PAST PAPERS

Q15 PAST PAPER Pearson IAL WME01 • January 2020 • Q4

M1 • TOPIC 6 10 marks

4.

**Figure 2**

Two identical small rings, A and B , each of mass m , are threaded onto a rough horizontal wire. The rings are connected by a light inextensible string. A particle C of mass $3m$ is attached to the midpoint of the string. The particle C hangs in equilibrium below the wire with angle $BAC = \beta$, as shown in Figure 2.

The tension in each of the parts, AC and BC , of the string is T

(a) By considering particle C , find T in terms of m , g and β (2)

(b) Find, in terms of m and g , the magnitude of the normal reaction between the wire and A . (3)

The coefficient of friction between each ring and the wire is $\frac{4}{5}$

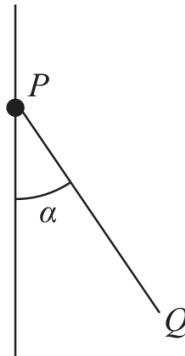
The two rings, A and B , are on the point of sliding along the wire towards each other.

(c) Find the value of $\tan \beta$ (5)

Q16 PAST PAPER Pearson IAL WME01 • October 2021 • Q5

M1 • TOPIC 6 10 marks

5.

**Figure 2**

A small bead of mass 0.2 kg is attached to the end P of a light rod PQ . The bead is threaded onto a fixed vertical rough wire.

The bead is held in equilibrium with the rod PQ inclined to the wire at an angle α , where $\tan \alpha = \frac{4}{3}$, as shown in Figure 2.

The thrust in the rod is T newtons.

The bead is modelled as a particle.

- (a) Find the magnitude and direction of the friction force acting on the bead when $T = 2.5$
- (3)

The coefficient of friction between the bead and the wire is μ .

Given that the greatest possible value of T is 6.125

- (b) find the value of μ .
- (7)

5.

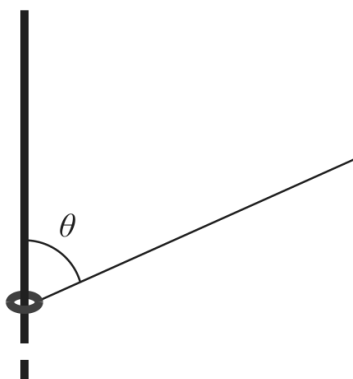


Figure 3

A small ring of mass 0.2 kg is attached to one end of a light inextensible string.

The ring is **threaded** onto a fixed rough vertical rod.

The string is taut and makes an angle θ with the rod, as shown in Figure 3,
where $\tan \theta = \frac{12}{5}$

Given that the ring is in equilibrium and that the tension in the string is 10 N ,

(a) find the magnitude of the frictional force acting on the ring, (3)

(b) state the direction of the frictional force acting on the ring. (1)

The coefficient of friction between the ring and the rod is $\frac{1}{4}$

Given that the ring is in equilibrium, and that the tension in the string, T newtons, can now vary,

(c) (i) find the minimum value of T
(ii) find the maximum value of T (8)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|--|
| 01 | Equilibrium | Resolve in two directions; two equations. |
| 02 | Friction | R first; $F \leq \mu R$; $= \mu R$ when limiting or moving. |
| 03 | Rings and beads | R perpendicular to the rod; friction against the slide. |

BOTTOM LINE Diagram $\rightarrow$ resolve $\rightarrow R \rightarrow F \leq \mu R$. Limiting means $F = \mu R$.

Answers

Forces and Friction • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • EQUILIBRIUM OF A PARTICLE: RESOLVE TWICE

- Q1** Dr Eslam • Two strings $T_{30} = 10\sqrt{3} \approx 17.3 \text{ N}, T_{60} = 10 \text{ N}$
- Q2** Dr Eslam • Vector forces $\mathbf{F} = -2\mathbf{i} - 8\mathbf{j}; \sqrt{68} \approx 8.25 \text{ N}$
- Q3** Pearson IAL WME01 • January 2022 • Q1 (a) $F = \frac{5\sqrt{3}}{2} \approx 4.3 \text{ N}$ (b) $T = 2.5 \text{ N}$
- Q4** Pearson IAL WME01 • January 2023 • Q6 (i) $\alpha \approx 31.9^\circ$ (ii) $P \approx 609 \text{ N}$
- Q5** Pearson IAL WME01 • January 2024 • Q1 (a) $T = 2.5 \text{ N}$ (b) $M \approx 0.408$

IDEA 02 • FRICTION ON A HORIZONTAL SURFACE

- Q6** Dr Eslam • Does it move? $\mu R = 39.2 > 30$: it stays; $F = 30 \text{ N}$
- Q7** Dr Eslam • Pulled at an angle $P \cos 30^\circ = 0.5(49 - 0.5P) \Rightarrow P \approx 22.0 \text{ N}$
- Q8** Pearson IAL WME01 • January 2021 • Q3 $T = 192 \text{ N}$
- Q9** Pearson IAL WME01 • June 2022 • Q4 (a) $F = 12.1 \text{ N}$ (less than μR): the block stays at rest (b) $P \approx 19.4$
- Q10** Pearson IAL WME01 • June 2023 • Q6 $R = W + \frac{4}{5}P$, $\frac{3}{5}P \leq \frac{1}{4}(W + \frac{4}{5}P) \Rightarrow P \leq \frac{5W}{8}$
- Q11** Pearson IAL WME01 • June 2024 • Q6 (a) $T = \frac{mg}{3}$ (b) $d = \frac{9u^2}{8g}$
- Q12** Pearson IAL WME01 • October 2025 • Q5 (a) $P \approx 21.8$ (b) 0.963 m s^{-2} (c) 1.70 m s^{-1}

IDEA 03 • RINGS AND BEADS ON RODS AND WIRES

- Q13** Dr Eslam • Ring on a horizontal rod $T \cos 40^\circ = 0.2(4.9 - T \sin 40^\circ) \Rightarrow T \approx 1.10 \text{ N}$
- Q14** Dr Eslam • Bead on a vertical wire $R = 12$; $3 \leq 12\mu \Rightarrow \mu \geq 0.25$
- Q15** Pearson IAL WME01 • January 2020 • Q4 (a) $T = \frac{3mg}{2 \sin \beta}$ (b) $R = \frac{5mg}{2}$ (c) $\tan \beta = \frac{3}{4}$
- Q16** Pearson IAL WME01 • October 2021 • Q5 (a) 0.46 N upwards (b) $\mu = 0.35$
- Q17** Pearson IAL WME01 • October 2023 • Q5 (a) $F \approx 1.89 \text{ N}$ (b) downwards (c)(i) $T_{\min} \approx 3.19 \text{ N}$ (ii) $T_{\max} \approx 12.7 \text{ N}$

Newton's Second Law

Resultant force = ma , in the direction of the acceleration. Connected particles: one equation per particle, or one for the whole system.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Newton's Second Law

01 One body

$F = ma$ lifts falling with resistance

02 Connected in a line

towing whole system, then one part

03 Pulleys: hanging, table

one equation each force on the pulley string breaks

04 Pulleys with a slope

friction $\tan \alpha = \frac{3}{4}$ direction of **S**

NEWTON'S SECOND LAW

$$(\text{forces WITH the acceleration}) - (\text{forces AGAINST it}) = ma$$

in equilibrium or at constant speed $a = 0$

“light, inextensible string”: same

tension, same a

01 One Body: Resultant Force = ma

Pick the direction of the acceleration as positive. Add the forces that way, subtract the forces the other way, and set it equal to ma . Then SUVAT if a speed, time or distance is wanted.

WHEN YOU SEE → YOU DO

“driving force ... resistance”

→ $D - R = ma$

a lift or a bucket moving up / down

→ $T - mg = ma$ (up) — use the direction of a

“the force on the man from the floor”

→ treat the man alone: $R - mg = ma$

“falls against a constant resistance”

→ $mg - R = ma$

a rough horizontal surface

→ friction μR against the motion

a collision, then it slides

→ momentum first, then $F = ma$ with friction

WORKED MODEL • OCTOBER 2022

Lift 250 kg; cable tension 3616 N; the children push down on the floor with $565 + 226$ N. Find a .

$$1 \quad 3616 - 250g - 791 = 250a$$

forces on the lift only

$$2 \quad 375 = 250a$$

$$\text{ANSWER} \quad a = 1.5 \text{ m s}^{-2}$$

× **TRAP** Using $F = ma$ with the forces on the *children* when the question says “forces acting on the lift only”.

IDEA 01

Practise: One Body: Resultant Force = ma

My questions first, then real past papers.

Q1 DR ESLAM Driving force and resistance

M1 • TOPIC 7 2 marks

A car of mass 1200 kg on a level road has driving force 3000 N and resistance 600 N. Find its acceleration.

Q2 DR ESLAM A lift

M1 • TOPIC 7 4 marks

A lift of mass 500 kg carries a man of mass 60 kg and accelerates upwards at 0.5 m s^{-2} . Find the tension in the cable and the reaction on the man. Whole system for T ; the man alone for R .

Q3 **DR ESLAM** Falling with resistance**M1 · TOPIC 7** **3 marks**

A stone of mass 0.5 kg falls from rest against a constant resistance of 1.4 N. Find its acceleration and its speed after 3 s.

PAST PAPERS

Q4 PAST PAPER Pearson IAL WME01 · June 2021 · Q4

M1 · TOPIC 7 6 marks

4.

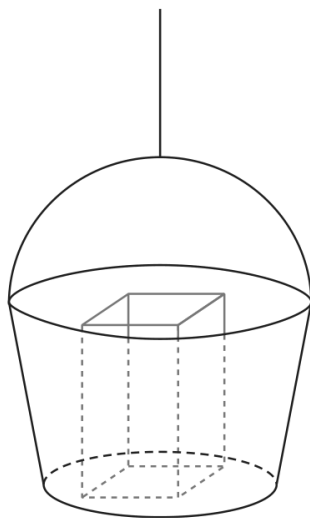
**Figure 1**

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration 0.2 ms^{-2} . Air resistance is modelled as being negligible.

(a) Find the tension in the cable.

(3)

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration 0.1 ms^{-2} . Air resistance is again modelled as being negligible.

(b) Find the magnitude of the normal reaction between the bucket and the box of tools.

(3)

Q5**PAST PAPER**

Pearson IAL WME01 · October 2021 · Q2

M1 · TOPIC 7 10 marks

2. A particle P of mass $2m$ is moving on a rough horizontal plane when it collides directly with a particle Q of mass $4m$ which is at rest on the plane. The speed of P immediately before the collision is $3u$. The speed of Q immediately after the collision is $2u$.

(a) Find, in terms of u , the speed of P immediately after the collision.

(3)

(b) State clearly the direction of motion of P immediately after the collision.

(1)

Following the collision, Q comes to rest after travelling a distance $\frac{6u^2}{g}$ along the plane.

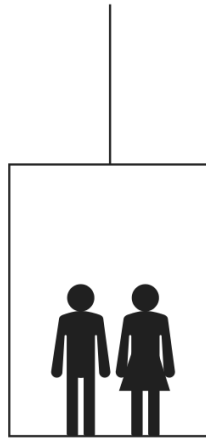
The coefficient of friction between Q and the plane is μ .

(c) Find the value of μ .

(6)

Q6 PAST PAPER Pearson IAL WME01 • October 2022 • Q4**M1 • TOPIC 7 6 marks**

4.

**Figure 3**

Two children, Alan and Bhavana, are standing on the horizontal floor of a lift, as shown in Figure 3.

The lift has mass 250 kg. The lift is raised vertically upwards with constant acceleration by a vertical cable which is attached to the top of the lift. The cable is modelled as being light and inextensible. While the lift is accelerating upwards, the tension in the cable is 3616 N.

As the lift accelerates upwards, the floor of the lift exerts a force of magnitude 565 N on Alan and a force of magnitude 226 N on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(a) By considering the forces acting on the lift only, find the acceleration of the lift.

(3)

(b) Find the mass of Alan.

(3)

Q7 PAST PAPER Pearson IAL WME01 · June 2023 · Q3

M1 · TOPIC 7 8 marks

3. Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student A suggests that the book is modelled as a particle falling freely under gravity.

- (a) Use student A 's model to find the time taken for the book to reach the floor.

(3)

Student B suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude R newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

- (b) use student B 's model to find the value of R

(5)

Q8 PAST PAPER Pearson IAL WME01 · October 2023 · Q3

M1 · TOPIC 7 10 marks

3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg .

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed 10 m s^{-1}

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed $v\text{ m s}^{-1}$

- (a) Find the value of v (2)

- (b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer. (3)

The ground exerts a constant vertical resistive force of magnitude R newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm .

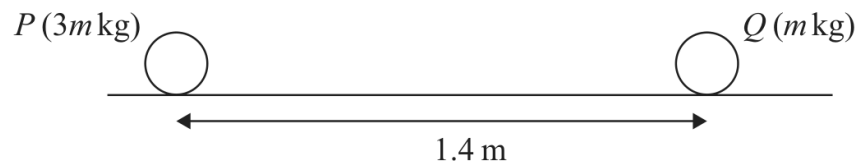
- (c) Find the value of R . (5)

Q9**PAST PAPER**

Pearson IAL WME01 • January 2025 • Q3

M1 • TOPIC 7**12 marks**

3.

**Figure 1**

A particle P of mass $3m \text{ kg}$ and a particle Q of mass $m \text{ kg}$ are at rest on a rough horizontal surface. The distance between P and Q is 1.4 m , as shown in Figure 1.

An impulse of magnitude $\lambda \text{ N s}$ is now applied to P in the direction PQ . Immediately after the impulse is applied, the speed of P is 5 m s^{-1} .

(a) Find λ in terms of m .

(2)

Immediately before P collides with Q , the speed of P is 2.5 m s^{-1} . The coefficient of friction between P and the surface is μ .

(b) Find the value of μ .

(7)

Immediately after P collides with Q , the speed of Q is 2.1 m s^{-1} .

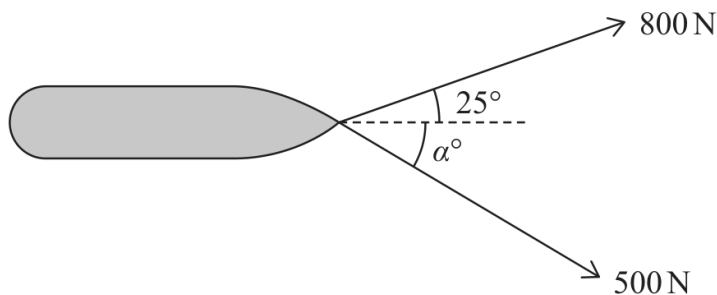
(c) Find the speed of P immediately after the collision.

(3)

Q10 PAST PAPER Pearson IAL WME01 · June 2025 · Q4

M1 · TOPIC 7 7 marks

4.

**Figure 2**

Two ropes are attached to a point on the front of a barge.
The barge is being pulled horizontally in a straight line along the centre of a long straight canal.

One rope makes an angle of 25° with the direction of motion of the barge and has a tension of 800 N.

The other rope makes an angle of α° with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

(a) Find the value of α

(3)

The mass of the barge is 15 tonnes and the resistance to the motion of the barge is a constant force of magnitude 750 N.

(b) Find the acceleration of the barge.

(4)

02 Connected in a Line: Towing and Lifts

Everything moves with the same acceleration. First treat the whole system as one body (the towbar or string tension is internal and cancels) to find a . Then take **one part alone** to find the tension.

WHEN YOU SEE → YOU DO

a car towing a trailer

→ whole: $D - R_1 - R_2 = (m_1 + m_2)a$; trailer: $T - R_2 = m_2a$

“up a road inclined at α ”

→ add $mg \sin \alpha$ against the motion

“the towbar is modelled as a light rod”

→ same tension along it; it can push (thrust)

boxes stacked in a lift

→ whole for the cable; top box alone for the contact force

“state how you used ... inextensible”

→ both parts have the same acceleration

Whole system for a

One part for T

Check with the other part

WORKED MODEL • JUNE 2023

Car 1200 kg tows trailer 600 kg up $\sin \alpha = \frac{1}{12}$; $D = 3000$ N, resistances $2R$ and R , $a = 0.75$. Show $R = 60$.

$$1 \quad 3000 - 3R - 1800g \cdot \frac{1}{12} = 1800(0.75)$$

whole system

$$2 \quad 3000 - 3R - 1470 = 1350$$

ANSWER $R = 60$

IDEA 02

Practise: Connected in a Line: Towing and Lifts

My questions first, then real past papers.

Q11 DR ESLAM Car and trailer

M1 · TOPIC 7 4 marks

A car (1000 kg) tows a trailer (500 kg) on a level road. Driving force 2400 N; resistances 300 N (car) and 150 N (trailer). Find the acceleration and the tension in the towbar.

Whole system for a ; then the trailer alone for T .

Q12 DR ESLAM Two particles on one string

M1 · TOPIC 7 4 marks

A force of 60 N pulls a 2 kg particle vertically upwards; a 3 kg particle hangs below it on a light string. Find the acceleration and the tension in the string.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · October 2019 · Q3

M1 · TOPIC 7 7 marks

3. A car of mass 800 kg is towing a trailer of mass 400 kg up a straight road using a towbar. The towbar is parallel to the road and parallel to the direction of motion of the car. The road is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{7}$. The engine of the car produces a constant driving force of magnitude D newtons. The resistance to the motion of the car from non-gravitational forces is modelled as a single force of magnitude 420 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a single force of magnitude 300 N. The car and trailer are modelled as particles and the towbar is modelled as a light rod.

Given that the tension in the towbar is 2060 N, find the value of D .

(7)

Q14 PAST PAPER Pearson IAL WME01 · June 2022 · Q3

M1 · TOPIC 7 7 marks

3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

The driving force of the tractor is 7400 N and the resistance to the motion of the tractor is 200 N. The resistance to the motion of the block is R newtons, where R is a constant.

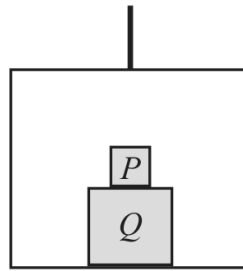
Given that the tension in the cable is 6000 N and the tractor is accelerating,

- (a) find the value of R . (6)
- (b) State how you have used the fact that the cable is modelled as being inextensible. (1)

Q15 PAST PAPER Pearson IAL WME01 · January 2023 · Q7

M1 · TOPIC 7 7 marks

7.

**Figure 4**

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass m

A box Q is placed on the floor of the lift.

A box P is placed directly on top of box Q , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration $\frac{2g}{5}$

Using the model,

(a) find, in terms of m , the combined mass of boxes P and Q

(4)

During the motion of the lift, the force exerted on box P by box Q is $\frac{14mg}{5}$

Using the model,

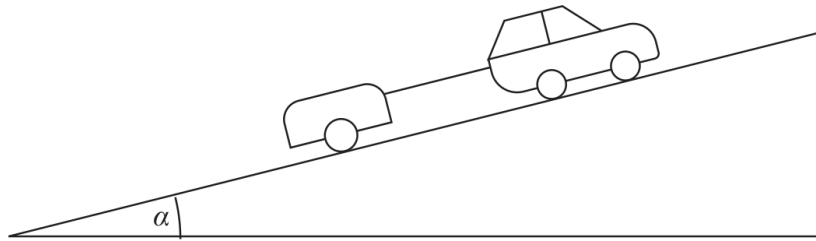
(b) find, in terms of m , the mass of box P

(3)

Q16 PAST PAPER Pearson IAL WME01 • June 2023 • Q7

M1 • TOPIC 7 11 marks

7.

**Figure 4**

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration 0.75 m s^{-2}

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude $2R$ newtons
- the **trailer** is modelled as a constant force of magnitude R newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of R is 60 (4)

(b) find the tension in the tow bar. (3)

When the car and trailer are moving at a speed of 12 m s^{-1} , the tow bar breaks.

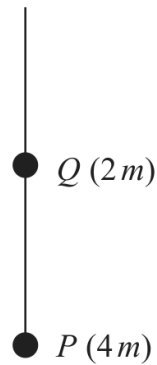
Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest. (4)

Q17 PAST PAPER Pearson IAL WME01 · June 2024 · Q3

M1 · TOPIC 7 6 marks

3.

**Figure 1**

Two particles, P and Q , have masses $4m$ and $2m$ respectively. The particles are connected by a light inextensible string. A second light inextensible string has one end attached to Q . Both strings are taut and vertical, as shown in Figure 1.

The particles are **accelerating** vertically **downwards**.

Given that the tension in the string connecting the two particles is $3mg$, find, in terms of m and g , the tension in the upper string.

(6)

03 Pulleys: Hanging and on a Table

Over a smooth pulley the tension is the same on both sides and both particles have the same size of acceleration. Write one equation for each particle, each in its **own** direction of motion, and add them to lose T .

WHEN YOU SEE → YOU DO

two particles hanging over a pulley

→ heavy: $m_1g - T = m_1a$; light: $T - m_2g = m_2a$

one on a table, one hanging over the edge

→ table: $T - F = m_1a$; hanging: $m_2g - T = m_2a$

“force exerted on the pulley”

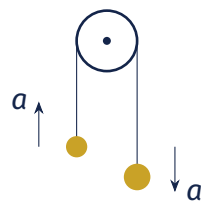
→ two tensions: $2T$ (vertical strings) or $T\sqrt{2}$ (at 90°)

“B hits the floor and does not rebound”

→ string slack: A now moves alone, new a

“the string breaks”

→ each particle moves freely under gravity



light: $T - m_2g = m_2a$

heavy: $m_1g - T = m_1a$

add: $(m_1 - m_2)g = (m_1 + m_2)a$

WORKED MODEL • OCTOBER 2024

P (3 kg) and Q (5 kg) hang over a smooth pulley. Find T and the force on the pulley.

1 $5g - T = 5a, T - 3g = 3a \Rightarrow a = \frac{g}{4}$

add

2 $T = 3(g + \frac{g}{4}) = \frac{15g}{4}$

ANSWER $T \approx 36.8 \text{ N}, F = 2T \approx 73.5 \text{ N}$

IDEA 03

Practise: Pulleys: Hanging and on a Table

My questions first, then real past papers.

Q18 DR ESLAM Two hanging masses

M1 • TOPIC 7 4 marks

Particles of 3 kg and 5 kg hang over a smooth pulley and are released from rest. Find the acceleration and the tension.

One equation for each particle, each in its own direction of motion.

Q19 DR ESLAM Table and edge

M1 • TOPIC 7 4 marks

A 2 kg block on a smooth table is joined over a pulley at the edge to a 1 kg particle hanging freely. Find the acceleration, the tension and the force on the pulley.

PAST PAPERS

Q20 PAST PAPER Pearson IAL WME01 · January 2020 · Q7

M1 · TOPIC 7 18 marks

7.

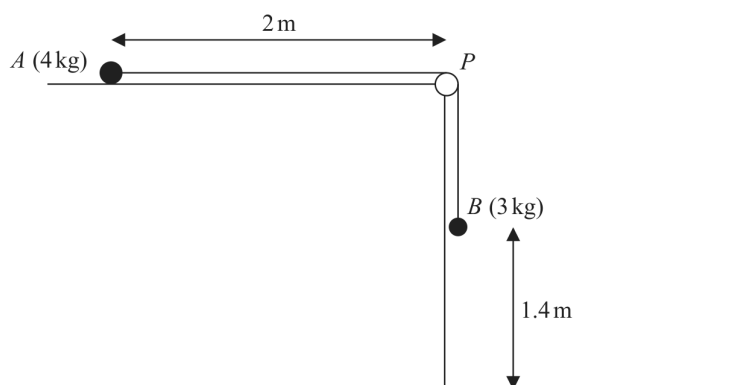


Figure 3

A particle A of mass 4 kg is held at rest on a rough horizontal table. Particle A is attached to one end of a string that passes over a pulley P . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle B , of mass 3 kg , which hangs freely below P .

The part of the string from A to P is perpendicular to the edge of the table and A , P and B all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release, A is 2 m from the edge of the table and B is 1.4 m above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for 2 seconds , B hits the floor and does not rebound.

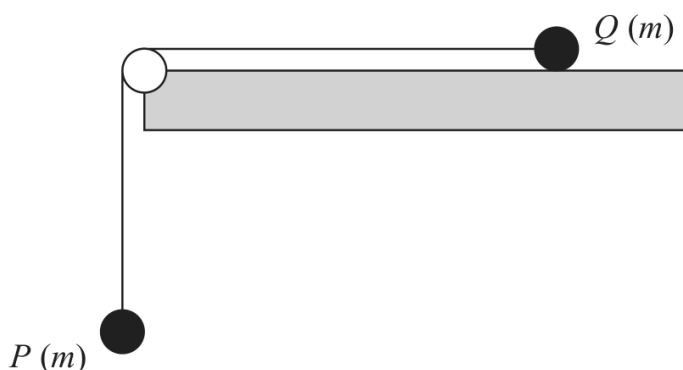
- (a) Show that the acceleration of A before B hits the floor is 0.7 m s^{-2} (2)
 - (b) State which of the modelling assumptions you have used in order to answer part (a). (1)
 - (c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)
- The coefficient of friction between A and the table is μ .
- (d) Find the value of μ . (6)
 - (e) Determine, by calculation, whether or not A reaches the pulley. (5)

(Total 18 marks)

Q21 PAST PAPER Pearson IAL WME01 · October 2022 · Q7

M1 · TOPIC 7 13 marks

7.

**Figure 4**

A particle P of mass m is attached to one end of a light inextensible string. Another particle Q , also of mass m , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle Q is held at rest on the table and particle P hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley, P and Q all lie in the same vertical plane.

The coefficient of friction between Q and the table is μ , where $\mu < 1$

Particle Q is released from rest.

The tension in the string before Q hits the pulley is kmg , where k is a constant.

(a) Find k in terms of μ .

(7)

Given that Q is initially a distance d from the pulley,

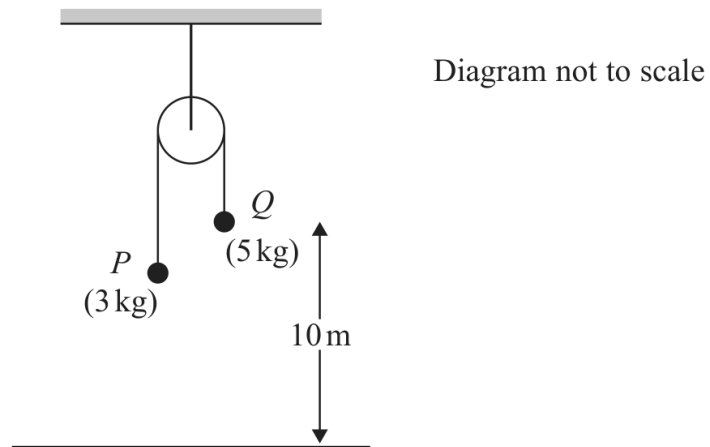
(b) find, in terms of d , g and μ , the time taken by Q , after release, to reach the pulley.

(4)

(c) Describe what would happen if $\mu \geq 1$, giving a reason for your answer.

(2)

5.

**Figure 3**

Two particles, P and Q , have masses 3 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a small smooth fixed pulley.

The particles are released from rest with the string taut and the hanging parts of the string vertical, as shown in Figure 3.

Immediately after the particles are released from rest, P moves upwards with acceleration $a\text{ ms}^{-2}$ and the tension in the string is T newtons.

(a) Write down an equation of motion for P . (2)

(b) Find the value of T . (4)

The total force acting on the pulley due to the string has magnitude F newtons.

(c) Find the value of F . (2)

Initially, Q is 10 m above horizontal ground and P is more than 2 m below the pulley.

At the instant when Q has descended a distance of 2 m , the string breaks and Q falls to the ground.

(d) Find the speed of Q at the instant it hits the ground. (5)

Q23 PAST PAPER Pearson IAL WME01 · January 2025 · Q7

M1 · TOPIC 7 15 marks

7.

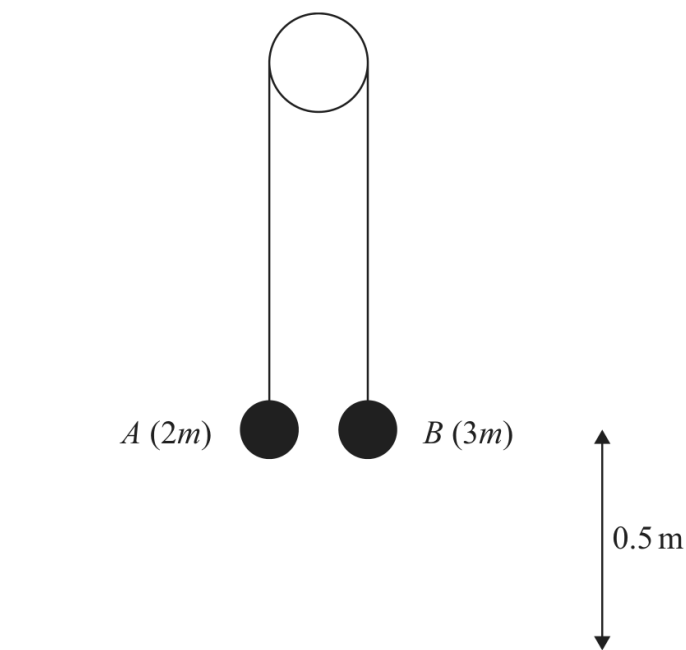


Figure 4

One end of a light inextensible string is attached to a particle A of mass $2m$.
 The other end is attached to a particle B of mass $3m$.
 The string passes over a small smooth fixed pulley.
 The string is taut and both straight parts of the string are vertical.
 Both particles are held at a distance 0.5 m above a horizontal surface, as shown in Figure 4.

The system is released from rest and B moves downwards.

(a) Find the tension in the string in terms of m and g . (5)

(b) Find the speed of B at the instant it strikes the surface. (4)

In the subsequent motion, A does not reach the pulley and B does not rebound after it strikes the surface.

(c) Find the time from the instant when the system is released from rest to the instant when A first reaches a height of 1.06 m above the surface. (6)

Q24 PAST PAPER Pearson IAL WME01 · June 2025 · Q8

M1 · TOPIC 7 17 marks

8.

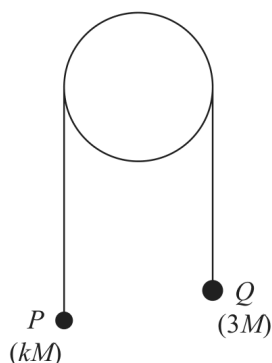


Figure 5

Two small balls, P and Q , have masses kM and $3M$ respectively, where $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion, P moves with an acceleration of magnitude $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for P .

(2)

(b) Find the value of k .

(3)

Given that $M = 0.5 \text{ kg}$,

(c) find the magnitude of the force exerted on the pulley by the string while Q is moving downwards.

(3)

At the instant when the system is released, P is more than 2.5 m from the pulley and Q is 2.5 m above horizontal ground.

After hitting the ground, Q rebounds with a speed of 0.4 ms^{-1}

(d) Find the magnitude of the impulse received by Q when it hits the ground.

(5)

In the subsequent motion, P does not hit the pulley.

(e) Find the total time from when the balls are released until P first comes to rest.

(4)

04 Pulleys with a Slope

Same method, with the slope rules from Topic 8: along the plane the weight gives $mg \sin \alpha$, perpendicular gives $R = mg \cos \alpha$, friction μR opposes the motion. One equation for the particle on the slope, one for the hanging particle.

WHEN YOU SEE → YOU DO

A on a rough slope, B hanging

$$\rightarrow T - mg \sin \alpha - \mu R = ma; Mg - T = Ma$$

"A comes to rest before the pulley" after B lands

→ new stage: A decelerates with weight and friction

"force exerted on the pulley by the string"

→ angle between the strings is $90^\circ - \alpha$ (vertical and slope): $2T \cos \frac{90^\circ - \alpha}{2}$

a horizontal string at the top (table on the other side)

→ angle between the strings $180^\circ - \alpha$

blocks added until it moves

→ limiting equilibrium, friction reverses

WORKED MODEL • OCTOBER 2021

A ($2m$) on a plane $\tan \alpha = \frac{3}{4}$, $\mu = \frac{1}{2}$; B ($3m$) hangs. Find a .

1 $R = \frac{8}{5}mg$, $F = \frac{4}{5}mg$

perpendicular, then μR

2 A: $T - \frac{6}{5}mg - \frac{4}{5}mg = 2ma$; B: $3mg - T = 3ma$

one equation each

ANSWER $mg = 5ma \Rightarrow a = \frac{g}{5}$

× **TRAP** Friction up the slope when A moves up. Friction always opposes the motion of A.

IDEA 04

Practise: Pulleys with a Slope

My questions first, then real past papers.

Q25 DR ESLAM Smooth slope and pulley

M1 • TOPIC 7 4 marks

A 4 kg particle on a smooth plane at 30° is joined over a pulley at the top to a 3 kg particle hanging freely. Find the acceleration.

Q26 DR ESLAM Rough slope and pulley

M1 • TOPIC 7 5 marks

The same system, but now $\mu = 0.2$ between the 4 kg particle and the plane. Find the acceleration.

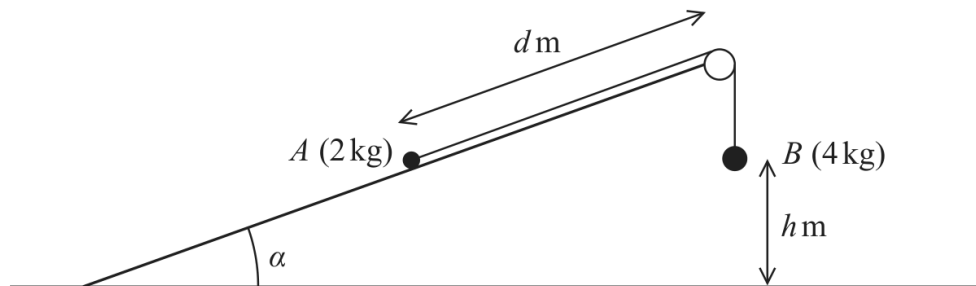
Friction acts down the slope, against the motion.

PAST PAPERS

Q27 PAST PAPER Pearson IAL WME01 · January 2021 · Q8

M1 · TOPIC 7 17 marks

8.

**Figure 3**

Two particles, A and B , have masses 2 kg and 4 kg respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle α where $\tan \alpha = \frac{3}{4}$. The particle A is held at rest on the plane at a distance d metres from the pulley. The particle B hangs freely at rest, vertically below the pulley, at a distance h metres above the ground, as shown in Figure 3. The part of the string between A and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between A and the plane is $\frac{1}{4}$.

The system is released from rest with the string taut and B descends.

- (a) Find the tension in the string as B descends. (9)

On hitting the ground, B immediately comes to rest.

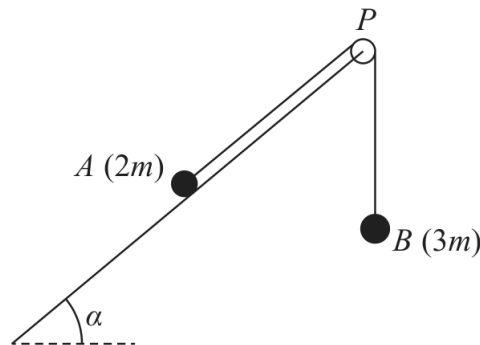
Given that A comes to rest before reaching the pulley,

- (b) find, in terms of h , the range of possible values of d . (7)
- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic. (1)

(Total 17 marks)

Q28 PAST PAPER Pearson IAL WME01 · October 2021 · Q7**M1 · TOPIC 7** 14 marks

7.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. The string passes over a small, smooth, light pulley P which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

Particle A is held at rest on the plane with the string taut and B hanging freely below P , as shown in Figure 4. The section of the string AP is parallel to a line of greatest slope of the plane.

The coefficient of friction between A and the plane is $\frac{1}{2}$

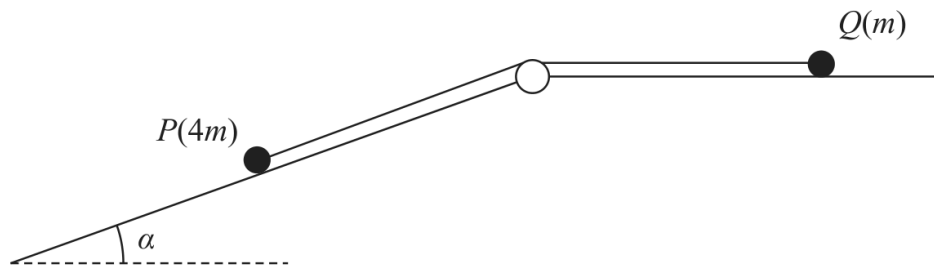
Particle A is released and begins to move up the plane.

For the motion before A reaches the pulley,

- (a) (i) write down an equation of motion for A ,
- (ii) write down an equation of motion for B ,
- (4)
- (b) find, in terms of g , the acceleration of A ,
- (5)
- (c) find the magnitude of the force exerted on the pulley by the string.
- (4)
- (d) State how you have used the information that P is a smooth pulley.
- (1)

Q29 PAST PAPER Pearson IAL WME01 · January 2022 · Q7**M1 · TOPIC 7** 13 marks

7.

**Figure 4**

A particle P of mass $4m$ lies on the surface of a fixed rough inclined plane.

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The particle P is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle Q of mass m which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

The system is released from rest with the string taut, as shown in Figure 4, and P moves down the plane.

The coefficient of friction between P and the plane is $\frac{1}{4}$

For the motion before Q reaches the pulley

(a) write down an equation of motion for Q , (1)

(b) find, in terms of m and g , the tension in the string, (7)

(c) find the magnitude of the force exerted on the pulley by the string. (4)

(d) State where in your working you have used the information that the string is light. (1)

Q30 PAST PAPER Pearson IAL WME01 · October 2023 · Q7**M1 · TOPIC 7** 13 marks

7.

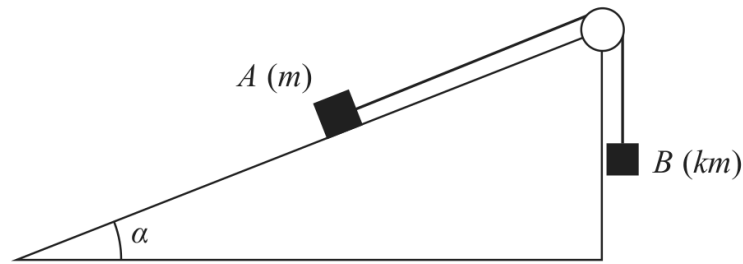
**Figure 4**

Figure 4 shows a block A of mass m held at rest on a rough plane. The plane is inclined at an angle α to the horizontal and the coefficient of friction between the block and the plane is μ .

One end of a light inextensible string is now attached to A . The string passes over a small smooth pulley which is fixed at the top of the plane. The other end of the string is attached to a block B of mass km . Block B hangs vertically below the pulley, with the string taut.

The string from A to the pulley lies along a line of greatest slope of the plane.

Both A and B are modelled as particles.

When the system is released from rest, A moves up the plane and the tension in the string is $\frac{4mg}{3}$

Given that $\mu = \frac{1}{3}$ and $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of A , giving your answer in terms of g ,
 (ii) find the value of k .

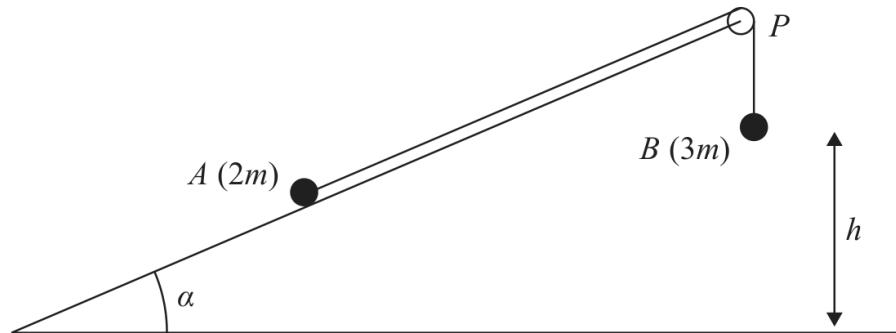
(9)

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of m and g .

(4)

Q31 PAST PAPER Pearson IAL WME01 · June 2024 · Q8**M1 · TOPIC 7** 15 marks

8.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. Particle A is held at rest on a rough plane which is inclined to horizontal ground at an angle α , where $\tan \alpha = \frac{5}{12}$.

The string passes over a small smooth pulley P which is fixed at the top of the plane. Particle B hangs vertically below P with the string taut, at a height h above the ground, as shown in Figure 4.

The part of the string between A and P lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between A and the plane is $\frac{11}{36}$.

The particle A is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on A as it moves up the plane is $\frac{22mg}{39}$ (3)

(b) Write down an equation of motion for B . (2)

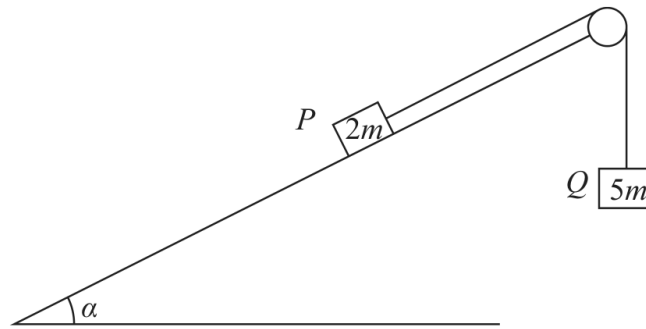
(c) Show that the acceleration of A immediately after its release is $\frac{1}{3}g$ (4)

In the subsequent motion, A comes to rest before it reaches the pulley.

(d) Find, in terms of h , the total distance travelled by A from when it was released from rest to when it first comes to rest again. (6)

Q32 PAST PAPER Pearson IAL WME01 · October 2025 · Q7**M1 · TOPIC 7 14 marks**

7.

**Figure 5**

A block P of mass $2m$ is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to P .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block Q of mass $5m$.

Block Q hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block P moves up the plane.

Immediately after the system is released, the tension in the string is T and the acceleration of the blocks is a .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for Q .

(2)

The coefficient of friction between P and the plane is $\frac{1}{8}$

(b) Find T in terms of m and g .

(7)

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

The magnitude of the force exerted on the pulley by the string is kmg .

(d) Find the value of k to 3 significant figures.

(4)

Q33 PAST PAPER Pearson IAL WME01 · January 2026 · Q7

M1 · TOPIC 7 14 marks

7.

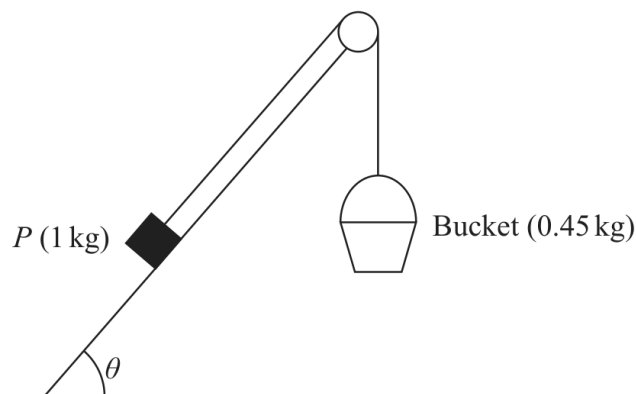


Figure 3

A rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package P .
 The string passes over a smooth pulley that is fixed at the top of the plane.
 The string from P to the pulley lies along a line of greatest slope of the plane.
 The other end of the string is attached to a bucket.
 The package P has mass 1 kg and is **held** at rest on the plane.
 The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between P and the plane is μ

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with P on the point of slipping down the plane.

(a) Show that $\mu = \frac{1}{4}$ (7)

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until P starts to move **up** the plane.

At the instant when P starts to move up the plane, there are a **total** of n blocks in the bucket.

(b) Find (7)

- the value of n
- the magnitude of the initial acceleration of P

Q34 PAST PAPER Pearson IAL WME01 · June 2026 · Q8

M1 · TOPIC 7 14 marks

8.

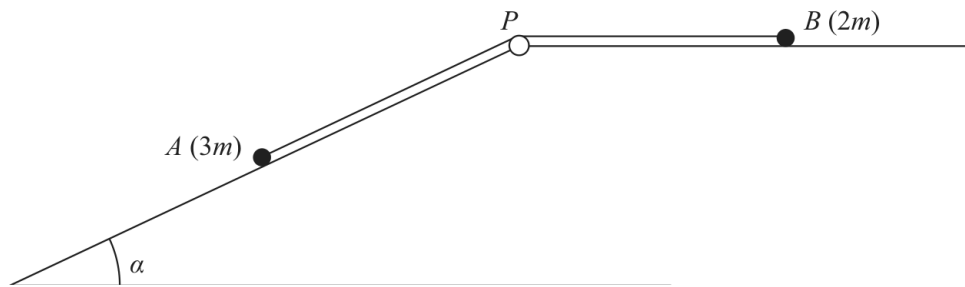


Figure 4

One end of a light inextensible string is attached to a particle A of mass $3m$. The other end of the string is attached to a particle B of mass $2m$. Particle A lies on a **rough** plane which is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$. Particle B lies on a **smooth** horizontal plane. The string passes over a small smooth pulley P which is fixed at the top of the inclined plane, as shown in Figure 4.

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between A and P lies along a line of greatest slope of the inclined plane.

The coefficient of friction between A and the inclined plane is $\frac{1}{4}$.

The system is released from rest with the string taut and A begins to move down the inclined plane.

(a) Find, in terms of m and g , the magnitude of the frictional force acting on A . (3)

(b) Write down an equation of motion for B . (1)

(c) Show that the tension in the string is $\frac{12mg}{25}$. (4)

(d) State how you have used the fact that the string is inextensible. (1)

The force exerted on the pulley by the string is S .

(e) State clearly, giving your answer in terms of α , the direction of S . (1)

(f) Find, in terms of m and g , the magnitude of S . (4)

(Total for Question 8 is 14 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------------|--|
| 01 | One body | With – against = ma , in the direction of a . |
| 02 | In a line | Whole system for a ; one part for the tension. |
| 03 | Pulleys | One equation each; add to lose T ; pulley force from two tensions. |
| 04 | Slope and pulley | $mg \sin \alpha$, $\mu mg \cos \alpha$ against the motion. |

BOTTOM LINE Diagram → direction of a → one equation per particle → add → SUVAT.

Answers

Newton's Second Law • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ONE BODY: RESULTANT FORCE = ma

- Q1** Dr Eslam • Driving force and resistance $2400 = 1200a \Rightarrow a = 2 \text{ m s}^{-2}$
- Q2** Dr Eslam • A lift $T = 560(10.3) = 5768 \text{ N}; R = 60(10.3) = 618 \text{ N}$
- Q3** Dr Eslam • Falling with resistance $0.5g - 1.4 = 0.5a \Rightarrow a = 7; v = 21 \text{ m s}^{-1}$
- Q4** Pearson IAL WME01 • June 2021 • Q4 (a) $T = 400 \text{ N}$ (b) $R \approx 119 \text{ N}$
- Q5** Pearson IAL WME01 • October 2021 • Q2 (a) u (b) reversed (c) $\mu = \frac{1}{3}$
- Q6** Pearson IAL WME01 • October 2022 • Q4 (a) 1.5 m s^{-2} (b) 50 kg
- Q7** Pearson IAL WME01 • June 2023 • Q3 (a) 0.553 s (b) $R \approx 0.293 \text{ N}$
- Q8** Pearson IAL WME01 • October 2023 • Q3 (a) $v = 9$ (b) 1.8 N s (c) $R \approx 695 \text{ N}$
- Q9** Pearson IAL WME01 • January 2025 • Q3 (a) $\lambda = 15m$ (b) $\mu \approx 0.68$ (c) 1.8 m s^{-1}
- Q10** Pearson IAL WME01 • June 2025 • Q4 (a) $\alpha \approx 43^\circ$ (b) 0.023 m s^{-2}

IDEA 02 • CONNECTED IN A LINE: TOWING AND LIFTS

- Q11** Dr Eslam · Car and trailer $a = 1.3 \text{ m s}^{-2}; T - 150 = 650 \Rightarrow T = 800 \text{ N}$
- Q12** Dr Eslam · Two particles on one string $a = \frac{60-5g}{5} = 2.2; T = 3(9.8 + 2.2) = 36 \text{ N}$
- Q13** Pearson IAL WME01 · October 2019 · Q3 $D = 6000$
- Q14** Pearson IAL WME01 · June 2022 · Q3 (a) $R = 5600$ (b) the tractor and the block have the same acceleration
- Q15** Pearson IAL WME01 · January 2023 · Q7 (a) $5m$ (b) $2m$
- Q16** Pearson IAL WME01 · June 2023 · Q7 (a) $R = 60$ (b) 1000 N (c) 78.5 m
- Q17** Pearson IAL WME01 · June 2024 · Q3 $T = \frac{9}{2}mg$

IDEA 03 · PULLEYS: HANGING AND ON A TABLE

- Q18** Dr Eslam • Two hanging masses
 $a = \frac{2g}{8} = 2.45 \text{ m s}^{-2}; T = 36.75 \text{ N}$
- Q19** Dr Eslam • Table and edge
 $a = \frac{g}{3} \approx 3.27; T \approx 6.53 \text{ N}; \text{ pulley: } T\sqrt{2} \approx 9.24 \text{ N}$
- Q20** Pearson IAL WME01 • January 2020 • Q7
 (a) $1.4 = \frac{1}{2}a(2^2) \Rightarrow a = 0.7$ (b) the string is inextensible (c) 38.6 N (d) $\mu = 0.625$ (e) A does not reach the pulley
- Q21** Pearson IAL WME01 • October 2022 • Q7
 (a) $k = \frac{1+\mu}{2}$ (b) $t = \sqrt{\frac{4d}{g(1-\mu)}}$ (c) the system does not move
- Q22** Pearson IAL WME01 • October 2024 • Q5
 (a) $T - 3g = 3a$ (b) $T = \frac{15g}{4} \approx 36.8 \text{ N}$ (c) $F = \frac{15g}{2} = 73.5 \text{ N}$ (d) $\sqrt{17g} \approx 12.9 \text{ m s}^{-1}$
- Q23** Pearson IAL WME01 • January 2025 • Q7
 (a) $T = \frac{12mg}{5}$ (b) 1.4 m s^{-1} (c) 0.767 s
- Q24** Pearson IAL WME01 • June 2025 • Q8
 (a) $T - kMg = \frac{1}{5}kMg$ (b) $k = 2$ (c) $\frac{12g}{5} \approx 23.5 \text{ N}$ (d) $1.5(\sqrt{g} + 0.4) \approx 5.30 \text{ N s}$ (e) $\frac{6}{\sqrt{g}} \approx 1.92 \text{ s}$

IDEA 04 • PULLEYS WITH A SLOPE

- Q25** Dr Eslam • Smooth slope and pulley
 $3g - 4g \sin 30^\circ = 7a \Rightarrow a = 1.4 \text{ m s}^{-2}$
- Q26** Dr Eslam • Rough slope and pulley
 $F = 0.2(4g \cos 30^\circ) \approx 6.79$; $a = \frac{29.4 - 19.6 - 6.79}{7} \approx 0.43 \text{ m s}^{-2}$
- Q27** Pearson IAL WME01 • January 2021 • Q8
 (a) $T = \frac{12g}{5} \approx 23.5 \text{ N}$ (b) $d > \frac{3h}{2}$ (c) e.g. friction at the pulley
- Q28** Pearson IAL WME01 • October 2021 • Q7
 (a)(i) $T - 2mg \sin \alpha - F = 2ma$ (ii) $3mg - T = 3ma$
 (b) $a = \frac{g}{5}$ (c) $\frac{48\sqrt{5}}{25}mg$ (d) same tension on both sides
- Q29** Pearson IAL WME01 • January 2022 • Q7
 (a) $T = ma$ (b) $T = \frac{8mg}{25}$ (c) $\frac{8\sqrt{10}}{125}mg$ (d) the tension is the same throughout
- Q30** Pearson IAL WME01 • October 2023 • Q7
 (a)(i) $\frac{11g}{15}$ (ii) $k = 5$ (b) $\frac{32mg}{15}$
- Q31** Pearson IAL WME01 • June 2024 • Q8
 (a) $F = \frac{11}{36} \cdot \frac{12}{13}(2mg) = \frac{22mg}{39}$ (b) $3mg - T = 3ma$
 (c) $a = \frac{g}{3}$ (d) $\frac{3h}{2}$
- Q32** Pearson IAL WME01 • October 2025 • Q7
 (a) $5mg - T = 5ma$ (b) $a = \frac{4g}{7}$, $T = \frac{15mg}{7}$ (c) same magnitude of acceleration (d) $k \approx 3.57$
- Q33** Pearson IAL WME01 • January 2026 • Q7
 (a) $\mu = \frac{1}{4}$ (b)(i) $n = 3$ (ii) $\frac{2g}{41} \approx 0.478 \text{ m s}^{-2}$
- Q34** Pearson IAL WME01 • June 2026 • Q8
 (a) $\frac{3mg}{5}$ (b) $T = 2ma$ (c) $T = \frac{12mg}{25}$ (d) same magnitude of acceleration (e) below the horizontal at $90^\circ - \frac{\alpha}{2}$ to it (f) $\frac{12\sqrt{10}}{125}mg$

Resolving Forces and Inclined Planes

On a slope, resolve along the plane and perpendicular to the plane. Weight splits into $mg \sin \alpha$ down the slope and $mg \cos \alpha$ into it.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Resolving Forces and Inclined Planes

01 — Equilibrium on a slope

parallel and perpendicular

horizontal force

force at an angle

02 — Up or down?

about to slide up

about to slide down

find μ

03 — Moving on a slope

$F = ma$ along the plane

projected up

will it slide back?

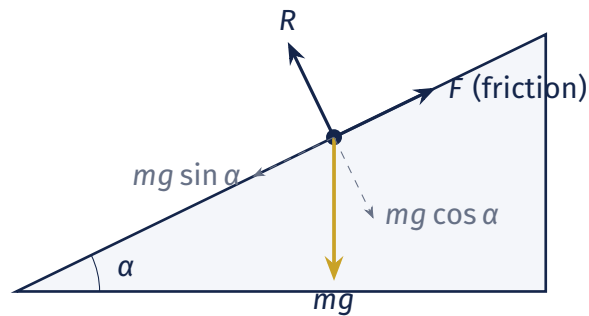
THE SLOPE TRIANGLE

weight: $mg \sin \alpha$ down the slope
slope

$mg \cos \alpha$ into the

sin goes *down* the slope (the steeper, the bigger)
pushes into the plane

$R = mg \cos \alpha$ only if no other force



01 Equilibrium on a Slope: Parallel and Perpendicular

Tilt your axes: one along the slope, one perpendicular. Every force splits into these two. Perpendicular gives R ; parallel gives the balance (and the friction, if any).

WHEN YOU SEE → YOU DO

“inclined at α where $\tan \alpha = \frac{3}{4}$ ”

→ $\sin \alpha = \frac{3}{5}$, $\cos \alpha = \frac{4}{5}$

a horizontal force X on the slope

→ along: $X \cos \alpha$; into the plane: $X \sin \alpha$

a force at 40° to the plane

→ along: $X \cos 40^\circ$; away from the plane: $X \sin 40^\circ$

“show there is no frictional force”

→ balance along the slope without F

“smallest value of X ”

→ on the point of sliding down: friction up the slope

WORKED MODEL • OCTOBER 2022

2 kg on a plane with $\tan \alpha = \frac{3}{4}$, held by a horizontal force $X = 14.7$ N. Show there is no friction.

1 up the slope: $14.7 \times \frac{4}{5} = 11.76$

$X \cos \alpha$

2 down the slope: $2g \times \frac{3}{5} = 11.76$

$mg \sin \alpha$

ANSWER balanced, so $F = 0$

IDEA 01

Practise: Equilibrium on a Slope: Parallel and Perpendicular

My questions first, then real past papers.

Q1 **DR ESLAM** Smooth slope, force up the plane

M1 · TOPIC 8 3 marks

A 4 kg particle rests on a smooth plane inclined at 30° , held by a force P up the line of greatest slope. Find P and the normal reaction.

Resolve along the plane and perpendicular to it.

Q2 **DR ESLAM** Horizontal force on a slope

M1 · TOPIC 8 3 marks

A 2 kg particle rests on a smooth plane inclined at 25° , held by a horizontal force H . Find H .

A horizontal force has a part along the plane ($H \cos 25^\circ$) and a part into the plane.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 · October 2019 · Q4

M1 · TOPIC 8 12 marks

4.

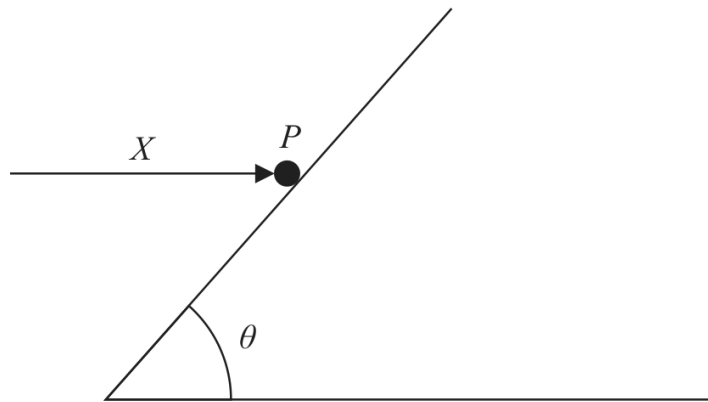


Figure 1

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

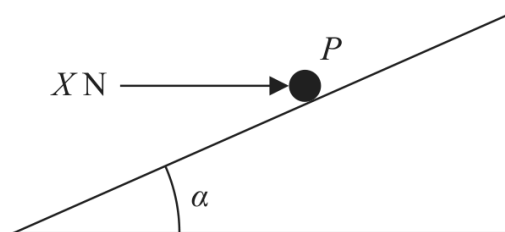
(a) Show that $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (10)

(b) Deduce that, when $k = 1$, θ must be greater than 45° (2)

Q4 PAST PAPER Pearson IAL WME01 • October 2022 • Q3

M1 • TOPIC 8 11 marks

3.

**Figure 2**

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

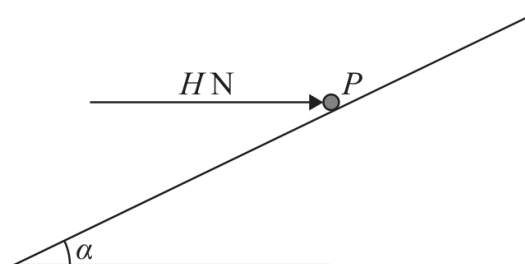
(a) Show that when $X = 14.7$ there is no frictional force acting on P (3)

The coefficient of friction between P and the plane is 0.5

(b) Find the smallest possible value of X . (8)

Q5 PAST PAPER Pearson IAL WME01 · October 2024 · Q6**M1 · TOPIC 8** 12 marks

6.

**Figure 4**

A particle P of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude H newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

(a) Find the smallest possible value of H .

(6)

The horizontal force is now removed, and P starts to slide down the slope.

In the first T seconds after P is released from rest, P slides 1.5 m down the slope.

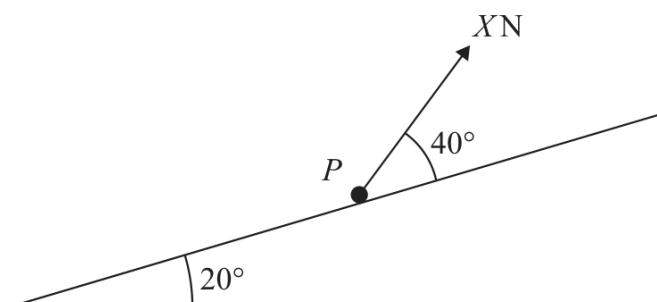
(b) Find the value of T .

(6)

Q6 PAST PAPER Pearson IAL WME01 · June 2026 · Q5

M1 · TOPIC 8 6 marks

5

**Figure 3**

A particle P , whose weight is 50 N , is at rest on a rough plane. The plane is inclined to the horizontal at 20° and the coefficient of friction between P and the plane is 0.3

The particle P is held in equilibrium by a force of magnitude X newtons.

The line of action of the force makes an angle of 40° with the plane, as shown in Figure 3. The line of action of the force lies in a vertical plane which contains a line of greatest slope of the plane.

Given that P is on the point of sliding **down** the plane, find the value of X .

(6)**(Total for Question 5 is 6 marks)**

02 Up or Down? The Two Limiting Cases

Friction always acts against the way the particle is about to move. About to slide **up**: friction points down the slope. About to slide **down**: friction points up the slope. Two situations give two equations.

WHEN YOU SEE → YOU DO

“on the point of sliding up the plane”

→ $F = \mu R$ down the slope

“on the point of sliding down the plane”

→ $F = \mu R$ up the slope

“when $X = 2P$... up; when $X = P$... down”

→ two equations, eliminate P

“the string breaks — does it stay at rest?”

→ compare $\tan \alpha$ with μ

nothing but weight, R , friction

stays at rest when

compare

$\tan \alpha \leq \mu$

WORKED MODEL • JANUARY 2022

$\tan \alpha = \frac{3}{4}$. $X = 2P$: about to slide up; $X = P$: about to slide down. Find μ .

1 up: $2P = mg \sin \alpha + \mu mg \cos \alpha$

friction down

2 down: $P = mg \sin \alpha - \mu mg \cos \alpha$

friction up

3 $0.6 + 0.8\mu = 2(0.6 - 0.8\mu)$

divide the first by the second

ANSWER $\mu = \frac{1}{4}$

IDEA 02

Practise: Up or Down? The Two Limiting Cases

My questions first, then real past papers.

Q7**DR ESLAM**

Does it slide?

M1 · TOPIC 8**3 marks**

A 3 kg block rests on a rough plane inclined at 20° , $\mu = 0.5$. Does it slide? Find the friction.

Q8**DR ESLAM**

Range of a force

M1 · TOPIC 8**4 marks**

A 2 kg particle on a rough plane with $\tan \alpha = \frac{3}{4}$, $\mu = 0.25$, is held by a force P up the plane. Find the range of P for equilibrium.

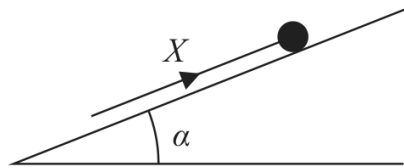
About to slide down: friction up. About to slide up: friction down.

PAST PAPERS

Q9 PAST PAPER Pearson IAL WME01 • January 2022 • Q5

M1 • TOPIC 8 8 marks

5.

**Figure 3**

A particle of mass m rests in equilibrium on a fixed rough plane under the action of a force of magnitude X . The force acts up a line of greatest slope of the plane, as shown in Figure 3.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The coefficient of friction between the particle and the plane is μ .

- When $X = 2P$, the particle is on the point of sliding up the plane.
- When $X = P$, the particle is on the point of sliding down the plane.

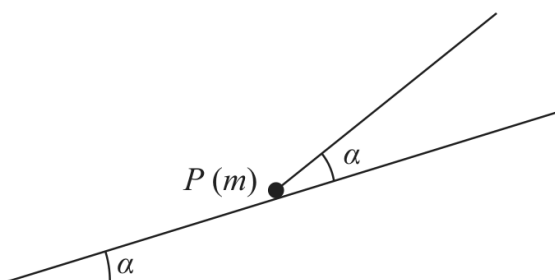
Find the value of μ .

(8)

Q10 PAST PAPER Pearson IAL WME01 · January 2025 · Q6

M1 · TOPIC 8 12 marks

6.

**Figure 3**

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is $0.75mg$, P is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on P , giving your answer in terms of m , g and α .

(3)

The coefficient of friction between P and the plane is $\frac{1}{2}$

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether P remains at rest. You must justify your reasoning.

(3)

03 Moving on a Slope: $F = ma$ Along the Plane

Moving means friction is μR , against the motion. Resolve perpendicular for R , then write resultant force = ma along the slope. Then SUVAT takes over.

WHEN YOU SEE → YOU DO

“projected up the plane”

→ weight part and friction both down: big deceleration

“slides down the plane”

→ $mg \sin \alpha - \mu R = ma$

“comes to instantaneous rest ... then slides down”

→ new friction direction, new acceleration

“determine whether it remains at rest at C”

→ compare $mg \sin \alpha$ with μR

a pulley at the top of the slope

→ one equation for each particle, same a

R from perpendicular

Friction = μR against motion

$F = ma$ along the slope

SUVAT

WORKED MODEL • JUNE 2025

Projected up a plane with $\tan \theta = \frac{3}{4}$, $\mu = \frac{1}{4}$. Show the deceleration is $\frac{4}{5}g$.

1 $R = mg \cos \theta = \frac{4}{5}mg$; $F = \frac{1}{4} \cdot \frac{4}{5}mg = \frac{1}{5}mg$

perpendicular, then μR

2 $-\frac{3}{5}mg - \frac{1}{5}mg = ma$

both down the slope

ANSWER $a = -\frac{4}{5}g$

× **TRAP** Keeping friction up the slope after the particle turns round. Friction always opposes the motion.

IDEA 03

Practise: Moving on a Slope: $F = ma$ Along the Plane

My questions first, then real past papers.

Q11 DR ESLAM Sliding down

M1 · TOPIC 8 3 marks

A 5 kg particle slides down a rough plane inclined at 30° , $\mu = 0.2$. Find its acceleration.

Q12 DR ESLAM Projected up the slope

M1 · TOPIC 8 4 marks

A particle is projected up a rough plane ($\tan \alpha = \frac{3}{4}$, $\mu = 0.5$) at 8 m s^{-1} . How far up does it go? Will it slide back?

Going up, weight AND friction both act down the plane.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · June 2021 · Q6

M1 · TOPIC 8 13 marks

6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane. (3)

The particle comes to instantaneous rest at the point B .

- (b) Find the distance AB . (5)

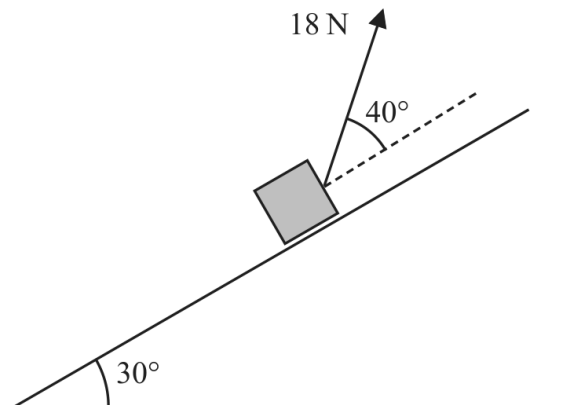
The particle now slides down the plane from B . At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 m s^{-1}

- (c) Find the distance BC . (5)

Q14 PAST PAPER Pearson IAL WME01 · January 2023 · Q8

M1 · TOPIC 8 15 marks

8.

**Figure 5**

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude 18 N and acts at an angle of 40° to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle P

(a) Find the acceleration of P (8)

The points A and B lie on a line of greatest slope of the plane, where $AB = 5$ m and B is above A . Particle P passes through A with speed 2 m s^{-1} in the direction AB .

(b) Find the speed of P as it passes through B . (3)

The force of 18 N is removed at the instant P passes through B . As a result, P comes to rest at the point C .

(c) Determine whether P will remain at rest at C . You must show all stages of your working clearly. (4)

Q15 PAST PAPER Pearson IAL WME01 · January 2024 · Q8

M1 · TOPIC 8 12 marks

8.

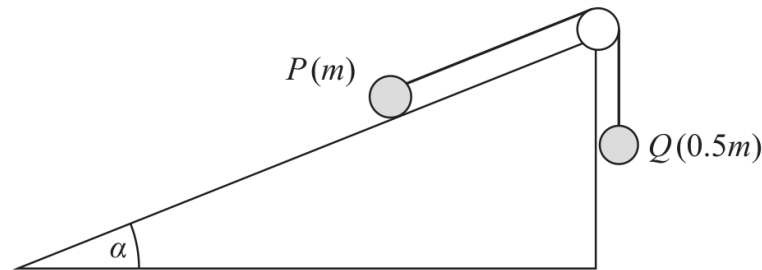


Figure 5

A fixed rough plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle P which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle Q which hangs vertically below the pulley, as shown in Figure 5.

Particle P has mass m and particle Q has mass $0.5m$

The string from P to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between P and the plane is μ .

The system is in **limiting equilibrium** with the string taut and P is on the point of slipping **up** the plane.

(a) Find the value of μ .

(8)

The string breaks and P begins to move down the plane.

When particle P has travelled a distance of 0.8 m down the plane, the speed of P is $V \text{ m s}^{-1}$

(b) Find the value of V .

(4)

Q16 PAST PAPER Pearson IAL WME01 · June 2025 · Q6**M1 · TOPIC 8** 11 marks

6.

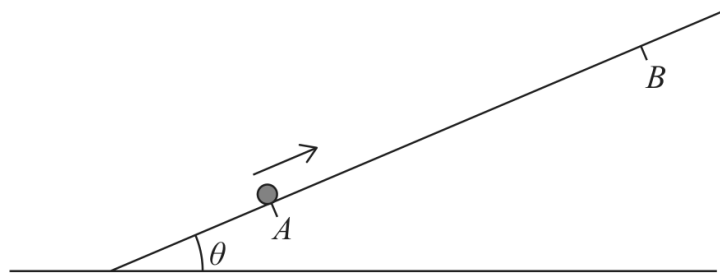
**Figure 4**

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

(a) Show that the **deceleration** of P , as it moves from A to B , is $\frac{4}{5}g$ (6)

The package P comes to rest at B .

Given that P is projected from A with speed $U \text{ ms}^{-1}$ and that $AB = 1.5 \text{ m}$,

(b) find the value of U . (2)

On reaching B , P is held at rest there by a force of magnitude X newtons acting up the plane in the direction AB .

Given that the mass of P is 2 kg ,

(c) find the smallest possible value of X . (3)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|--|
| 01 | Equilibrium | Axes along and perpendicular to the plane. |
| 02 | Up or down | Friction opposes the slide; two cases, two equations. |
| 03 | Moving | R , then μR against the motion, then $F = ma$. |

BOTTOM LINE $mg \sin \alpha$ down the slope · $mg \cos \alpha$ into it · friction against the motion · stays at rest if $\tan \alpha \leq \mu$.

Answers

Resolving Forces and Inclined Planes · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · EQUILIBRIUM ON A SLOPE: PARALLEL AND PERPENDICULAR

- Q1** Dr Eslam · Smooth slope, force up the plane $P = 4g \sin 30^\circ = 19.6 \text{ N}; R = 4g \cos 30^\circ \approx 33.9 \text{ N}$
- Q2** Dr Eslam · Horizontal force on a slope $H \cos 25^\circ = 2g \sin 25^\circ \Rightarrow H \approx 9.14 \text{ N}$
- Q3** Pearson IAL WME01 · October 2019 · Q4 (a) $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (b) $\mu > 0 \Rightarrow \tan \theta > 1 \Rightarrow \theta > 45^\circ$
- Q4** Pearson IAL WME01 · October 2022 · Q3 (a) $X \cos \alpha = 2g \sin \alpha \Rightarrow X = 14.7$: no friction needed (b) $X = \frac{4g}{11} \approx 3.56 \text{ N}$
- Q5** Pearson IAL WME01 · October 2024 · Q6 (a) $H = \frac{40g}{19} \approx 20.6 \text{ N}$ (b) $T = \sqrt{\frac{15}{2g}} \approx 0.875 \text{ s}$
- Q6** Pearson IAL WME01 · June 2026 · Q5 $X \approx 5.24$

IDEA 02 · UP OR DOWN? THE TWO LIMITING CASES

- Q7** Dr Eslam · Does it slide? down-slope part $10.1 \text{ N} < \mu R = 13.8 \text{ N}$: it stays; $F = 10.1 \text{ N}$ up the plane
- Q8** Dr Eslam · Range of a force $7.84 \leq P \leq 15.68 \text{ N}$
- Q9** Pearson IAL WME01 · January 2022 · Q5 $\mu = \frac{1}{4}$
- Q10** Pearson IAL WME01 · January 2025 · Q6 (a) $F = 0.75mg \cos \alpha - mg \sin \alpha$ (b) $\tan \alpha = \frac{2}{5}$
(c) $\tan \alpha < \mu$: P remains at rest

IDEA 03 · MOVING ON A SLOPE: $F = ma$ ALONG THE PLANE

- Q11** Dr Eslam • Sliding down $a = g(\sin 30^\circ - 0.2 \cos 30^\circ) \approx 3.20 \text{ m s}^{-2}$
- Q12** Dr Eslam • Projected up the slope deceleration $g(0.6 + 0.4) = 9.8$: $s \approx 3.27 \text{ m}$; $\tan \alpha > \mu$, so it slides back
- Q13** Pearson IAL WME01 • June 2021 • Q6 (a) $F \approx 13.6 \text{ N}$ (b) $AB \approx 2.07 \text{ m}$ (c) $BC \approx 8.29 \text{ m}$
- Q14** Pearson IAL WME01 • January 2023 • Q8 (a) 1.18 m s^{-2} up the plane (b) 3.98 m s^{-1} (c) $\tan 30^\circ > 0.3$: it does not stay at rest
- Q15** Pearson IAL WME01 • January 2024 • Q8 (a) $\mu = \frac{1}{8}$ (b) $V \approx 2.05$
- Q16** Pearson IAL WME01 • June 2025 • Q6 (a) $g(\frac{3}{5} + \frac{1}{4} \cdot \frac{4}{5}) = \frac{4}{5}g$ (b) $U = \sqrt{\frac{12g}{5}} \approx 4.85$ (c) $X = \frac{4g}{5} \approx 7.84 \text{ N}$

Momentum, Impulse & Collisions

Draw before and after. Choose a positive direction. Then only two equations: momentum is conserved, and impulse is change in momentum.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Momentum, Impulse & Collisions

01 - Momentum and impulse on one body

$$I = mv - mu$$

rebounds

km h⁻¹

02 - Collisions

before = after

direction reversed

03 - One common velocity

coalesce

string taut

ejected

04 - Impulse in a collision

equal and opposite

two possible values

05 - With SUVAT, friction, slopes

dropped / bounces

μ

inclined plane

THE TWO EQUATIONS OF THE WHOLE TOPIC

$$I = mv - mu \quad (= Ft)$$

impulse on **one** particle

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

conservation for the **whole** collision

01 Momentum and Impulse on One Body

Momentum = mv . Impulse = change in momentum = force $\times$ time. Units: **N s**.

WHEN YOU SEE $\rightarrow$ YOU DO

“momentum of ...”

$\rightarrow mv$ — units N s (or kg m s⁻¹)

“impulse given by the force” / “ F applied for t seconds”

$\rightarrow I = Ft = mv - mu$

“rebounds” / “bounces”

$\rightarrow$ the velocity changes sign

km h⁻¹ or grams

$\rightarrow \div 3.6 \cdot \div 1000$ first

1

Choose a positive direction

Draw an arrow. Every velocity the other way is **negative**.

2

One particle only

$I = m(v - u)$ with signs. The sign of I is its direction.

WORKED MODEL • THE BALL REBOUNDS

Ball 0.075 kg hits a ceiling at 5 m s⁻¹ and rebounds at 3 m s⁻¹. Find the impulse.

1 positive = downwards (after): $u = -5$, $v = +3$

signs

2 $I = 0.075(3 - (-5)) = 0.075 \times 8$

$I = m(v - u)$

ANSWER $I = 0.6$ N s

× **TRAP** $0.075(3 - 5)$ is the classic mistake. A rebound **adds** the speeds.

IDEA 01

Practise: Momentum and Impulse on One Body

Questions written to practise exactly this idea.

Q1 DR ESLAM Momentum and units

M1 · TOPIC 9 2 marks

A car of mass 1200 kg moves at 54 km h^{-1} . Find its momentum.

Change km h^{-1} to m s^{-1} first: $\div 3.6$.

Q2 DR ESLAM Rebound: the speeds add

M1 · TOPIC 9 2 marks

A ball of mass 0.2 kg hits a wall at 6 m s^{-1} and rebounds at 4 m s^{-1} . Find the impulse on the ball.

Q3 DR ESLAM Impulse = force $\times$ time

M1 · TOPIC 9 2 marks

A force of 8 N acts for 1.5 s on a 2 kg particle moving at 5 m s^{-1} in the direction of the force. Find its new speed.

02 Collisions: Conservation of Momentum

Total momentum before = total momentum after. Draw it — the diagram gives you the signs.

WHEN YOU SEE → YOU DO

“collide directly”

→ before/after diagram, $\sum mu = \sum mv$

“moving towards each other”

→ one velocity is negative before

“direction of motion is reversed”

→ its sign changes after

“direction unchanged”

→ same sign as before

“state the direction of motion of ...”

→ look at the **sign** of your answer



WORKED MODEL • JUNE 2023

1 $4(2u) + 2(-3u) = 4v_A + 2(2u)$

$\sum mu = \sum mv$

2 $2u = 4v_A + 4u \Rightarrow v_A = -\frac{1}{2}u$

solve

ANSWER speed $\frac{1}{2}u$, A is reversed (negative)

× **TRAP** Draw every “after” arrow in the positive direction and let the sign decide. Guessing directions is where the marks are lost.

IDEA 02

Practise: Collisions: Conservation

My questions first, then real past papers.

Q4 DR ESLAM Draw, then conserve

M1 · TOPIC 9 3 marks

A (4 kg) moves at 6 m s^{-1} towards B (2 kg) moving at 3 m s^{-1} . After the collision A moves at 1 m s^{-1} in its original direction. Find the velocity of B.

Positive = A's direction. B starts at -3 .

Q5 DR ESLAM Let the sign decide

M1 · TOPIC 9 3 marks

P (2 kg) at 5 m s^{-1} hits Q (3 kg) at rest. After, Q moves at 4 m s^{-1} . Find the velocity of P and state its direction.

PAST PAPERS

Q6 PAST PAPER Pearson IAL WME01 · January 2022 · Q2

M1 · TOPIC 9 7 marks

2. A particle P has mass km and a particle Q has mass m . The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P has speed $3u$ and Q has speed u .

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

- (a) Find the value of k .

(4)

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on Q in the collision.

(3)

Q7**PAST PAPER**

Pearson IAL WME01 · January 2023 · Q2

M1 · TOPIC 9**8 marks**

2. Two particles, A and B , are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle A has mass $3m$ kg and particle B has mass m kg.

Immediately before the collision, both particles have a speed of 1.5 m s^{-1}

Immediately after the collision, the direction of motion of A is unchanged and the difference between the speed of A and speed of B is 1 m s^{-1}

(a) Find (i) the speed of A immediately after the collision,

(ii) the speed of B immediately after the collision.

(5)

(b) Find, in terms of m , the magnitude of the impulse exerted on B in the collision.

(3)

Q8 PAST PAPER Pearson IAL WME01 • June 2023 • Q1

M1 • TOPIC 9 7 marks

1. A particle A has mass 4 kg and a particle B has mass 2 kg .

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\text{ ms}^{-1}$ and the speed of B is $3u\text{ ms}^{-1}$

Immediately after the collision, the speed of B is $2u\text{ ms}^{-1}$

The direction of motion of B is reversed by the collision.

- (a) Find, in terms of u , the speed of A immediately after the collision. (3)
- (b) State the direction of motion of A immediately after the collision. (1)
- (c) Find, in terms of u , the magnitude of the impulse received by B in the collision.
State the units of your answer. (3)

Q9**PAST PAPER**

Pearson IAL WME01 · January 2024 · Q2

M1 · TOPIC 9 6 marks

2.

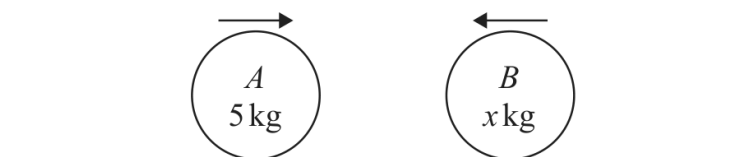
**Figure 2**

Figure 2 shows two particles, A and B , moving in opposite directions on a smooth horizontal surface. Particle A has mass 5 kg and particle B has mass $x\text{ kg}$.

The particles collide directly.

Immediately before the collision, the speed of A is 3 ms^{-1} and the speed of B is $x\text{ ms}^{-1}$

Immediately after the collision, the speed of A is 1 ms^{-1} and its direction of motion is unchanged.

Immediately after the collision, the speed of B is 1.5 ms^{-1}

(a) Find the value of x .

(3)

(b) Find the magnitude of the impulse exerted on A by B in the collision.

(3)

Q10**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q1

M1 · TOPIC 9 6 marks

1. Particle A has mass $4m$ and particle B has mass $3m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Immediately **before** the collision, the speed of A is $2x$ and the speed of B is x .

Immediately **after** the collision, the speed of A is y and the speed of B is $5y$.

The direction of motion of each particle is reversed as a result of the collision.

- (a) Show that $y = \frac{5}{11}x$. (3)
- (b) Find, in terms of m and x , the magnitude of the impulse received by A in the collision. (3)

03 One Common Velocity: Coalesce, Couple, String Taut

When the particles move **together** after, there is only one unknown: the common velocity.

WHEN YOU SEE → YOU DO

“coalesce” / “join together” / “couple together”

→ one mass $m_1 + m_2$, one velocity v

“string becomes taut”

→ common speed — same equation as coalescing

“jumps off” / “is ejected” / “recoils”

→ started together: $(m_1 + m_2)u = m_1v_1 + m_2v_2$

1

Momentum

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

2

Impulse

Take **one** particle: $I = m(v - u)$.

WORKED MODEL • JUNE 2026

Truck A ($5m$) at u , truck B ($3m$) at $1.5u$ towards each other; they couple.

1 $5m(u) + 3m(-1.5u) = 8m v$

one common velocity

2 $0.5mu = 8mv \Rightarrow v = \frac{u}{16}$

solve

3 $I_B = 3m\left(\frac{u}{16} - (-1.5u)\right)$

impulse on B only

ANSWER $v = \frac{u}{16}, I = \frac{75}{16}mu$

IDEA 03

Practise: One Common Velocity

My questions first, then real past papers.

Q11 DR ESLAM Coalesce

M1 · TOPIC 9 2 marks

A (3 kg) at 4 m s^{-1} hits B (2 kg) at rest and they coalesce. Find the common speed.

Q12 DR ESLAM String becomes taut

M1 · TOPIC 9 3 marks

A (m) and B ($2m$) are joined by a slack string. A is projected away from B at $3u$. Find the common speed when the string becomes taut, and the impulse on B.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · October 2019 · Q1

M1 · TOPIC 9 6 marks

1. Two particles, P and Q , have masses $3m$ and $2m$ respectively. The particles are connected by a light inextensible string. Initially P and Q are at rest on a smooth horizontal plane with the string slack.

Particle P is then projected along the plane directly away from Q with speed $4u$. At the same instant, particle Q is projected along the plane in the opposite direction with speed $3u$.

Find

- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

Q14 PAST PAPER Pearson IAL WME01 · October 2022 · Q1

M1 · TOPIC 9 5 marks

1. A railway truck S of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck T of mass 30 tonnes which is at rest. Immediately before the collision the speed of S is 4 m s^{-1}

As a result of the collision, the two railway trucks join together.

Find

- (a) the common speed of the railway trucks immediately after the collision,

(2)

- (b) the magnitude of the impulse exerted on S in the collision, stating the units of your answer.

(3)

Q15**PAST PAPER**

Pearson IAL WME01 · June 2024 · Q1

M1 · TOPIC 9 5 marks

1. Two particles, A and B , have masses m and $3m$ respectively. The particles are connected by a light inextensible string. Initially A and B are at rest on a smooth horizontal plane with the string slack.

Particle A is then projected along the plane away from B with speed U .

Given that the common speed of the particles immediately after the string becomes taut is S

- (a) find S in terms of U .

(2)

- (b) Find, in terms of m and U , the magnitude of the impulse exerted on A immediately after the string becomes taut.

(3)

Q16**PAST PAPER**

Pearson IAL WME01 · June 2026 · Q1

M1 · TOPIC 9**6 marks**

1. Railway truck A has mass $5m$ and railway truck B has mass $3m$. The trucks are moving towards each other in opposite directions along the same straight track when they collide directly. Immediately before the collision the speed of A is u and the speed of B is $1.5u$

As a result of the collision, the trucks couple together.

- (a) Find, in terms of u , the common speed of the trucks immediately after the collision.

(3)

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on B by A in the collision.

(3)

(Total for Question 1 is 6 marks)

04 Impulse in a Collision: Equal and Opposite

In a collision A gives B an impulse, and B gives A the same impulse the other way. Use it instead of conservation when the impulse is given.

WHEN YOU SEE → YOU DO

“the impulse exerted on Q by P is ...”

→ use it on Q: $I = m_Q(v - u)$

“impulse exerted on P by Q”

→ same size, opposite sign

“find the two possible values”

→ the speed after is $\pm$: solve twice

“hence show that $m_2 > \frac{3}{4}m_1$ ”

→ the speed you found must be > 0

1

Use the impulse on the particle it acts on

Positive direction = the direction the impulse pushes that particle.

2

The other particle

Same impulse, opposite sign — or use conservation.

WORKED MODEL • JUNE 2021

P (3m) at ku , Q (5m) at $2u$ towards each other. After: P speed u , Q speed $3u$ reversed.

$$1 \quad I_Q = 5m(3u - (-2u)) = 25mu$$

impulse on Q

$$2 \quad 3m(v_P - ku) = -25mu \Rightarrow v_P = ku - \frac{25}{3}u$$

opposite on P

$$3 \quad v_P = \pm u$$

speed u : two cases

ANSWER $k = \frac{28}{3}$ or $k = \frac{22}{3}$

IDEA 04

Practise: Impulse in a Collision

My questions first, then real past papers.

Q17 **DR ESLAM** Use the impulse given

M1 · TOPIC 9 **3 marks**

P (2 kg) moves at 5 m s^{-1} . In a collision the impulse on P has magnitude 6 N s. Find the two possible speeds of P after the collision.

$$2v - 10 = \pm 6.$$

PAST PAPERS

Q18 PAST PAPER Pearson IAL WME01 · January 2020 · Q1

M1 · TOPIC 9 8 marks

1. Two particles, P and Q , of mass m_1 and m_2 respectively, are moving on a smooth horizontal plane. The particles are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision, both particles are moving with speed u .

The direction of motion of each particle is reversed by the collision.

Immediately after the collision, the speed of Q is $\frac{1}{3}u$.

- (a) Find, in terms of m_2 and u , the magnitude of the impulse exerted by P on Q in the collision. (3)
- (b) Find, in terms of m_1 , m_2 and u , the speed of P immediately after the collision. (3)
- (c) Hence show that $m_2 > \frac{3}{4}m_1$ (2)

Q19**PAST PAPER**

Pearson IAL WME01 · January 2021 · Q2

M1 · TOPIC 9**6 marks**

2. Two particles, P and Q , have masses $2m$ and m respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $2u$.

The magnitude of the impulse exerted on Q by P in the collision is $5mu$.

Find

- (a) the speed of P immediately after the collision, (3)
- (b) the speed of Q immediately after the collision. (3)

Q20 PAST PAPER Pearson IAL WME01 • June 2021 • Q1

M1 • TOPIC 9 7 marks

1. A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

- (a) Find, in terms of m and u , the magnitude of the impulse exerted on Q by P in the collision. (2)

- (b) Find the two possible values of k . (5)

Q21 PAST PAPER Pearson IAL WME01 · June 2022 · Q1

M1 · TOPIC 9 4 marks

1. Two particles, P and Q , are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of Q is $2u$. The mass of Q is $3m$ and the magnitude of the impulse exerted by P on Q in the collision is $4mu$.

Find

(a) the speed of Q immediately after the collision, (3)

(b) the direction of motion of Q immediately after the collision. (1)

Q22 PAST PAPER Pearson IAL WME01 · June 2025 · Q1

M1 · TOPIC 9 6 marks

1.

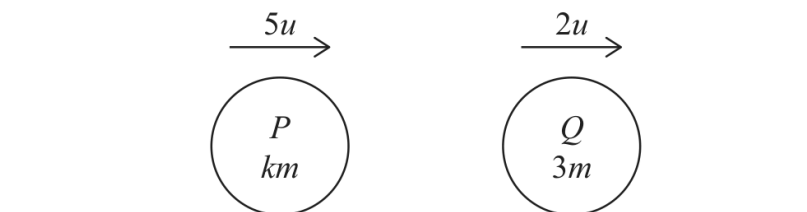
**Figure 1**

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(a) Find v in terms of u only.

(3)

(b) Find the value of k

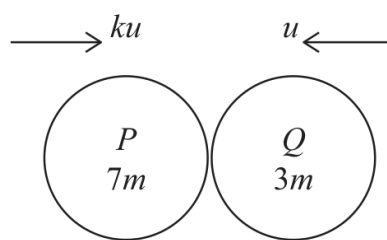
(3)

Q23**PAST PAPER**

Pearson IAL WME01 · October 2025 · Q2

M1 · TOPIC 9 8 marks

2.

**Figure 2**

Particle P of mass $7m$ and particle Q of mass $3m$ are moving in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is ku and the speed of Q is u , as shown in Figure 2.

Immediately **after** the collision, the speed of P is w and the speed of Q is $2w$.

The direction of motion of Q is reversed by the collision.

The impulse received by Q in the collision has magnitude $\frac{7}{2}mu$.

(a) Find w in terms of u .

(3)

(b) Find the two possible values of k .

(5)

Q24 **PAST PAPER** Pearson IAL WME01 · January 2026 · Q1**M1 · TOPIC 9** **6 marks**

1. A particle P of mass $3m$ and a particle Q of mass $5m$ are on a smooth horizontal surface. The particles move towards each other in opposite directions along the same straight line and collide.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $4u$

The magnitude of the impulse exerted on P by Q in the collision is $\frac{33}{2}mu$

Find

- (i) the speed of P immediately after the collision,
- (ii) the speed of Q immediately after the collision.

(6)

05 Momentum with SUVAT, Friction and Slopes

The hardest questions put a collision in the middle of a motion. Split the story into stages.

WHEN YOU SEE → YOU DO

“dropped from a height” / “bounces”

→ $v^2 = u^2 + 2as$ before and after the impulse

“rough”, “coefficient of friction μ ”

→ $F = \mu R$, then $F = ma$ gives the deceleration

“comes to rest after ... m / ... s”

→ SUVAT with $v = 0$

“inclined at θ ”, smooth slope

→ $a = g \sin \theta$ along the slope

WORKED MODEL • THE DROPPED BALL

0.16 kg dropped from 2 m; impulse on the ground 1.44 N s. Greatest height after?

1 $v^2 = 2(9.8)(2) \Rightarrow v = 6.26 \text{ m s}^{-1}$ down

stage 1: SUVAT

2 $1.44 = 0.16(w + 6.26) \Rightarrow w = 2.74 \text{ m s}^{-1}$ up

stage 2: impulse

3 $0 = 2.74^2 - 2(9.8)h$

stage 3: SUVAT

ANSWER $h = 0.383 \text{ m}$

IDEA 05

Practise: Momentum with SUVAT and Friction

Questions written to practise exactly this idea.

Q25**DR ESLAM**

Rough ground after a collision

M1 · TOPIC 9**4 marks**

After a collision a 3 kg block slides at 4 m s^{-1} on rough ground, $\mu = 0.2$. How far does it travel before it stops?

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|--|
| 00 | Always | Before/after diagram • positive direction • all “after” arrows positive. |
| 01 | One body | $I = mv - mu = Ft$. Rebound: the speeds add. Units N s. |
| 02 | Collision | $\sum mu = \sum mv$. The sign of the answer gives the direction. |
| 03 | Together | Coalesce / couple / string taut → one common velocity. |
| 04 | Impulse given | Use it on that particle; the other gets the opposite. “Two values” → $\pm$. |
| 05 | In a story | Before (SUVAT) → collision (momentum) → after (SUVAT). |

BOTTOM LINE Diagram → positive direction → conservation for the collision, impulse for one particle → read the direction from the sign.

Answers

Momentum, Impulse & Collisions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MOMENTUM AND IMPULSE ON ONE BODY

- Q1** Dr Eslam • Momentum and units $15 \text{ m s}^{-1}, 18\,000 \text{ N s}$
- Q2** Dr Eslam • Rebound: the speeds add $0.2(4 + 6) = 2 \text{ N s}$
- Q3** Dr Eslam • Impulse = force $\times$ time $I = 12; 2(v - 5) = 12 \Rightarrow v = 11 \text{ m s}^{-1}$

IDEA 02 • COLLISIONS: CONSERVATION

- Q4** Dr Eslam • Draw, then conserve $24 - 6 = 4 + 2v \Rightarrow v = 7 \text{ m s}^{-1}$, in A's original direction
- Q5** Dr Eslam • Let the sign decide $10 = 2v + 12 \Rightarrow v = -1$: speed 1 m s^{-1} , reversed
- Q6** Pearson IAL WME01 • January 2022 • Q2 (a) $k = \frac{1}{3}$ (b) $1.5mu$
- Q7** Pearson IAL WME01 • January 2023 • Q2 (a)(i) 0.5 (ii) 1.5 m s^{-1} (b) $3m \text{ N s}$
- Q8** Pearson IAL WME01 • June 2023 • Q1 (a) $0.5u$ (b) reversed (c) $10u \text{ N s}$
- Q9** Pearson IAL WME01 • January 2024 • Q2 (a) $x = 2.5$ (b) 10 N s
- Q10** Pearson IAL WME01 • October 2024 • Q1 (a) $5x = 11y$ (b) $\frac{108}{11}mx$

IDEA 03 • ONE COMMON VELOCITY

- Q11** Dr Eslam · Coalesce $12 = 5v \Rightarrow v = 2.4 \text{ m s}^{-1}$
- Q12** Dr Eslam · String becomes taut $v = u; I = 2mu$
- Q13** Pearson IAL WME01 · October 2019 · Q1 (a) $1.2u$ (b) $8.4mu$
- Q14** Pearson IAL WME01 · October 2022 · Q1 (a) 1.6 m s^{-1} (b) $48\,000 \text{ N s}$
- Q15** Pearson IAL WME01 · June 2024 · Q1 (a) $S = \frac{U}{4}$ (b) $\frac{3}{4}mU$
- Q16** Pearson IAL WME01 · June 2026 · Q1 (a) $\frac{u}{16}$ (b) $\frac{75}{16}mu$

IDEA 04 · IMPULSE IN A COLLISION

- Q17** Dr Eslam · Use the impulse given $v = 8 \text{ or } v = 2 \text{ (both in the original direction)}$
- Q18** Pearson IAL WME01 · January 2020 · Q1 (a) $\frac{4}{3}m_2u$ (b) $\frac{(4m_2 - 3m_1)u}{3m_1}$ (c) speed > 0
- Q19** Pearson IAL WME01 · January 2021 · Q2 (a) $0.5u$ (b) $3u$
- Q20** Pearson IAL WME01 · June 2021 · Q1 (a) $25mu$ (b) $k = \frac{28}{3} \text{ or } \frac{22}{3}$
- Q21** Pearson IAL WME01 · June 2022 · Q1 (a) $\frac{2}{3}u$ (b) unchanged
- Q22** Pearson IAL WME01 · June 2025 · Q1 (a) $v = 3.5u$ (b) $k = 1.5$
- Q23** Pearson IAL WME01 · October 2025 · Q2 (a) $w = \frac{u}{12}$ (b) $k = \frac{7}{12} \text{ or } \frac{5}{12}$
- Q24** Pearson IAL WME01 · January 2026 · Q1 (i) $2.5u$ (ii) $0.7u$

IDEA 05 · MOMENTUM WITH SUVAT AND FRICTION

Q25 Dr Eslam · Rough ground after a collision

$$a = 0.2g = 1.96; \quad s = \frac{16}{2(1.96)} = 4.08 \text{ m}$$

Moments

Draw every force on the rod. Take moments about the point you do not care about. Then resolve.
On the point of tilting, one reaction is zero.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Moments

01 - The moment of a force

force $\times$ perpendicular distance resultant moment

02 - A rod in equilibrium

two supports uniform rod $T_C = 2T_D$

03 - On the point of tilting

other reaction = 0 range of M

04 - Non-uniform rods

find the centre of mass two tilting cases

THE TWO EQUATIONS OF EVERY MOMENTS QUESTION

$\uparrow$ forces = $\downarrow$ forces clockwise moments =
anticlockwise moments

resolve vertically

about any point you choose

01 The Moment of a Force

Moment = force $\times$ **perpendicular** distance from the point to the line of the force. Units: N m. Always say clockwise or anticlockwise.

WHEN YOU SEE $\rightarrow$ YOU DO

“moment of the force about P ”

$\rightarrow F \times d_{\perp}$, with a direction

the distance is not perpendicular (angle θ)

$\rightarrow F \times d \sin \theta$ (or use the perpendicular component of F)

“resultant moment” / several forces

$\rightarrow$ choose a positive sense, add with signs

force passes through the point

$\rightarrow$ its moment is zero



moment about P
 $= F \times d \sin \theta$ (anticlockwise)

WORKED MODEL • RESULTANT MOMENT

Light rod: 4 N up at 1.2 m left of P ; 2 N and 1 N up at 1.2 m and 4.8 m right of P .

1 anticlockwise $+$: $-4(1.2) + 2(1.2) + 1(4.8)$

signs from the sense

2 $-4.8 + 2.4 + 4.8$

each $F \times d$

ANSWER 2.4 N m anticlockwise

IDEA 01

Practise: The Moment of a Force

Questions written to practise exactly this idea.

Q1 DR ESLAM Perpendicular distance

M1 · TOPIC 10 2 marks

A force of 12 N acts at the end of a 3 m rod, at 30° to the rod. Find its moment about the other end.

Q2 DR ESLAM Resultant with signs

M1 · TOPIC 10 3 marks

A light rod AB of length 4 m. Forces: 5 N down at A , 3 N down at B , 2 N up at the midpoint M . Find the resultant moment about M .

02 A Rod in Equilibrium

Two unknown forces, two equations. Take moments about one unknown so it disappears, then resolve for the other.

WHEN YOU SEE → YOU DO

“rests on two supports” / “held by two ropes”

→ two unknown reactions / tensions

“uniform rod”

→ weight at the **midpoint**

“the tension at C is twice the tension at D”

→ write both as T and $2T$ first

“in terms of g ” / masses in kg

→ forces are mg ; keep g or use 9.8

“state how you used ... rod / particle”

→ rod: stays straight · particle: weight at one point

WORKED MODEL · JUNE 2026

Rod 6 m, 20 kg, centre of mass 4 m from A; strings at P (0.5 m) and Q (4.5 m).

1 $M(P): T_Q(4) = 20g(3.5)$

moments about P

2 $T_Q = 17.5g$

solve

ANSWER $T_Q = 171.5 \text{ N}$

IDEA 02

Practise: A Rod in Equilibrium

My questions first, then real past papers.

Q3 DR ESLAM Moments about a support

M1 · TOPIC 10 3 marks

A uniform rod AB , 4 m, weight 60 N, rests on supports at A and C where $AC = 3$ m. Find both reactions.

Take moments about A first.

Q4 DR ESLAM Ratio of tensions

M1 · TOPIC 10 4 marks

A uniform rod AB , 6 m, mass 10 kg, hangs from ropes at A and at C . The tension at C is 4 times the tension at A . Find AC .

$AC =$ _____

PAST PAPERS

Q5 PAST PAPER Pearson IAL WME01 · June 2023 · Q4

M1 · TOPIC 10 12 marks

4.

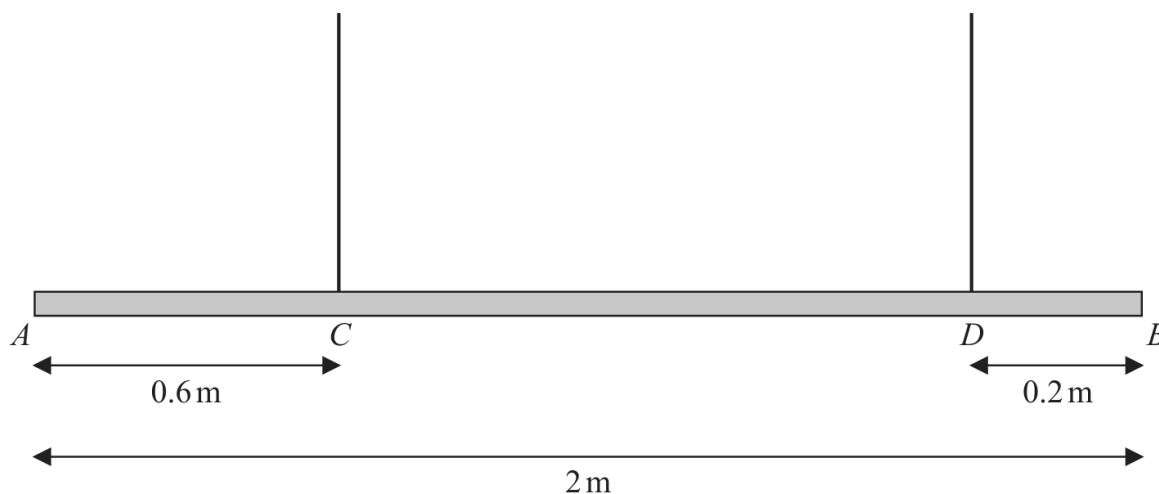


Figure 1

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

- (a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D , use the model to find the exact value of p .

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A .

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

- (b) Using the model, find the exact value of q

(4)

- (c) State how you have used the modelling assumption that the beam is uniform.

(1)

Q6 PAST PAPER Pearson IAL WME01 · October 2023 · Q1

M1 · TOPIC 10 5 marks

1.

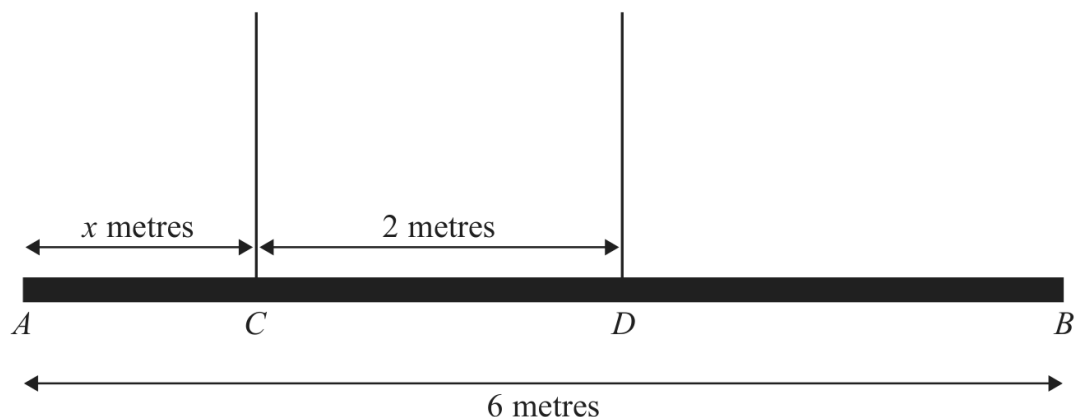
**Figure 1**

Figure 1 shows a beam AB with weight 24 N and length 6 m .

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2\text{ m}$.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D .

The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

- (i) the tension in the rope attached to the beam at D .
- (ii) the value of x .

(5)

Q7 PAST PAPER Pearson IAL WME01 · January 2024 · Q5

M1 · TOPIC 10 10 marks

5.

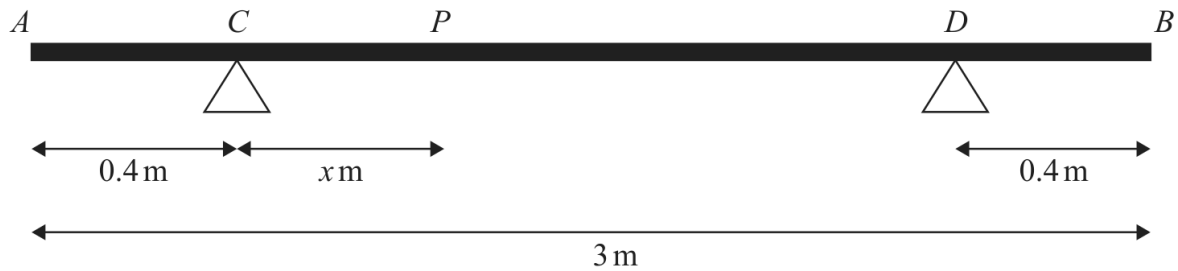


Figure 4

A beam AB has mass 30 kg and length 3 m.

The beam rests on supports at C and D where $AC = 0.4$ m and $DB = 0.4$ m, as shown in Figure 4.

A person of mass 55 kg stands on the beam between C and D .

The person is modelled as a particle at the point P , where $CP = x$ metres and $0 < x < 2.2$

The beam is modelled as a uniform rod resting in equilibrium in a horizontal position.

Using the model,

(a) show that the magnitude of the reaction at C is $(686 - 245x)$ N. (3)

The magnitude of the reaction at C is **four** times the magnitude of the reaction at D .

Using the model,

(b) find the value of x (4)

The person steps off the beam and places a package of mass M kg at A .

The package is modelled as a particle at the point A .

The beam is now on the point of tilting about C .

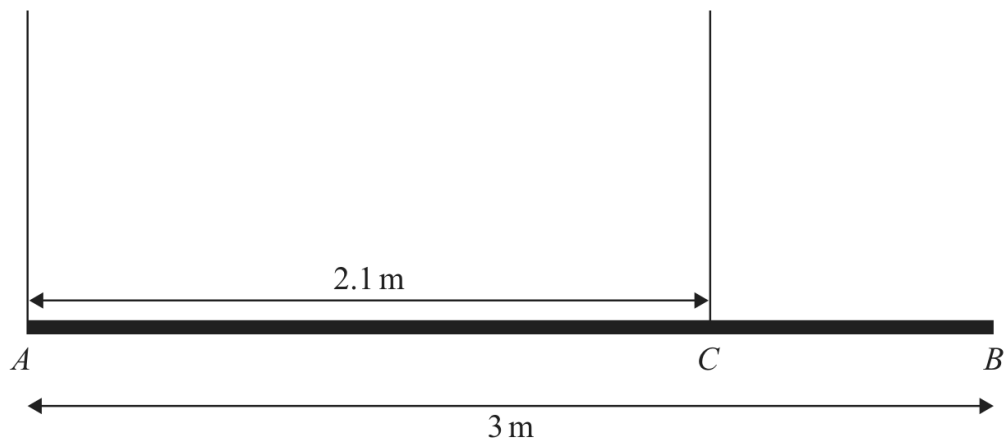
Using the model,

(c) find the value of M (3)

Q8 PAST PAPER Pearson IAL WME01 · January 2025 · Q4

M1 · TOPIC 10 10 marks

4.

**Figure 2**

A uniform rod AB has length 3 m and weight W newtons.

The rod is suspended by two light vertical ropes.

One rope is attached to the rod at A and the other rope is attached to the rod at C , where $AC = 2.1$ m.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

The tension in the rope at C is 350 N.

(a) Show that $W = 490$

(3)

A particle P of weight 210 N is attached to the rod at a distance d metres from A .

The tension in the rope at C is now 600 N.

The rod remains in equilibrium in a horizontal position.

(b) Find the value of d .

(3)

Particle P is removed from the rod.

A particle Q of weight X newtons is now attached at B .

The rod remains in equilibrium in a horizontal position and is now on the point of tilting.

(c) Find the value of X .

(4)

Q9 PAST PAPER Pearson IAL WME01 · October 2025 · Q1**M1 · TOPIC 10** 7 marks

1.

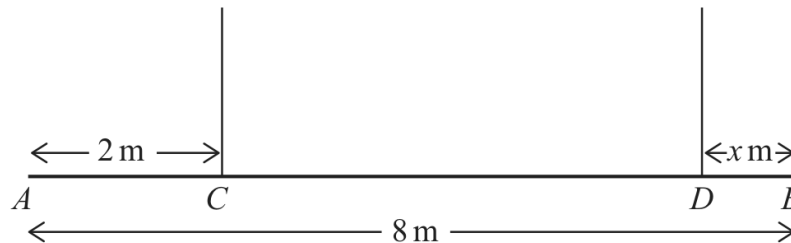
**Figure 1**

Figure 1 shows a sketch of a beam AB , with weight 240 N and length 8 m .

The beam is held in equilibrium in a horizontal position by two vertical ropes. The ropes are attached to the beam at the points C and D , where $AC = 2\text{ m}$ and $DB = x\text{ metres}$.

The beam is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

The tension in the rope at D is 90 N .

(a) Show that $x = \frac{2}{3}$ (3)

The rope at C will break if its tension exceeds 183 N . The rope at D cannot break. A package of weight W newtons is now attached to the beam at A . The beam remains horizontal and in equilibrium.

The package is modelled as a particle.

It is given that the rope at C does not break.

(b) Find the greatest possible value of W . (4)

Q10 PAST PAPER Pearson IAL WME01 · January 2026 · Q4

M1 · TOPIC 10 9 marks

4.

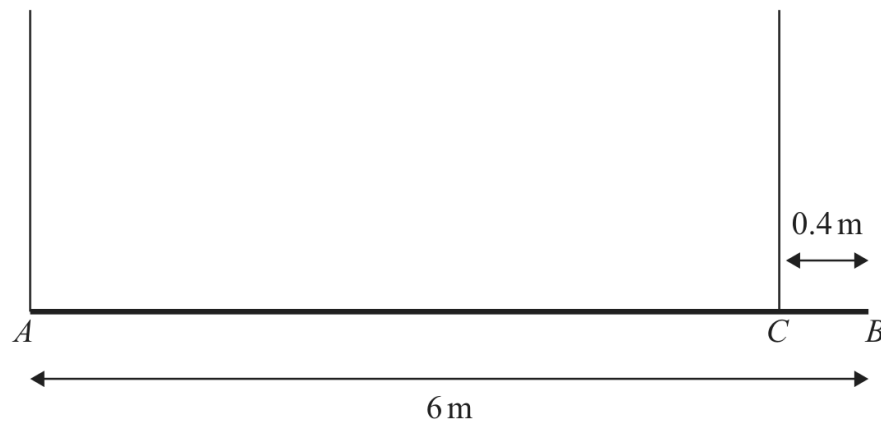
**Figure 2**

Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

The rod has mass 30 kg.

A particle of mass 20 kg is now attached to the rod at B .

The rod is in equilibrium in a horizontal position.

(a) Find

- (i) the tension in the rope attached to the rod at C ,
- (ii) the tension in the rope attached to the rod at A .

(6)

The particle of mass 20 kg at B is removed and replaced by a particle of mass M kg.

The rod remains in equilibrium in a horizontal position.

(b) Find the exact maximum value of M .

(3)

03 On the Point of Tilting

The rod is about to turn about one support. The other support has lifted: its reaction is **zero**.

WHEN YOU SEE → YOU DO

“on the point of tilting about D ”

→ reaction at the other support = 0 • moments about D

“largest / smallest mass so the rod stays horizontal”

→ it is on the point of tilting

“a rope is about to go slack”

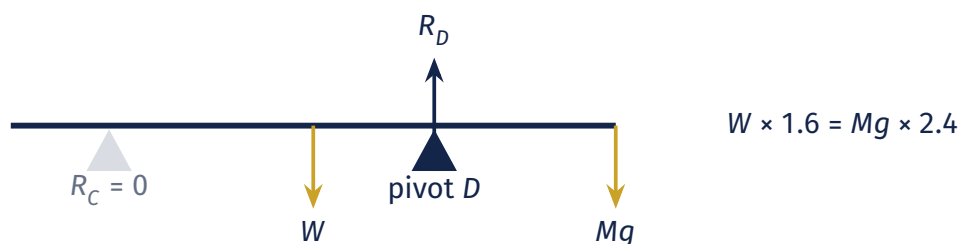
→ that tension = 0

“hence find the range of M ”

→ each reaction ≥ 0

overhangs a step / ledge

→ the edge is the pivot



WORKED MODEL • JANUARY 2022

Uniform beam 5 m on a step, edge D with $DB = 1.3$ m; a 30 kg boy at C ($CB = 0.5$ m) — on the point of tilting.

- 1 pivot D ; weight of the beam 1.2 m on the other side

tilting: A lifts

- 2 $M(1.2) = 30(0.8)$

moments about D

ANSWER $M = 20$ kg

IDEA 03

Practise: On the Point of Tilting

My questions first, then real past papers.

Q11 DR ESLAM Reaction becomes zero

M1 · TOPIC 10 3 marks

A uniform plank AB , 5 m, mass 20 kg, rests on supports at C ($AC = 1$ m) and B . How far from C , towards A , can a 40 kg child stand before the plank tilts?

Q12 DR ESLAM Range of a mass

M1 · TOPIC 10 4 marks

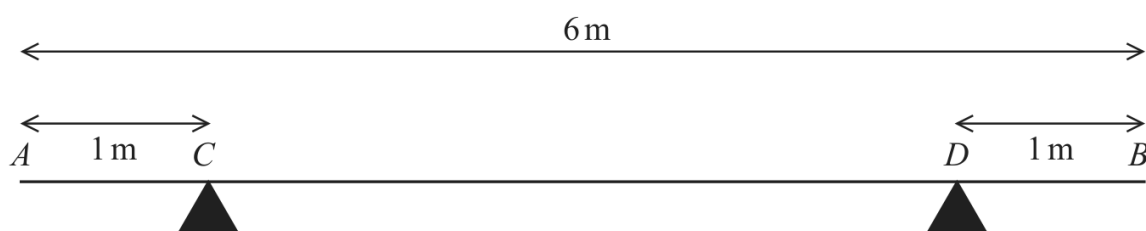
Uniform rod AB , 4 m, 10 kg, on supports 1 m from each end. A mass M kg hangs at B . Find the largest M so that the rod stays horizontal.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · January 2021 · Q4

M1 · TOPIC 10 6 marks

4.

**Figure 2**

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

When a force of magnitude 900 N is applied vertically upwards to the girder at A , the girder is about to tilt about D .

When a force of magnitude 1500 N is applied vertically upwards to the girder at B , the girder is about to tilt about C .

The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A .

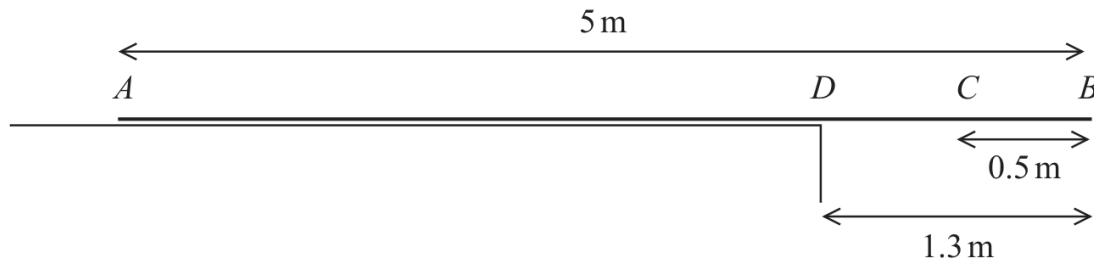
Find the value of x .

(6)

Q14 PAST PAPER Pearson IAL WME01 · January 2022 · Q3

M1 · TOPIC 10 7 marks

3.

**Figure 2**

A beam $ADCB$ has length 5 m . The beam lies on a horizontal step with the end A on the step and the end B projecting over the edge of the step. The edge of the step is at the point D where $DB = 1.3\text{ m}$, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at C , where $CB = 0.5\text{ m}$, the beam is on the point of tilting.

The boy is modelled as a particle and the beam is modelled as a uniform rod.

(a) Find the mass of the beam.

(3)

A block of mass $X\text{ kg}$ is now placed on the beam at A .

The block is modelled as a particle.

(b) Find the smallest value of X that will enable the boy to stand on the beam at B without the beam tilting.

(3)

(c) State how you have used the modelling assumption that the block is a particle in your calculations.

(1)

Q15 PAST PAPER Pearson IAL WME01 · June 2022 · Q5

M1 · TOPIC 10 9 marks

5.

**Figure 2**

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

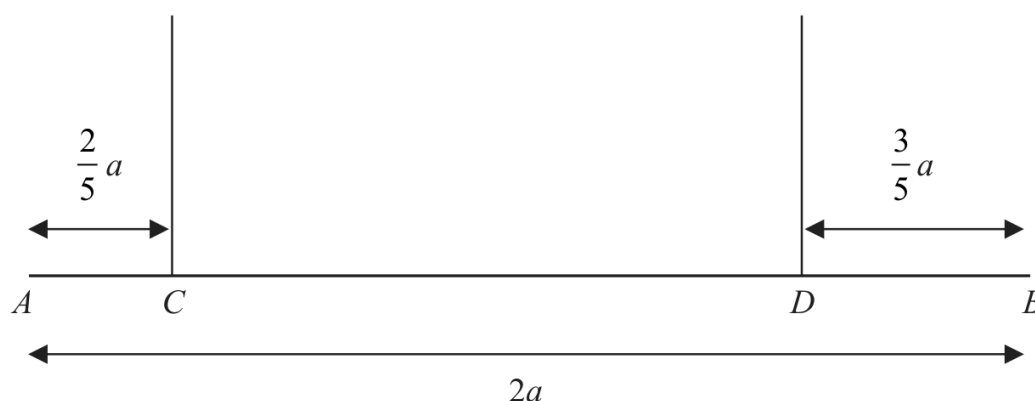
A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

- (a) Find, in terms of M , the magnitude of the reaction on the rod at C . (3)
- (b) Find, in terms of M , the magnitude of the reaction on the rod at D . (3)
- (c) Hence, or otherwise, find the range of possible values of M . (3)

Q16 PAST PAPER Pearson IAL WME01 · October 2022 · Q2

M1 · TOPIC 10 6 marks

2.

**Figure 1**

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the largest possible mass of the particle P (3)

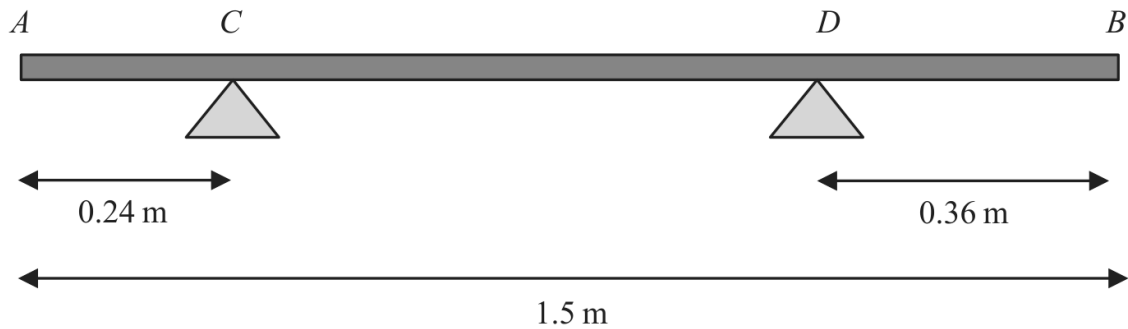
Given that the mass of P is $\frac{1}{2}M$

(b) find, in terms of M and g , the tension in the string that is attached to the rod at C . (3)

Q17 PAST PAPER Pearson IAL WME01 · January 2023 · Q4

M1 · TOPIC 10 8 marks

4.

**Figure 1**

A branch AB , of length 1.5 m, rests horizontally in equilibrium on two supports.

The two supports are at the points C and D , where $AC = 0.24$ m and $DB = 0.36$ m, as shown in Figure 1.

When a force of 150 N is applied vertically upwards at B , the branch is on the point of tilting about C .

When a force of 225 N is applied vertically downwards at B , the branch is on the point of tilting about D .

The branch is modelled as a non-uniform rod AB of weight W newtons.

The distance from the point C to the centre of mass of the rod is x metres.

Use the model to find

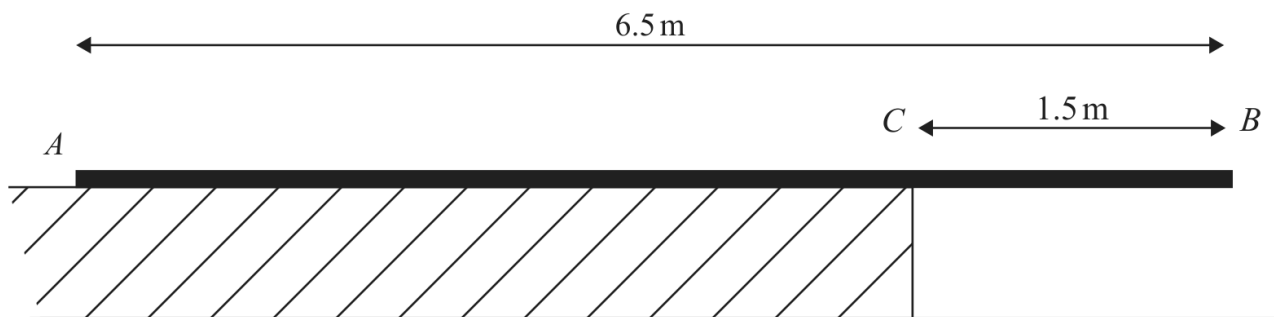
- (i) the value of W
- (ii) the value of x

(8)

Q18 PAST PAPER Pearson IAL WME01 · June 2024 · Q4

M1 · TOPIC 10 6 marks

4.

**Figure 2**

A non-uniform rod AB has length 6.5 m and mass 1.2 kg . The centre of mass of the rod is 3 m from A . The rod rests on a horizontal step and overhangs the end of the step C by 1.5 m , as shown in Figure 2.

The rod is perpendicular to the edge of the step.

A particle of mass 4 kg is placed on the rod at B and another particle, whose mass is $M\text{ kg}$, is placed on the rod at D , where $AD = 0.5\text{ m}$.

The rod remains in equilibrium in a horizontal position.

(a) Find the smallest possible value of M .

(3)

The particle at B **and** the particle at D are now **removed**.

A new particle is placed on the rod at the point E , where $EB = 0.9\text{ m}$.

The rod remains in equilibrium in a horizontal position but is on the point of tilting about C .

(b) Find the magnitude of the force acting on the rod at C .

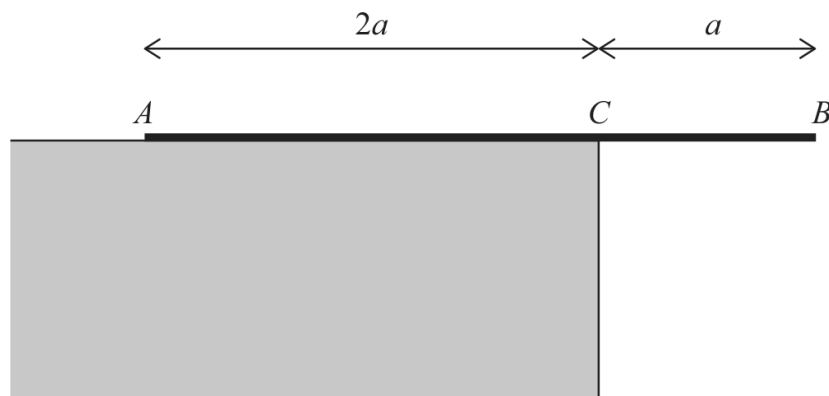
(3)

Q19**PAST PAPER**

Pearson IAL WME01 · June 2025 · Q5

M1 · TOPIC 10**6 marks**

5.

**Figure 3**

A non-uniform rod AB of length $3a$ rests in equilibrium on a horizontal ledge and overhangs the edge of the ledge at C .

The point C is such that $AC = 2a$ and $CB = a$, as shown in Figure 3.

The rod has weight W .

The distance of the centre of mass of the rod from A is d .

The rod is perpendicular to the edge of the ledge.

When a force of magnitude P , **acting vertically upwards**, is applied to the rod at B , the rod is on the point of tilting about A .

When the force applied at B is replaced by a force of magnitude $1.25P$, **acting vertically downwards** at B , the rod is on the point of tilting about C .

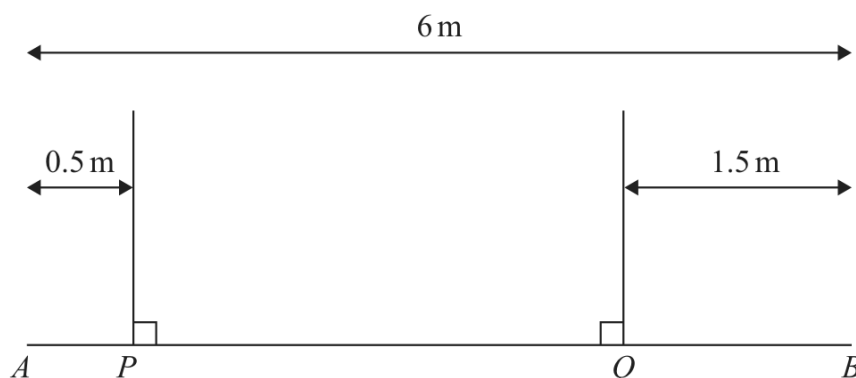
Find d in terms of a .

(6)

Q20 PAST PAPER Pearson IAL WME01 · June 2026 · Q2

M1 · TOPIC 10 6 marks

2.

**Figure 1**

A non-uniform rod AB has mass 20 kg and length 6 m. The rod is held in equilibrium in a horizontal position by two light vertical strings. One string is attached to the rod at P , where $AP = 0.5$ m and the other is attached to the rod at Q , where $QB = 1.5$ m, as shown in Figure 1.

The centre of mass of the rod is 4 m from A .

- (a) Find the tension in the string attached to the rod at Q . (3)

A particle is now attached to the rod at B .

The rod remains in equilibrium in a horizontal position but is on the point of tilting.

The tension in the string attached to the rod at Q is now X newtons.

- (b) Find the value of X . (3)

(Total for Question 2 is 6 marks)

04 Non-Uniform Rods: Find the Centre of Mass

The centre of mass is an unknown distance x . Two situations give two equations — solve them together.

WHEN YOU SEE → YOU DO

“non-uniform rod”

→ centre of mass at x from A — an unknown

“the reaction at D is three times the reaction at C ”

→ R and $3R$, then moments

two “point of tilting” situations

→ two equations → divide to cancel W

“show that $x = 1.38$ ”

→ finish with the exact number given

WORKED MODEL • JANUARY 2021

Girder 6 m, supports at 1 m and 5 m. 900 N up at A → about to tilt about D ; 1500 N up at B → about to tilt about C .

$$1 \quad 900(5) = W(5 - x)$$

about D , $R_D = 0$

$$2 \quad 1500(5) = W(x - 1)$$

about C , $R_C = 0$

$$3 \quad \frac{4500}{7500} = \frac{5 - x}{x - 1}$$

divide: W cancels

ANSWER $x = 3.5 \text{ m}$

IDEA 04

Practise: Non-Uniform Rods

My questions first, then real past papers.

Q21 DR ESLAM Find x from a ratio

M1 · TOPIC 10 4 marks

A non-uniform rod AB , 5 m, weight W , rests on supports at A and B . The reaction at B is 1.5 times the reaction at A . Find the distance of the centre of mass from A .

PAST PAPERS

Q22 PAST PAPER Pearson IAL WME01 · October 2019 · Q5

M1 · TOPIC 10 8 marks

5.

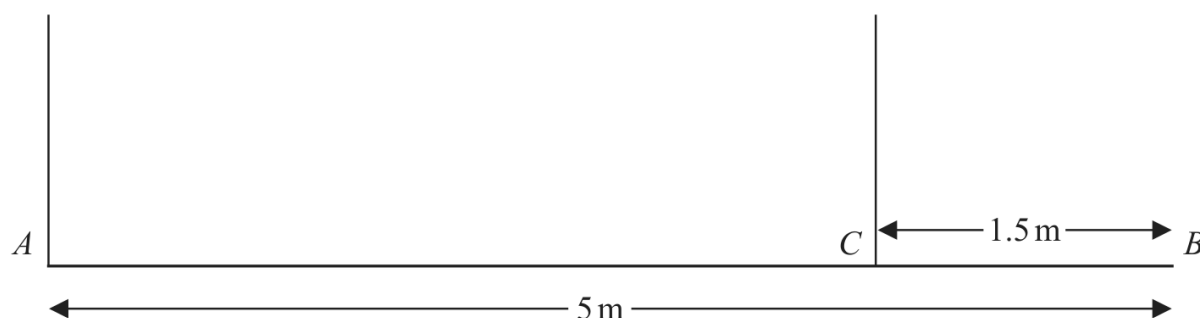


Figure 2

A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

A particle of mass M kg is now placed on the beam at B and the beam remains in equilibrium in a horizontal position.

(a) Find the largest possible value of M .

(3)

The particle at B is now removed and a particle of mass 15 kg is now placed on the beam at the point D , where $AD = x$ metres. The beam remains in equilibrium in a horizontal position.

Given that the tension in the rope attached to the beam at C is now twice the tension in the rope attached to the beam at A ,

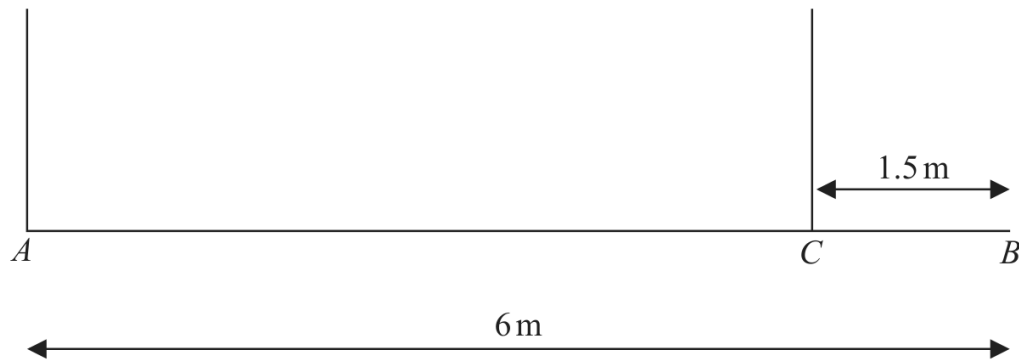
(b) find the value of x .

(5)

Q23 PAST PAPER Pearson IAL WME01 · January 2020 · Q2

M1 · TOPIC 10 7 marks

2.

**Figure 1**

A non-uniform beam AB has length 6 m and weight W newtons. The beam is supported in equilibrium in a horizontal position by two vertical ropes, one attached to the beam at A and the other attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 1.

The centre of mass of the beam is 2.625 m from A .

The ropes are modelled as light strings. The beam is modelled as a non-uniform rod.

Given that the tension in the rope attached at C is 20 N greater than the tension in the rope attached at A ,

(a) find the value of W .

(6)

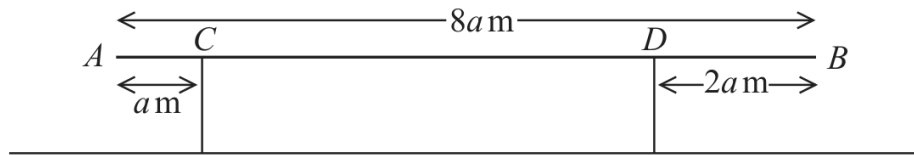
(b) State how you have used the fact that the beam is modelled as a rod.

(1)

Q24 PAST PAPER Pearson IAL WME01 · June 2021 · Q7

M1 · TOPIC 10 10 marks

7.

**Figure 2**

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A ,

(a) find an expression for x in terms of a .

(5)

A box of mass M kg is placed on the beam at E , where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D .

By modelling the box as a particle,

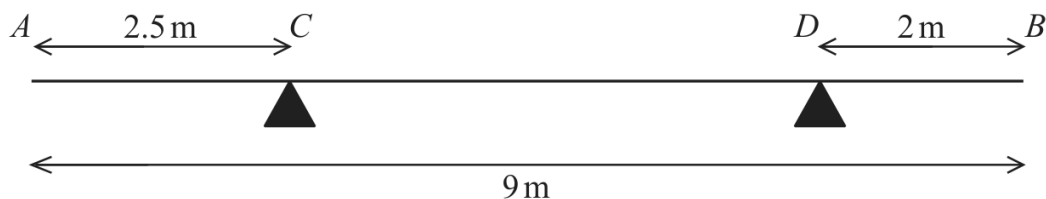
(b) find the value of M .

(5)

Q25 PAST PAPER Pearson IAL WME01 · October 2021 · Q1

M1 · TOPIC 10 6 marks

1.

**Figure 1**

A non-uniform rod AB has length 9 m and mass M kg.

The rod rests in equilibrium in a horizontal position on two supports, one at C where $AC = 2.5$ m and the other at D where $DB = 2$ m, as shown in Figure 1.

The magnitude of the force acting on the rod at D is twice the magnitude of the force acting on the rod at C .

The centre of mass of the rod is d metres from A .

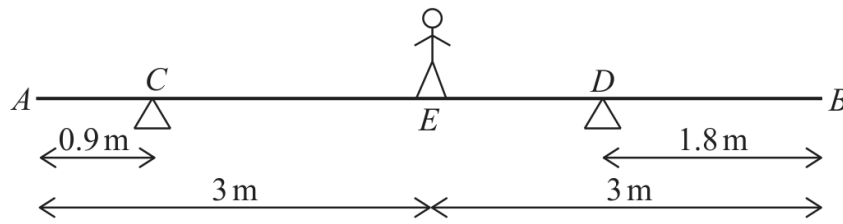
Find the value of d .

(6)

Q26 PAST PAPER Pearson IAL WME01 · October 2024 · Q2

M1 · TOPIC 10 9 marks

2.

**Figure 1**

A non-uniform beam AB has length 6 m and mass 50 kg. The beam rests horizontally on two supports at C and D , where $AC = 0.9$ m and $DB = 1.8$ m.

A child of mass 25 kg stands on the beam at E , where $AE = EB = 3$ m, as shown in Figure 1.

The beam is in equilibrium.

The magnitude of the normal reaction between the beam and the support at C is R_C newtons.

The magnitude of the normal reaction between the beam and the support at D is R_D newtons.

The beam is modelled as a rod and the child is modelled as a particle.

The centre of mass of the beam is between C and D and is a distance x metres from D .

Given that $2R_D = 3R_C$

(a) show that $x = 1.38$

(6)

The child remains at E and a block of mass M kg is placed on the beam at B .

The block is modelled as a particle.

Given that the beam is on the point of tilting,

(b) find the value of M .

(3)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|---|
| 00 | Always | Diagram with every force and every distance.
Uniform $\rightarrow$ midpoint. |
| 01 | Moment | $F \times$ perpendicular distance ($d \sin \theta$). Say the direction. |
| 02 | Equilibrium | Moments about an unknown force, then resolve. |
| 03 | Tilting | The other reaction is 0. Largest/smallest/range $\rightarrow$ tilting, reactions ≥ 0 . |
| 04 | Non-uniform | $\times$ unknown $\cdot$ two situations $\cdot$ two equations $\cdot$ divide. |

BOTTOM LINE Diagram $\rightarrow$ moments about the force you do not need $\rightarrow$ resolve $\rightarrow$ tilting means a zero reaction.

Answers

Moments • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE MOMENT OF A FORCE

- Q1** Dr Eslam • Perpendicular distance
 $12 \times 3 \sin 30^\circ = 18 \text{ N m}$
- Q2** Dr Eslam • Resultant with signs
 $5(2) - 3(2) = 4 \text{ N m}$ (in the sense of the 5 N force)

IDEA 02 • A ROD IN EQUILIBRIUM

- Q3** Dr Eslam • Moments about a support
 $R_C = 40 \text{ N}, R_A = 20 \text{ N}$
- Q4** Dr Eslam • Ratio of tensions
 $T_A = 2g, T_C = 8g; 8g(AC) = 10g(3) \Rightarrow AC = 3.75 \text{ m}$
- Q5** Pearson IAL WME01 • June 2023 • Q4
 (a) $p = \frac{4}{21}$ (b) $q = \frac{2}{3}$ (c) weight at the midpoint
- Q6** Pearson IAL WME01 • October 2023 • Q1
 (i) 8 N (ii) $x = \frac{7}{3}$
- Q7** Pearson IAL WME01 • January 2024 • Q5
 (b) $x = 0.08$ (c) $M = 82.5$
- Q8** Pearson IAL WME01 • January 2025 • Q4
 (b) $d = 2.5$ (c) $X = \frac{980}{3} \approx 327$
- Q9** Pearson IAL WME01 • October 2025 • Q1
 (b) $W = 24$
- Q10** Pearson IAL WME01 • January 2026 • Q4
 (a)(i) 367.5 N (ii) 122.5 N (b) $M = 195$

IDEA 03 • ON THE POINT OF TILTING

- Q11** Dr Eslam · Reaction becomes zero
- Q12** Dr Eslam · Range of a mass
- Q13** Pearson IAL WME01 · January 2021 · Q4
- Q14** Pearson IAL WME01 · January 2022 · Q3
- Q15** Pearson IAL WME01 · June 2022 · Q5
- Q16** Pearson IAL WME01 · October 2022 · Q2
- Q17** Pearson IAL WME01 · January 2023 · Q4
- Q18** Pearson IAL WME01 · June 2024 · Q4
- Q19** Pearson IAL WME01 · June 2025 · Q5
- Q20** Pearson IAL WME01 · June 2026 · Q2
- $40d = 20(1.5) \Rightarrow d = 0.75 \text{ m}$
- tilting about the support near B: $M(1) = 10(1) \Rightarrow M = 10$
 $x = 3.5$
- (a) 20 kg (b) $X = 4.05$ (c) its weight acts exactly at A
- (a) $(16.25 - M)g$ (b) $(2M - 1.25)g$ (c) $0.625 \leq M \leq 16.25$
- (a) $\frac{2}{3}M$ (b) $0.1Mg$
- (i) $W = 300$ (ii) $x = 0.63$
- (a) $M = 0.8$ (b) $5.2g = 51.0 \text{ N}$
- $d = \frac{24}{17}a$
- (a) 171.5 N (b) $X = \frac{80}{3}g \approx 261$

IDEA 04 · NON-UNIFORM RODS

- Q21** Dr Eslam · Find x from a ratio
 $R_B = 0.6W$; $0.6W(5) = Wx \Rightarrow x = 3 \text{ m}$
- Q22** Pearson IAL WME01 · October 2019 · Q5
(a) $M = 14$ (b) $x = 2.8$
- Q23** Pearson IAL WME01 · January 2020 · Q2
(a) $W = 120$ (b) the beam stays straight
- Q24** Pearson IAL WME01 · June 2021 · Q7
(a) $x = 4.75a$ (b) $M = 50$
- Q25** Pearson IAL WME01 · October 2021 · Q1
 $d = 5.5$
- Q26** Pearson IAL WME01 · October 2024 · Q2
(a) $x = 1.38$ (b) $M = 55$